Sparse Discrepancy Beyond Additive Functions: Acyclic, Bounded-Degree and Degenerate Sparsity Patterns

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Abstract

Hollender, Manurangsi, Meka and Suksompong recently initiated the study of discrepancy for non-additive set functions: given 1-Lipschitz functions $f_1, \dots, f_n$ on the subsets of a ground set of m items, two-colour the items so that every f_i takes nearly the same value on the two colour classes. They asked for an analogue of the Beck–Fiala theorem: if every item is relevant to at most t of the functions, is the discrepancy bounded by a function of t alone? This is open even for $t = 2$, where the sparsity pattern is a multigraph whose parallel classes of items (*bundles*) may be arbitrarily large.

For $t = 2$ we give bounds that depend only on the sparsity pattern, and neither on the number of functions nor on the bundle sizes. Our main result is that the discrepancy is at most $\max(2, d)$, where d is the degeneracy of the pattern, and at most $p + 1$, where p is its pseudoarboricity, independently of the number of functions and items, the vertex degrees and the bundle sizes. Hence families whose pattern is 2-degenerate (for example outerplanar or series–parallel) have discrepancy at most 2, which is tight, and every proper minor-closed class of patterns has bounded discrepancy. Refinements of the method give the bound 2 for wheels, 3 for planar patterns, and $\lceil n/2 \rceil$ for every 2-sparse family of $n \geq 3$ functions ($\lfloor n/2 \rfloor$ with a computer-assisted base case). The proof deletes the functions one at a time. When a function is deleted, each of its neighbours receives a local constraint of the same type on its remaining items: either one level lower (a *hub-transfer lemma*), or, for as many neighbours as the level of the deleted function, with no loss at all (a *near-loop transfer*). Both rest on dichotomy and separation statements for sets that block all antipodal paths, respectively all *near-loops*, in the Hamming cube, which we prove on the coned 2-skeleton of the cube using its simple connectivity and the Borsuk–Ulam theorem for a two-cut necklace sphere mapped into it.

For general t, if the bundle hypergraph is Berge-acyclic, the discrepancy is at most $\mathcal{K}_t$, the worst-case discrepancy of t arbitrary 1-Lipschitz functions, and this is tight; if every function meets at most Δ shared bundles, it is at most $t\Delta \max(1, 2^{\Delta-2})$. We also determine $\mathcal{K}_1, \dots, \mathcal{K}_6 = 1, 2, 2, 3, 3, 4$ (the upper bounds for $n \geq 3$ are computer-assisted, in exact arithmetic), and show $\mathcal{K}_8 \geq 5$. The same computer-assisted method, run at a threshold just below an integer, shows that every 2-sparse family of at most six functions has discrepancy at most 2, and that eight functions give at most 3. The question remains open for $t = 2$ on every pattern class of unbounded degeneracy, already for complete graphs with single-item bundles. For n functions, a weighted union bound gives $\sqrt{2\ln(2t)\, n}$ for every t-sparse family, which is below $1.666\sqrt{n}$ for $t = 2$ with arbitrary bundle sizes, and a deterministic rounding gives $\sqrt{2n} + O(1)$ for single-item bundles. These results, and exhaustive SAT computations, support the conjecture that every 2-sparse family has discrepancy at most 2.

1 Introduction

Discrepancy. In combinatorial discrepancy one colours the elements of a ground set red and blue so that every set of a given set system is split as evenly as possible. Two classical results frame the area. Spencer's theorem [Spe85] bounds the discrepancy of any n sets by $O(\sqrt{n})$, independently of the number of elements. The Beck–Fiala theorem [BF81] is a statement about *sparse* systems: if every element lies in at most t sets, then the discrepancy is at most $2t-1$, independently of both the number of sets and the number of elements. Improving the Beck–Fiala bound to $O(\sqrt{t})$ is a central open problem. Banaszczyk [Ban98] proved $O(\sqrt{t\log n})$, and Bansal and Jiang [BJ25b, BJ25a] recently made substantial progress. The proofs of these results are *additive* at heart. The Beck–Fiala proof iteratively rounds an extreme point of a linear system with few tight constraints. Spencer's theorem and its algorithmic versions [Ban10, LM15] rely on the partial colouring method of Beck [Bec81] and on the triangle inequality to combine partial colourings.

Discrepancy beyond additive functions. Hollender, Manurangsi, Meka and Suksompong [HMMS25] and, independently, Dupré la Tour and Fujii [DF25] initiated the study of discrepancy for *non-additive* functions. Let $f_1,\dots,f_n\colon 2^{[m]}\to\mathbb{R}$ be set functions on m items that are 1-*Lipschitz*: adding or removing one item changes each value by at most 1. Following [HMMS25], the discrepancy of the family $\mathcal{F}=\{f_1,\dots,f_n\}$ is

$$\operatorname{disc}(\mathcal{F}) \;=\; \min_{\chi\colon [m]\to[2]}\ \max_{i\in[n]} |f_i(\chi^{-1}(1)) - f_i(\chi^{-1}(2))|.$$

Additive functions $f(S)=\sum_{j\in S} a_j$ with $|a_j|\le 1$ are the classical case. The motivation comes from fair division. A *consensus halving* splits a set of goods into two parts that every agent values equally. When the goods are indivisible, one asks for a split that is fair up to a small number of goods, and the agents' valuations need not be additive. The main theorem of [HMMS25] (also proved in [DF25]) states that every 1-Lipschitz family of n functions has discrepancy $O(\sqrt{n\log n})$; precisely, $\operatorname{disc}(\mathcal{F}) < \sqrt{2n(1+\ln 2n)}$ [HMMS25, Theorem 8]. As a consequence, a consensus halving up to $O(\sqrt{n\log n})$ goods exists for n agents with monotone valuations [HMMS25, Theorem 3]. Since partial colouring and linear algebra are unavailable, the proof is topological. A consensus-halving theorem for non-additive valuations [SS03] (a Borsuk–Ulam type statement) yields a fractional colouring with at most n fractional items on which all multilinear extensions are balanced [HMMS25, Lemma 9], and randomized rounding finishes the proof.

The sparse question. An item is *relevant* to f if it affects the value of f, and $\mathcal{F}$ is t-*sparse* if every item is relevant to at most t functions. Hollender, Manurangsi, Meka and Suksompong asked for a non-additive Beck–Fiala theorem.

Question 1.1 ([HMMS25, Question 14])**.** *Let $\mathcal{F}$ be a t-sparse family of 1-Lipschitz functions. Is* $\operatorname{disc}(\mathcal{F}) = O_t(1)$*?*

They note that the question is open even for $t=2$, and they prove the bound $O((mt)^{1/3}\log^{1/3}(mt))$ [HMMS25, Theorem 15]. The difficulty is twofold. First, the iterated-rounding and partial-colouring arguments behind the Beck–Fiala theorem have no analogue for non-additive functions. Second, the topological lemma of [HMMS25] controls only the *total* number of fractional items, which is n, while a single sparse function may read all of them.

The case $t=2$ already contains the essence of the problem. View the functions as the vertices of a multigraph and every item relevant to two functions as an edge between them; items relevant to one function are *private*. For additive functions (counting functions of stars), alternating colours

along an Euler tour gives discrepancy at most 2. For arbitrary 1-Lipschitz functions on the edges at each vertex, no such argument is known. Each vertex may read its edges through an arbitrary non-additive function, and the parallel classes of edges (which we call *bundles*) may be arbitrarily large.

Several natural approaches fail already for $t = 2$. Fixing items to break all cycles costs the sizes of the fixed bundles (Corollary 1.5). The fractional point of [HMMS25] has up to n fractional items, and when the pattern is a complete graph a single function may read all of them. A topological statement with one cut per bundle is false already for cycles (Proposition 1.7). Finally, local constraints cannot all be held at the tightest level: for two functions sharing a bundle of two items, the constraints $|x_1 + x_2| \leq 1$ and $|x_1 - x_2| \leq 1$ each meet every antipodal path of the square $\{\pm 1\}^2$, but they have no common solution. Our arguments therefore track, for every function, how much slack its constraint has left, and spend this slack carefully along an ordering of the pattern.

1.1 Our results

Our main result is Theorem 1.9: for $t = 2$ the discrepancy is at most $\max(2, d)$ for patterns of degeneracy d, and at most the pseudoarboricity plus one, with no dependence on the bundle sizes. More generally, we make progress on Question 1.1 by identifying structural conditions on the *sparsity pattern* under which the discrepancy is bounded by a function of t and the pattern, independently of n, m and the sizes of the bundles (the exceptions are Theorems 1.10 and 8.37, bounds of order $\sqrt{n}$ for dense patterns, and the bounds for n functions in Theorem 1.9(d) and Theorem 8.34). Informally:

(1) For $t = 2$, vertex degrees and bundle sizes are irrelevant: the discrepancy is at most $\max(2, d)$ for patterns of degeneracy $d \geq 1$, which is tight for $d \leq 2$, and at most $p+1$ for pseudoarboricity p; so it is at most 2, which is tight, for 2-degenerate patterns (cycles, cacti, outerplanar and series–parallel patterns), and it is bounded on every proper minor-closed class. Refinements give at most 2 for wheels, 3 for planar patterns and $\lceil n/2 \rceil$ for $n \geq 3$ functions ($\lfloor n/2 \rfloor$ with a computer-assisted base case). For n functions it is at most $1.666\sqrt{n}$ for arbitrary bundles and $\sqrt{2n} + O(1)$ for single-item bundles, and at most 2 if $n \leq 6$ (computer-assisted). For $d \geq 3$ the only lower bound we know is 2. Question 1.1 for $t = 2$ remains open on every pattern class of unbounded degeneracy, already for complete graphs with single-item bundles.

(2) For *acyclic* sparsity patterns the answer to Question 1.1 is positive, with the best possible bound $\mathcal{K}_t = O(\sqrt{t \log t})$, where $\mathcal{K}_t$ is the worst discrepancy of t arbitrary functions. We show $\mathcal{K}_2 = \mathcal{K}_3 = 2$, $\mathcal{K}_4 = \mathcal{K}_5 = 3$ and $\mathcal{K}_6 = 4$ (computer-assisted for $n \geq 3$), so for $t \leq 3$ the bound is 2. Cyclic patterns can exceed $\mathcal{K}_t$, but every 2-sparse family with at most six functions has discrepancy at most 2 (computer-assisted).

(3) For patterns in which every function meets *few bundles*, the answer is positive, with a bound depending only on t and the number of bundles per function.

(4) The two mechanisms combine: heavy functions can be eliminated when they have few ports in each of their lobes.

(5) Exhaustive computations on small instances suggest that every family with $t = 2$ has discrepancy at most 2.

We now make these statements precise; Table 1 summarizes them.

Bundles. For a nonempty set S of functions, the *bundle* B_S is the set of items whose set of relevant functions is *exactly* S. The *bundle hypergraph* $H(\mathcal{F})$ has the functions as vertices and a hyperedge S for every S with $|S| \geq 2$ and $B_S \neq \emptyset$. The family is t-sparse if and only if all hyperedges have size at most t. For $t = 2$, $H(\mathcal{F})$ is the simple graph underlying the item multigraph. The sizes $|B_S|$ are unrestricted; they are the "multiplicities" of the hyperedges.

Small families. Let $\mathcal{K}_n$ be the supremum of $\operatorname{disc}(\mathcal{F})$ over all families of n arbitrary 1-Lipschitz functions, with any number of items. By [HMMS25, Theorem 8], $\mathcal{K}_n < \sqrt{2n(1+\ln 2n)}$, and a Hadamard matrix shows $\mathcal{K}_n \geq \sqrt{n/2}$ (Lemma 2.5). The constant $\mathcal{K}_t$ is the natural benchmark for t-sparse families: a single bundle shared by t functions is t-sparse.

Theorem 1.2. *$\mathcal{K}_1 = 1$, $\mathcal{K}_2 = \mathcal{K}_3 = 2$ and $\mathcal{K}_4 = 3$. The upper bounds $\mathcal{K}_3 \leq 2$ and $\mathcal{K}_4 \leq 3$ are computer-assisted; the computations use exact arithmetic.*

The same method shows that sparsity helps already for few functions: every 2-sparse family of at most five functions has discrepancy at most 2 (Theorem 3.8). Run at a threshold just below the next integer, which suffices by an integrality argument (Lemma 3.9), the method gives more: every 2-sparse family of at most six functions has discrepancy at most 2, and $\mathcal{K}_5 = 3$, $\mathcal{K}_6 = 4$, while additive examples from covering codes give $\mathcal{K}_8 \geq 5$ (Theorem 3.10). Hence a 2-sparse family of discrepancy greater than 2, if one exists, has at least seven functions, and every 2-sparse family whose blocks have at most six vertices (for example K_6, $K_{3,3}$ or the octahedron with arbitrary bundle sizes) has discrepancy at most 2. Combined with the near-loop transfer below, every 2-sparse family of $n \geq 4$ functions has discrepancy at most $\lfloor n/2 \rfloor$ (Theorem 1.9(d)).

Acyclic patterns. A hypergraph is *Berge-acyclic* if its bipartite incidence graph (vertices versus hyperedges) is a forest [Ber89]. For $t = 2$, this means that the simple graph underlying the item multigraph is a forest; the edge multiplicities and private items are arbitrary.

Theorem 1.3. *If $\mathcal{F}$ is t-sparse and $H(\mathcal{F})$ is Berge-acyclic, then $\operatorname{disc}(\mathcal{F}) \leq \mathcal{K}_t$. This is tight: for every $\varepsilon > 0$ there is such a family (with a single hyperedge) whose discrepancy exceeds $\mathcal{K}_t - \varepsilon$.*

Corollary 1.4. *If $\mathcal{F}$ is t-sparse and $H(\mathcal{F})$ is Berge-acyclic, then $\operatorname{disc}(\mathcal{F}) < \sqrt{2t(1+\ln 2t)}$. If $t \in \{2, 3\}$, then $\operatorname{disc}(\mathcal{F}) \leq 2$, and this is tight.*

Acyclicity cannot simply be dropped from Theorem 1.3: there is a 3-sparse family of four additive functions on four items with discrepancy $3 > \mathcal{K}_3$ (Proposition 4.4).

Discrepancy can also be traded against fixing items. The following corollary bounds the price of breaking all cycles.

Corollary 1.5. *Let $\mathcal{F}$ be t-sparse and let F be a set of items such that the bundle hypergraph of the items outside F (with their original relevance sets) is Berge-acyclic. Then $\operatorname{disc}(\mathcal{F}) \leq \mathcal{K}_t + \max_i |F \cap R(f_i)|$. In particular, for $t = 2$, if F is a union of bundles whose removal leaves a forest, then $\operatorname{disc}(\mathcal{F}) \leq 2 + \max_i |F \cap R(f_i)|$.*

Bounded bundle-degree. For a function v let d_v be the number of hyperedges of $H(\mathcal{F})$ containing v, and let $\Delta = \max_v d_v$. Functions that share no item with another function have discrepancy at most $\mathcal{K}_1 = 1$ on their own and do not interact with the rest, so we ignore them. Thus $d_v \geq 1$.

Theorem 1.6. *Let $\mathcal{F}$ be t-sparse. For a hyperedge e of $H(\mathcal{F})$ put $k_e = \sum_{v \in e} \max(1, 2^{d_v - 2})$. Then*

$$\operatorname{disc}(\mathcal{F}) \;\leq\; \max_v \sum_{e \ni v} k_e \;\leq\; t\,\Delta \max(1, 2^{\Delta - 2}).$$

In particular, if $t = 2$ and $\Delta \geq 2$, then $\operatorname{disc}(\mathcal{F}) \leq \Delta 2^{\Delta-1}$. If $t = 2$ and the simple graph underlying the item multigraph is a cycle, with arbitrary bundle sizes and private items, then $\operatorname{disc}(\mathcal{F}) \leq 4$.

Theorem 1.6 is proved with one necklace sphere per bundle, in the spirit of the multidimensional necklace splitting of de Longueville and Živaljević [dLŽ08], and a Fadell–Husseini index computation. The step specific to our setting is a *parity decomposition* of the test map into characters of $(\mathbb{Z}_2)^{\text{bundles}}$ (Section 1.2), a device familiar from the Grünbaum–Hadwiger–Ramos problem [Ram96, BFHZ18]. The argument needs at least two cuts per bundle, and this cannot be improved within the approach:

Proposition 1.7 (informal; see Proposition 6.1)**.** *The "one cut per bundle" version of the topological statement behind Theorem 1.6 is false already for cycles.*

Combining the two mechanisms: bounded block-degree. Fix $\Delta \geq 2$ and call a function *heavy* if it lies in more than Δ hyperedges of $H(\mathcal{F})$. For a function v, delete v from every hyperedge of $H(\mathcal{F})$. The *lobes* of v are the connected components of the resulting hypergraph that share a hyperedge with v, and the *ports* of a lobe are the hyperedges containing v and meeting it. For $t = 2$ the lobes of v correspond to the blocks of the simple graph $H(\mathcal{F})$ containing v, and the number of ports of a lobe is the degree of v inside the corresponding block (Lemma 7.7). Here the blocks are the maximal 2-connected subgraphs and the bridges.

Theorem 1.8. *Let $\Delta \geq 2$ and let $\mathcal{F}$ be t-sparse. If every heavy function has at most Δ ports in each of its lobes, then $\operatorname{disc}(\mathcal{F}) \leq t\Delta \max(1, 2^{\Delta-2})$. In particular, for $t = 2$: if every block of the simple graph underlying the item multigraph has maximum degree at most Δ, then $\operatorname{disc}(\mathcal{F}) \leq \Delta 2^{\Delta-1}$. Every 2-sparse family whose underlying simple graph is a* cactus *(every block is an edge or a cycle), with arbitrary degrees and bundle sizes, has discrepancy at most 4.*

For $t = 2$ the bounds of Theorems 1.6 and 1.8 are superseded by Theorem 1.9 below; they matter for $t \geq 3$. Theorem 1.8 contains Theorem 1.6 (no heavy functions), up to the finer bound stated there. It contains Theorem 1.3 with the constant $2t$ in place of $\mathcal{K}_t$: in a Berge-acyclic pattern every lobe has a single port. For $t = 2$ it covers every pattern in which no function of degree larger than Δ lies on a cycle (Corollary 7.8). Heavy functions are eliminated one at a time by a *closing-function* reduction, which combines the tree argument behind Theorem 1.3 with the topological Theorem 1.6.

Degenerate patterns for $t = 2$: hub transfer and near-loops. The closing-function reduction fails at a function of large degree inside a block, such as the hub of a wheel: the closing function is then as heavy as the function it replaces. For $t = 2$ we overcome this obstacle whenever the pattern is sparse in the sense of degeneracy. Recall that a graph is *d-degenerate* if its vertices can be ordered so that each has at most d neighbours that come later; the *degeneracy* $\operatorname{dgn}(G)$ is the least such d. Planar graphs are 5-degenerate, graphs of treewidth k are k-degenerate, and graphs of maximum degree Δ are Δ-degenerate.

Theorem 1.9. *Let $\mathcal{F}$ be 2-sparse, let $H = H(\mathcal{F})$ be the simple graph underlying the item multigraph, let $d = \operatorname{dgn}(H)$, and let $p(H) = \max_{H'} \lceil |E(H')|/|V(H')| \rceil$ (over nonempty subgraphs H') be its pseudoarboricity. Then, whatever the vertex degrees, bundle sizes and private items,*

$$\operatorname{disc}(\mathcal{F}) \leq \max(2, d) \quad (d \geq 1), \qquad \operatorname{disc}(\mathcal{F}) \leq 1 \quad (d = 0), \qquad \operatorname{disc}(\mathcal{F}) \leq p(H) + 1.$$

The first two bounds are tight for $d \leq 2$. Moreover:

(a) planar patterns have $\operatorname{disc}(\mathcal{F}) \leq 3$, patterns of treewidth k have $\operatorname{disc}(\mathcal{F}) \leq \max(2, k)$, and K_r-minor-free patterns have $\operatorname{disc}(\mathcal{F}) = O(r\sqrt{\log r})$;

(b) *2-degenerate patterns, among them cycles, cacti, outerplanar and series–parallel patterns, planar patterns of girth at least 6, and patterns obtained by subdividing every edge, as well as wheels, have* $\operatorname{disc}(\mathcal{F}) \le 2$*, which is tight; subgraphs of* $K_{q,n}$ *have* $\operatorname{disc}(\mathcal{F}) \le \max(2, q)$*;*

(c) *if every block has maximum degree at most* $\Delta \ge 2$*, in particular if every function has at most* Δ *distinct neighbours, then* $\operatorname{disc}(\mathcal{F}) \le \lceil \Delta/2 \rceil + 1$*; if the functions with more than* Δ *distinct neighbours form an independent set, then* $\operatorname{disc}(\mathcal{F}) \le \max(2, \Delta)$*;*

(d) *every 2-sparse family of* $n \ge 3$ *functions has* $\operatorname{disc}(\mathcal{F}) \le \lceil n/2 \rceil$*, and* $\operatorname{disc}(\mathcal{F}) \le \lfloor n/2 \rfloor$ *if* $n \ge 4$*.*

Only the second bound in (d) uses the computer-assisted Theorem 3.8.

For dense patterns, where the degeneracy is large, a deterministic rounding of the fractional point of [HMMS25] removes the logarithm from their bound when bundles are thin.

Theorem 1.10. *Let* $\mathcal{F}$ *be 2-sparse with* n *functions, any two of which share at most* s *items. Then* $\operatorname{disc}(\mathcal{F}) \le 1 + 2\lceil \sqrt{sn/2}\,\rceil$*. In particular, every 2-sparse family whose item multigraph is simple, for example the complete graph* K_n *with single-item bundles, has* $\operatorname{disc}(\mathcal{F}) \le \sqrt{2n} + 3$*, compared with* $\sqrt{2n(1 + \ln 2n)}$ *for arbitrary families [HMMS25], an improvement for large* n*. For* t*-sparse families the bound is* $1 + 2s\sqrt{t(t-1)n}$*.*

For arbitrary bundle sizes, a union bound weighted by the number of fractional items of each function gives $\operatorname{disc}(\mathcal{F}) \le \sqrt{2\ln(2t)\, n}$ for every t-sparse family, so less than $1.666\sqrt{n}$ for $t = 2$ (Theorem 8.37).

For $t = 2$, Theorem 1.9 supersedes the bounds of Theorems 1.6 and 1.8: it gives 2 instead of 4 for cycles and cacti, and $\lceil \Delta/2 \rceil + 1$ instead of $\Delta 2^{\Delta-1}$ when every block has maximum degree at most Δ. Those theorems remain the relevant ones for $t \ge 3$. The key new ingredients are two transfer lemmas. When a function h is deleted, each of its neighbours receives a new local constraint of the same type as before, such that every solution of the smaller instance extends to h. The *hub-transfer lemma* (Lemma 8.15) asks the neighbour to leave room for a 1-loop of the shared bundle and costs it one level, which is optimal for this demand (Remark 8.38); on its own it gives $d + 1$ (Theorem 8.1). The *near-loop transfer* (Corollary 8.27) asks only for a *near-loop*, a path with steps of length at most 2 that ends within distance 1 of the antipode of its start, and costs nothing; a deleted function of level ℓ tolerates this weaker demand from up to ℓ neighbours (Lemma 8.28). Charging every edge to one of its endpoints along an ordering or an orientation gives Theorem 1.9. The hub-transfer lemma only uses that the items of h are shared pairwise, so for every t high-degree functions whose items are shared pairwise can be deleted from patterns covered by Theorems 1.3 and 1.6 (Corollary 8.18). The method relies on pairwise sharing: we show that a natural extension to items read by three functions fails with unbounded loss (Remark 8.38).

The hub-transfer lemma reduces to two statements about a single Hamming cube, which may be of independent interest. Call $B \subseteq \{\pm 1\}^S$ *thick* if every path along the edges of the cube from a vertex z to its antipode $-z$ meets B.

Theorem 1.11 (antipodal blocking sets; Lemmas 8.12 and 8.13)**.** *Let* $|S| \ge 3$*.*

(a) *Any two thick subsets of* $\{\pm 1\}^S$ *are at Hamming distance at most 1.*

(b) *If* $P, N \subseteq \{\pm 1\}^S$ *are at Hamming distance at least 3 and* $P \cup N$ *is thick, then* P *or* N *is thick. The distance 3 cannot be replaced by 2.*

Both parts are proved on the coned 2-skeleton of the cube, using its simple connectivity and a Borsuk–Ulam argument for a two-cut necklace sphere mapped into it (Section 8.3). For near-loops the analogous statements hold with distance 2 in place of 3 in (b), and any two sets meeting every near-loop intersect (Lemmas 8.24 and 8.25).

Sparsity pattern	Bound on disc	Reference
$t = 2$, degeneracy $d \geq 1$; pseudoarboricity p	$\max(2, d)$ (tight for $d \leq 2$); $p + 1$	Thm. 1.9
$t = 2$, planar; treewidth k; K_r-minor-free	3; $\max(2, k)$; $O(r\sqrt{\log r})$	Thm. 1.9
$t = 2$, 2-degenerate (outerplanar, series–parallel, cacti); wheels	2 (tight)	Cor. 8.33
$t = 2$, $n \geq 4$ functions; subgraphs of $K_{q,n}$	$\lceil n/2 \rceil$ ($\lfloor n/2 \rfloor$ with Thm. 3.8); $\max(2, q)$	Thm. 1.9
$t = 2$, at most 6 functions; 8 functions	2 (tight); 3 (both computer-assisted)	Thms. 3.10, 8.34
$t = 2$, blocks with at most 6 vertices	2 (tight; computer-assisted)	Thm. 3.10
$t = 2$, n functions, pairwise $\leq s$ shared items	$1 + 2\lceil\sqrt{sn/2}\,\rceil$	Thm. 1.10
t-sparse, n functions, any bundles	$\sqrt{2\ln(2t)\,n}$ ($< 1.666\sqrt{n}$ for $t = 2$)	Thm. 8.37
$t = 2$, general	open; conjectured 2	Conj. 1.12, App. B
n arbitrary functions	$\mathcal{K}_n$; $\mathcal{K}_1, \ldots, \mathcal{K}_6 = 1, 2, 2, 3, 3, 4$ (computer-assisted for $n \geq 3$); $\mathcal{K}_8 \geq 5$	Thms. 1.2, 3.10
t-sparse, Berge-acyclic	$\mathcal{K}_t < \sqrt{2t(1 + \ln 2t)}$ (tight)	Thm. 1.3, Cor. 1.4
3-sparse, cyclic	can exceed $\mathcal{K}_3$	Prop. 4.4
t-sparse, bundle-degree $\leq \Delta$, or $\leq \Delta$ ports per lobe	$t\Delta \max(1, 2^{\Delta-2})$	Thms. 1.6, 1.8

Table 1: Summary of the main results. Unless the bound itself involves n or s, it is independent of the number of functions, the number of items, and the sizes of the bundles. Further results (Corollaries 1.5 and 8.18, and the leveled results of Section 8.6) are stated in the text.

Computational evidence. For $t = 2$, our bounds are 2 for 2-degenerate patterns and wheels, and $\max(2, d)$ for degeneracy d, while the lower bound is $\mathcal{K}_2 = 2$. An exhaustive SAT-based search (Appendix B) found no 2-sparse family of discrepancy greater than 2. The search covered cycles of bundles, complete graphs, wheels, the graphs $K_{2,n}$, and 148 random multigraphs with up to 18 items. By now every class in the search is covered by a theorem, for all bundle sizes (computer-assisted for at most six functions): the instances with at most six functions, among them K_6, by Theorem 3.10, and cycles, $K_{2,5}$ and the wheel W_6 by Theorem 1.9(b). The smallest patterns not covered by our results are K_7 and a few dense graphs on seven vertices (Theorem 8.34).

Conjecture 1.12. *Every* 2*-sparse family of* 1*-Lipschitz functions has discrepancy at most* 2*.*

1.2 Technical overview

Reformulation. Encode a colouring as $x \in \{\pm 1\}^m$, with item j red if $x_j = 1$. For a 1-Lipschitz f, the *imbalance* $g(x) = f(\{j : x_j = 1\}) - f(\{j : x_j = -1\})$ is *odd*, meaning $g(-x) = -g(x)$, and 2*-Lipschitz* for the Hamming distance. Conversely, every odd 2-Lipschitz function arises in this way (Lemma 2.1). Hence $\operatorname{disc}(\mathcal{F}) = \min_x \max_i |g_i(x)|$: we look for a common approximate zero of odd Lipschitz functions on the cube. Oddness is the source of all the topology below.

Trees: feasible sets, separation and loops (Theorem 1.3). Root the incidence forest of $H(\mathcal{F})$ at a function and process it bottom-up. For a non-root function v with parent bundle P, let $F_v \subseteq \{\pm 1\}^P$ be the set of colourings of P that can be extended to v's subtree with all imbalances at most $c = \mathcal{K}_t$. A crude characterization does not propagate, so we track two dual invariants.

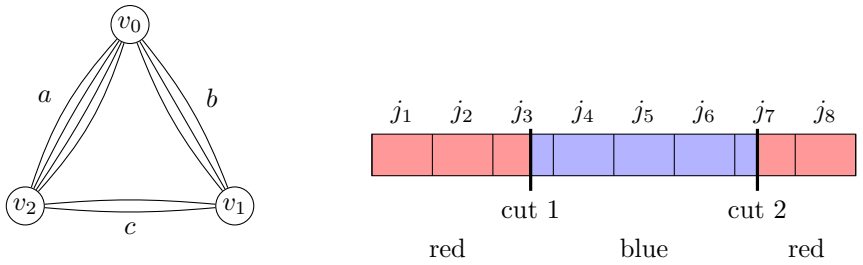


Figure 1: Left: a cycle of three bundles a, b, c of arbitrary sizes (here 4, 3 and 2 items); each function reads its two bundles through an arbitrary 1-Lipschitz function. Right: a necklace colouring of one bundle with 8 items and 2 cuts. Only the items containing a cut (j_3 and j_7) receive a fractional colour, so a function reading two bundles sees at most 4 fractional items.

- F_v is *c-separating*: its complement splits into two antipodal halves at Hamming distance greater than c. Such a set contains the "coarse zero set" $\{|\varphi| \le c\}$ of an odd 2-Lipschitz function φ, namely a signed distance function.
- For a bundle S with child functions $w_1, \dots, w_q$ ($q \le t-1$), the set $\bigcap_i F_{w_i}$ contains a *c-loop*, that is, a path from some z to $-z$ with steps of Hamming length at most c.

The second invariant follows from the first by a "loop-or-certificate" dichotomy. A symmetric set without a c-loop splits into two far-apart antipodal halves, and the signed distance to them is an odd function ψ that is larger than c in absolute value on the whole set. Applying the definition of $\mathcal{K}_{q+1} \le \mathcal{K}_t$ to $\varphi_1, \dots, \varphi_q$ and ψ gives a contradiction. The first invariant follows from the second at the parent: if $y \notin F_v$, then $g_v(\cdot, y)$ exceeds c in absolute value on the whole product of the children's loops, so by continuity along short steps it has a constant sign there, and this sign is odd in y. The constant $\mathcal{K}_t$ arises exactly because a bundle has at most $t-1$ children and one test function is added.

The topological approach and its bottleneck. Lemma 9 of [HMMS25] places all items on $[0,1]$ and applies a consensus-halving theorem: with at most n cuts, all multilinear extensions are balanced. After rounding, a function errs by at most the number of fractional items it reads, which may be all n of them. For sparse families we need *local* control, so we give every bundle e its own interval and allow it k_e cuts. Colourings of e with at most k_e cuts form a sphere S^{k_e} (the Hobby–Rice parametrization [HR65, Pin76]), and the colour swap is the antipodal map. The composite $\widehat{G}_v$ (the multilinear extension of g_v, evaluated on the necklace colourings) is a map on the product of spheres $X = \prod_e S^{k_e}$. It depends only on the bundles $N(v)$ at v and is odd under flipping all of them simultaneously. If all $\widehat{G}_v$ vanish simultaneously, rounding gives $|g_v| \le \sum_{e \in N(v)} k_e$.

The obvious symmetry is the *diagonal* $\mathbb{Z}_2$, which flips every bundle. It is far too weak. By itself it cannot force a common zero of more than $\min_e k_e$ odd functions (project X to its smallest factor), whereas we have n functions (Remark 5.11).

The parity decomposition (Theorem 1.6). The function $\widehat{G}_v$ is *not* odd in the individual bundles at v, but it can be split into pieces that are. For $S \subseteq N(v)$ let

$$\widehat{G}_v^S(w) = 2^{-d_v} \sum_{T \subseteq N(v)} (-1)^{|S \cap T|}\, \widehat{G}_v(\mathrm{flip}_T w),$$

where flip_T negates the coordinates of the bundles in T. Then $\widehat{G}_v = \sum_S \widehat{G}_v^S$, each piece transforms by the character $U \mapsto (-1)^{|S\cap U|}$ of $(\mathbb{Z}_2)^{N(v)}$, and joint oddness kills the pieces with $|S|$ even. For two bundles a, b at v, this is simply

$$E_v(a,b) = \tfrac{1}{2}\big(\widehat{G}_v(a,b) + \widehat{G}_v(a,-b)\big), \qquad O_v(a,b) = \tfrac{1}{2}\big(\widehat{G}_v(a,b) - \widehat{G}_v(a,-b)\big),$$

where E_v is odd in a and even in b, and O_v is even in a and odd in b. A common zero of all pieces is a common zero of all $\widehat{G}_v$. For the triangle of Figure 1, with two cuts per bundle, the six equations $E_0(a,b)$, $O_0(a,b)$, $E_1(b,c)$, $O_1(b,c)$, $E_2(c,a)$ and $O_2(c,a)$ are odd in a, b, b, c, c, a, respectively, and even in the other variable. Each bundle sphere S^2 thus carries exactly two equations that are odd in it, just as S^k carries k odd equations in the Borsuk–Ulam theorem. The equations are coupled, so this is not Borsuk–Ulam on each factor; the cohomological computation described next makes the heuristic precise. We have thus replaced the $\mathbb{Z}_2$-equivariant problem by a $(\mathbb{Z}_2)^{\text{bundles}}$-equivariant one, at the cost of 2^{d_v-1} equations per function. The Fadell–Husseini index [FH88] of X with the factorwise action is the monomial ideal $(t_e^{k_e+1})$. The Euler class of the target is a product of linear forms $\sum_{e\in S} t_e$, one for each piece. This product is nonzero, and each of its monomials has t_e-degree at most the number k_e of pieces that are odd in e. Hence it lies outside the monomial ideal, and an equivariant map must have a zero. For a cycle of bundles each bundle e receives exactly two equations, one E and one O, from its two endpoints; the Euler class is $\prod_e t_e^2$, two cuts per bundle suffice, and every function reads at most 4 fractional items.

Why one cut per bundle fails (Proposition 6.1). With one cut per bundle the spheres become circles. On a cycle the topological statement would read: continuous functions $f_i(\theta_i, \theta_{i+1})$ on tori, odd under the joint antipodal map, have a common zero. We give explicit counterexamples of the form $\sin(\theta_{i+1} - 2\theta_i - c_i)$ and $\sin(2\theta_{i+1} - \theta_i - c_i)$, whose zero sets wind inconsistently around the cycle. Hence two cuts per bundle are necessary for any argument that uses only the oddness and locality of the $\widehat{G}_v$.

Combining (Theorem 1.8). We eliminate heavy functions one at a time. Let v be heavy. Each lobe L of v interacts with the rest of the instance only through the colouring of its ports. As seen from L, we replace v by a new *closing function* c_L that reads exactly the ports through an *arbitrary* odd test function ψ. The closing function is light, because L has at most Δ ports, so the new instance has fewer heavy functions, and by induction it has a good colouring for every ψ. Hence the set T_L of port colourings that extend into L meets every coarse zero set $\{|\psi| \le C\}$, and by the loop-or-certificate dichotomy T_L contains a loop. As in the tree argument, the crossing lemma on the product of the lobes' loops then gives a colouring of v's items that is good for v and extends into every lobe. When no heavy function is left, Theorem 1.6 applies.

Hub transfer and near-loops (Theorem 1.9). For $t = 2$ we order the functions so that each has at most d later neighbours and delete them from last to first. When h is deleted, an earlier neighbour r loses the bundle $S = B_{hr}$. We give r the new constraint

$$\mathcal{G}_r = \{\rho \in \{\pm1\}^T : \text{ the slice } \{s \in \{\pm1\}^S : (s,\rho) \in Z_r\} \text{ contains a 1-loop}\},$$

where T is the set of remaining items of r and Z_r its current constraint. If every neighbour's new constraint is met, then h sees a product of 1-loops, one in each of its bundles, and the crossing lemma gives h a good value. The heart of the matter is that $\mathcal{G}_r$ is again a constraint of the same kind: if Z_r contains the coarse zero set $\{|\psi| \le \ell\}$ of an odd 2-Lipschitz function, then $\mathcal{G}_r$ contains such a set at

level $\ell - 1$. By the loop-or-certificate dichotomy, this amounts to the following hub-transfer lemma: along every antipodal $(\ell-1)$-path $\rho^0, \dots, \rho^L$ in $\{\pm 1\}^T$, some slice $\{s : |\psi(s, \rho^\tau)| \le \ell\}$ contains a 1-loop. Call $B \subseteq \{\pm 1\}^S$ *thick* if every antipodal 1-path meets B. For each τ the cube $\{\pm 1\}^S$ splits into $P_\tau = \{\psi(\cdot, \rho^\tau) > \ell\}$, the slice, and $N_\tau = \{\psi(\cdot, \rho^\tau) < -\ell\}$, and the Lipschitz condition gives $\operatorname{dist}(P_\tau, N_\tau) \ge 3$ and $\operatorname{dist}(P_\tau, N_{\tau\pm1}) \ge 2$. Two statements about a single cube finish the proof:

- (dichotomy) if $\operatorname{dist}(P, N) \ge 3$ and the rest of the cube contains no 1-loop, then P or N is thick;
- (separation) two thick sets are at distance at most 1.

If no slice contains a 1-loop, then for each τ exactly one of P_τ, N_τ is thick. Oddness forces the thick side to differ at $\tau = 0$ and $\tau = L$, and separation forbids a switch in one step. We prove both statements on the 2-skeleton of the cube coned off at the barycentres of its squares. This complex is simply connected and carries a free involution, and its components under a labelling correspond exactly to components in the cube, so nothing has to be rounded. Simple connectivity makes the components of the label classes form a tree. The dichotomy then follows from a separation lemma for graphs with a free involution. The separation follows from the Borsuk–Ulam theorem, applied to a two-cut necklace sphere mapped into the complex and composed with the map to the tree, embedded in $\mathbb{R}^2$. So each such deletion costs every earlier neighbour one level; since a function has at most d later neighbours, starting from level $d+1$ leaves level at least 1 for every function (Theorem 8.1). To do better we weaken the demand: instead of a 1-loop, the neighbour only guarantees a *near-loop* of the shared bundle, a path with steps of length at most 2 that ends within distance 1 of the antipode of its start. Dichotomy and separation hold for near-loops as well (the dichotomy now with $\operatorname{dist}(P, N) \ge 2$), so this demand costs the neighbour nothing, and a deleted function at level ℓ can absorb the resulting deficiency from ℓ neighbours. Deleting in a degeneracy order with only such demands gives $\max(2, d)$, and charging each edge to the endpoint it points to in an orientation of small in-degree gives $p + 1$ (Theorem 1.9). For planar patterns, a canonical ordering splits the pattern into a 2-degenerate graph, whose edges are absorbed, and a forest, whose edges are charged to their children; every function then pays at most one level, which gives 3. Keeping track of the individual levels, the process may stop at a forest whose level-1 vertices are pairwise non-adjacent: by an asymmetric version of $\mathcal{K}_2 = 2$ (any two odd 2-Lipschitz functions have a common point where one is at most 1 and the other at most 2 in absolute value), such a forest is always solvable. This gives results for systems in which some functions are held at level 1 (Section 8.6).

1.3 Related work

Additive discrepancy. Beyond the results above [BF81, Spe85, Ban98, BJ25b, BJ25a], constructive versions of partial colouring [Ban10, LM15] underlie most modern algorithmic discrepancy. As discussed in [HMMS25], partial colouring breaks down for non-additive functions because the discrepancies of partial colourings cannot be combined by the triangle inequality.

Non-additive discrepancy. Hollender, Manurangsi, Meka and Suksompong [HMMS25] (see [HMMS26] for the conference version) and Dupré la Tour and Fujii [DF25] proved $O(\sqrt{n \log n})$ bounds for n arbitrary 1-Lipschitz functions, with multicolour extensions; the bound of [DF25] for k colours is $O(\sqrt{n \log(nk)})$ when k is a prime power. Both works derive applications to consensus halving and envy-freeness up to c goods, and Dupré la Tour and Fujii also derive an application to envy-freeness with subsidies. Hollender, Manurangsi, Meka and Suksompong also proved the $O((mt)^{1/3} \log^{1/3}(mt))$ bound for t-sparse families and raised Question 1.1. Coverage-type

special cases of sparse non-additive discrepancy have also been studied: Avila, Rohwedder and Wennmann [ARW26] consider t-sparse families of coverage functions and prove constructive bounds polynomial in t, the number of colours, and $\log n$. Our results are of a different nature. We allow *arbitrary* 1-Lipschitz functions and restrict the *sparsity pattern*, and apart from Theorem 1.10 and Theorem 1.9(d) our bounds do not depend on n or m.

Consensus halving and necklaces. The existence of consensus halvings with n cuts for n non-additive valuations is due to Simmons and Su [SS03]; the additive case goes back to the Hobby–Rice theorem [HR65] (see [Pin76] for a Borsuk–Ulam proof) and to necklace splitting [Alo87]. The prime-k generalization used in [HMMS25] is due to Filos-Ratsikas, Hollender, Sotiraki and Zampetakis [FRHSZ21]. Consensus halving for indivisible items was studied by Goldberg, Hollender, Igarashi, Manurangsi and Suksompong [GHI+20]. On the computational side, approximate consensus halving is PPA-complete [FRG18], and its complexity has been mapped in detail [FRHSZ23]. Our necklace spheres are the configuration spaces of these proofs, used bundle by bundle; products of necklace spheres with prescribed numbers of cuts in each direction were used by de Longueville and Živaljević [dLŽ08] to split multidimensional necklaces, and the Borsuk–Ulam approach to necklace bisection goes back to Alon and West [AW86].

Equivariant topology. The ideal-valued index is due to Fadell and Husseini [FH88]; see Matoušek's book [Mat03] for topological methods in combinatorics and the survey of Blagojević and Ziegler [BZ17] for the ideal-valued index in combinatorial applications. Products of spheres with $(\mathbb{Z}_2)^k$-actions, and the question whether an Euler class that is a product of linear forms lies in the monomial ideal $(t_1^{d+1}, \dots, t_k^{d+1})$, are central in the Grünbaum–Hadwiger–Ramos hyperplane mass-partition problem [Ram96, MLVŽ06, BFHZ18]; the character decomposition of a test map used in our parity decomposition is a standard tool there [BFHZ18]. Antipodal paths in the Hamming cube also appear in work on cube labellings with monochromatic antipodal paths, see Feder and Subi [FS13]; our thick-set lemmas concern a different question (blocking all antipodal paths) and are closer in spirit to Tucker's lemma. In our setting the obstruction can be evaluated for every bundle hypergraph, because each monomial of the Euler class respects the degree bounds termwise.

Organization. Section 2 sets up the reformulation, bundles, the constants $\mathcal{K}_n$, multilinear extensions and the consensus-halving lemma, and the notions of loops and separating sets. Section 3 proves Theorem 1.2, Section 4 proves Theorem 1.3 and its corollaries, and Section 5 proves Theorem 1.6. Section 6 contains the torus counterexample, Section 7 proves Theorem 1.8, Section 8 proves Theorems 1.9 and 1.10 and their consequences, and Section 9 lists open problems. Appendix A documents the computer-assisted part of Theorem 1.2 and the ancillary files, and Appendix B reports the SAT computations behind Conjecture 1.12. Readers interested mainly in the main result can read Section 2 and then Section 8, which is self-contained apart from the constraint-system reformulation.

2 Preliminaries

Notation. $[r] = \{1, \dots, r\}$. For a finite set U let $\{\pm 1\}^U$ be the set of sign vectors indexed by U, with Hamming distance dist. For $A, A' \subseteq \{\pm 1\}^U$ write $-A = \{-z : z \in A\}$, $\operatorname{dist}(z, A) = \min_{a \in A} \operatorname{dist}(z, a)$ and $\operatorname{dist}(A, A') = \min_{a \in A} \operatorname{dist}(a, A')$, with $\operatorname{dist}(\cdot, \emptyset) = \infty$. A set A is *symmetric* if $A = -A$. A function $h\colon \{\pm 1\}^U \to \mathbb{R}$ is *odd* if $h(-z) = -h(z)$ and *L-Lipschitz* if $|h(z) - h(z')| \le L \operatorname{dist}(z, z')$. Since the Hamming distance is a path metric, it suffices to check the Lipschitz condition

for $\operatorname{dist}(z, z') = 1$. For disjoint U, W we write $(z, y) \in \{\pm 1\}^{U \cup W}$ for the concatenation of $z \in \{\pm 1\}^U$ and $y \in \{\pm 1\}^W$.

2.1 Imbalance functions and instances

For $f\colon 2^{[m]} \to \mathbb{R}$ define $g_f\colon \{\pm 1\}^{[m]} \to \mathbb{R}$ by $g_f(x) = f(\{j : x_j = 1\}) - f(\{j : x_j = -1\})$. The set $R(f)$ of items relevant to f [HMMS25, Definition 13] consists of the j such that $f(S) \neq f(S \cup \{j\})$ for some S. Analogously, coordinate j is relevant to $g\colon \{\pm 1\}^{[m]} \to \mathbb{R}$ if changing x_j changes $g(x)$ for some x.

Lemma 2.1. *(a) If f is 1-Lipschitz, then g_f is odd and 2-Lipschitz, and it depends only on the coordinates in $R(f)$.*

(b) If $g\colon \{\pm 1\}^{[m]} \to \mathbb{R}$ is odd and 2-Lipschitz, then $f(S) = \frac{1}{2} g(\mathbf{1}_S - \mathbf{1}_{[m]\setminus S})$ is 1-Lipschitz, $g_f = g$, and $R(f)$ is the set of coordinates relevant to g.

(c) For every family $\mathcal{F} = \{f_1, \dots, f_n\}$, $\operatorname{disc}(\mathcal{F}) = \min_{x \in \{\pm 1\}^{[m]}} \max_i |g_{f_i}(x)|$.

Proof. (a) Swapping the two colour classes negates g_f, so g_f is odd. Changing x_j moves item j from one colour class to the other, so each of the two terms of g_f changes by at most 1, and g_f is 2-Lipschitz. If $j \notin R(f)$, then $f(S) = f(S \cup \{j\})$ for all S, so moving j does not change either term. (b) If S, S' differ in one element, then the vectors $\mathbf{1}_S - \mathbf{1}_{\bar{S}}$ and $\mathbf{1}_{S'} - \mathbf{1}_{\bar{S}'}$ differ in one coordinate, so $|f(S) - f(S')| \leq 1$. Moreover $g_f(x) = \frac{1}{2} g(x) - \frac{1}{2} g(-x) = g(x)$, and $f(S) \neq f(S \cup \{j\})$ holds exactly when g changes in coordinate j at $\mathbf{1}_S - \mathbf{1}_{\bar{S}}$. (c) This is the definition of discrepancy, with $\chi^{-1}(1) = \{j : x_j = 1\}$. □

Definition 2.2 (instances)**.** An *instance* on the item set $[m]$ is a finite family $\mathcal{G} = (g_v, U_v)_{v \in V}$, where each $g_v\colon \{\pm 1\}^{[m]} \to \mathbb{R}$ is odd and 2-Lipschitz and depends only on the coordinates in its *support* $U_v \subseteq [m]$. We often regard g_v as a function on $\{\pm 1\}^{U_v}$. Its *discrepancy* is $\operatorname{disc}(\mathcal{G}) = \min_x \max_v |g_v(x)|$. The instance is *$t$-sparse* if every item lies in at most t supports.

By Lemma 2.1, a family $\mathcal{F}$ of 1-Lipschitz functions yields the instance $\mathcal{G}(\mathcal{F}) = (g_f, R(f))_{f \in \mathcal{F}}$ with $\operatorname{disc}(\mathcal{G}(\mathcal{F})) = \operatorname{disc}(\mathcal{F})$, and $\mathcal{G}(\mathcal{F})$ is t-sparse if and only if $\mathcal{F}$ is. Conversely, every instance with U_v equal to the set of relevant coordinates of g_v arises from a family. Supports may be larger than the sets of relevant coordinates. All our upper bounds are proved for instances under hypotheses on the supports, and they apply to families through $\mathcal{G}(\mathcal{F})$. An odd function with empty support is identically zero.

2.2 Bundles and the bundle hypergraph

Definition 2.3 (bundles)**.** Let $\mathcal{G} = (g_v, U_v)_{v \in V}$ be an instance. Items lying in no support are irrelevant, can be coloured arbitrarily, and are ignored. For nonempty $S \subseteq V$ let $B_S = \{j : \{v : j \in U_v\} = S\}$. The nonempty sets B_S are the *bundles*; they partition the relevant items. A bundle with $|S| = 1$ is *private*. The *bundle hypergraph* $H(\mathcal{G})$ has vertex set V and edge set $\{S : |S| \geq 2,\ B_S \neq \emptyset\}$. The *incidence graph* $I(\mathcal{G})$ is the bipartite graph with classes V and $\{S : B_S \neq \emptyset\}$, in which v is joined to S if $v \in S$. We call $H(\mathcal{G})$ *Berge-acyclic* if $I(\mathcal{G})$ is a forest. Private bundles are leaves of $I(\mathcal{G})$, so it does not matter whether they are included.

For a family $\mathcal{F}$ we use the bundles of $\mathcal{G}(\mathcal{F})$; then B_S is the set of items whose set of relevant functions is exactly S, as in Section 1.1. Berge-acyclicity forbids in particular two hyperedges sharing two vertices, since these would form a 4-cycle in $I(\mathcal{G})$.

The case $t = 2$. For a 2-sparse instance, the *item multigraph* has vertex set V; every item lying in two supports U_u, U_v is an edge uv, and items lying in one support are private. Bundles with $|S| = 2$ are the parallel classes, $H(\mathcal{G})$ is the simple graph underlying the item multigraph, and $H(\mathcal{G})$ is Berge-acyclic if and only if this simple graph is a forest. For a vertex v, the number d_v of hyperedges of $H(\mathcal{G})$ at v is its number of distinct neighbours.

2.3 The constants $\mathcal{K}_n$

Definition 2.4. For $n \geq 1$ let $\mathcal{K}_n$ be the supremum of $\mathrm{disc}(\mathcal{G})$ over all $m \geq 1$ and all instances $\mathcal{G}$ with n functions on the item set $[m]$. By Lemma 2.1, this is also the supremum of $\mathrm{disc}(\mathcal{F})$ over families of n 1-Lipschitz functions.

Adding the zero function shows $\mathcal{K}_n \leq \mathcal{K}_{n+1}$. By definition, every instance with at most n functions has a colouring with $|g_v| \leq \mathcal{K}_n$ for all v. The supremum is used only through this property, so it does not matter whether it is attained.

Lemma 2.5. *For every $n \geq 1$, $\sqrt{n/2} \leq \mathcal{K}_n < \sqrt{2n(1 + \ln(2n))}$.*

Proof. For $k = 2$, Theorem 8 of [HMMS25] states that every 1-Lipschitz family of n functions has $\mathrm{disc}(\mathcal{F}, 2) < \sqrt{2n(k-1)(1 + \ln(nk))} = \sqrt{2n(1 + \ln 2n)}$, and $\mathrm{disc}(\mathcal{F}, 2)$ is our $\mathrm{disc}(\mathcal{F})$. For the lower bound, let $N = 2^a$ and let $h_1, \ldots, h_N$ be the rows of a Sylvester Hadamard matrix (± 1 entries, pairwise orthogonal rows). The functions $g_i(x) = \langle h_i, x \rangle$ on $\{\pm 1\}^{[N]}$ are odd and 2-Lipschitz, and for every x we have $\max_i |g_i(x)|^2 \geq \frac{1}{N} \sum_i \langle h_i, x \rangle^2 = \frac{1}{N} \cdot N \|x\|_2^2 = N$. Hence $K_N \geq \sqrt{N}$. For general n take $N = 2^{\lfloor \log_2 n \rfloor} > n/2$ and use monotonicity. □

The next lemma is needed only to state tightness for families in which sparsity is measured by relevance.

Lemma 2.6. *Let $g_1, \ldots, g_n$ be odd 2-Lipschitz functions on $\{\pm 1\}^{[m]}$ and let $\varepsilon > 0$. There are odd 2-Lipschitz $g'_1, \ldots, g'_n$ on $\{\pm 1\}^{[m]}$ such that every coordinate is relevant to every g'_i and $\min_x \max_i |g'_i(x)| \geq \min_x \max_i |g_i(x)| - \varepsilon$.*

Proof. Let $\ell(x) = \frac{1}{m} \sum_j x_j$ and $g'_i = (1 - \delta) g_i + \delta \ell$ with $\delta \in (0, 1)$. Each g'_i is odd and satisfies $|g'_i(x) - g'_i(x^{(j)})| \leq 2(1 - \delta) + 2\delta/m \leq 2$, where $x^{(j)}$ is x with coordinate j flipped. Since $|g_i(x)| = \frac{1}{2}|g_i(x) - g_i(-x)| \leq m$, we have $|g'_i - g_i| \leq \delta(m + 1)$. Fix i and j. If g_i does not depend on x_j, then $g'_i(x) - g'_i(x^{(j)}) = 2\delta x_j / m \neq 0$. Otherwise pick x with $D = g_i(x) - g_i(x^{(j)}) \neq 0$; then $g'_i(x) - g'_i(x^{(j)}) = (1 - \delta) D + 2\delta x_j / m$, which vanishes for at most one value of δ. Choosing $\delta \leq \varepsilon / (m + 1)$ outside these finitely many values proves the claim. □

2.4 Multilinear extensions and the consensus-halving lemma

For $g \colon \{\pm 1\}^{[m]} \to \mathbb{R}$ and $x \in [-1, 1]^m$ let $\xi^x \in \{\pm 1\}^{[m]}$ be the random vector with independent coordinates and $\Pr[\xi^x_j = 1] = \frac{1 + x_j}{2}$, so that $\mathbb{E}\xi^x_j = x_j$. The *multilinear extension* of g is $G(x) = \mathbb{E}[g(\xi^x)]$. A coordinate j of x is *fractional* if $x_j \in (-1, 1)$.

Lemma 2.7. *Let $g \colon \{\pm 1\}^{[m]} \to \mathbb{R}$ be odd and 2-Lipschitz, depending only on the coordinates in U, and let G be its multilinear extension. Then:*

(a) $G = g$ on $\{\pm 1\}^{[m]}$, G depends only on the coordinates in U, and G is continuous;

(b) G is odd: $G(-x) = -G(x)$;

(c) *G is 1-Lipschitz with respect to ℓ_1: $|G(x) - G(x')| \le \|x - x'\|_1$;*

(d) *if $G(x) = 0$ and $x' \in \{\pm 1\}^{[m]}$ is obtained from x by replacing every fractional coordinate by a nearest sign, then $|g(x')|$ is at most the number of fractional coordinates of x in U.*

Proof. (a) If $x \in \{\pm 1\}^{[m]}$ then $\xi^x = x$. The distribution of $\xi^x|_U$ depends only on $x|_U$, and G is a polynomial in x. (b) The vector $-\xi^x$ has the distribution of ξ^{-x}, so $G(-x) = \mathbb{E}g(-\xi^x) = -G(x)$. (c) Let $h(S) = \frac{1}{2}g(\mathbf{1}_S - \mathbf{1}_{[m]\setminus S})$, which is 1-Lipschitz on $\{0,1\}^m$, and let $\bar{h}(p) = \mathbb{E}_{S\sim D_p} h(S)$ be its multilinear extension on $[0,1]^m$ in the sense of [HMMS25], where D_p includes each j independently with probability p_j. By [HMMS25, Lemma 16], $\bar{h}$ is 1-Lipschitz with respect to ℓ_1. Since $\xi^x = \mathbf{1}_S - \mathbf{1}_{\bar{S}}$ with $S \sim D_{(\mathbf{1}+x)/2}$, we have $G(x) = 2\bar{h}(\frac{\mathbf{1}+x}{2})$, and therefore $|G(x) - G(x')| \le 2\|\frac{x-x'}{2}\|_1 = \|x - x'\|_1$. (d) Rounding changes each fractional coordinate by at most 1 and leaves the others unchanged, so by (a) and (c), $|g(x')| = |G(x') - G(x)| \le \sum_{j\in U} |x_j - x'_j|$, which is at most the number of fractional coordinates in U. □

We now recall the consensus-halving lemma of [HMMS25] for two colours. Theorem 5 of [HMMS25], due to Simmons and Su [SS03] for $k = 2$ and to Filos-Ratsikas, Hollender, Sotiraki and Zampetakis [FRHSZ21] for prime k, reads as follows for $k = 2$: *for any utility functions $u_1, \dots, u_n$ on the Borel subsets of $[0,1]$ satisfying the continuity condition of [HMMS25, Assumption 4], it is possible to partition $[0,1]$ into two pieces A_1, A_2 using at most n cuts so that $u_i(A_1) = u_i(A_2)$ for all i.* Placing item j on the interval $[\frac{j-1}{m}, \frac{j}{m}]$ yields Lemma 9 of [HMMS25], which for $k = 2$ reads: *for any 1-Lipschitz $f_1, \dots, f_n \colon \{0,1\}^m \to \mathbb{R}$ there is a fractional 2-colouring χ with at most n fractional elements such that $\bar{f}_i(\chi_1) = \bar{f}_i(\chi_2)$ for all i.* Here $\chi_1 \in [0,1]^m$, $\chi_2 = \mathbf{1} - \chi_1$, and $\bar{f}$ is the multilinear extension. In our language:

Corollary 2.8. *For odd 2-Lipschitz $g_1, \dots, g_n$ on $\{\pm 1\}^{[m]}$ there is $x \in [-1,1]^m$ with at most n fractional coordinates such that $G_i(x) = 0$ for all i.*

Proof. Apply [HMMS25, Lemma 9] to the 1-Lipschitz functions $f_i(S) = \frac{1}{2}g_i(\mathbf{1}_S - \mathbf{1}_{\bar{S}})$ (Lemma 2.1(b)), obtaining $p = \chi_1$. If $S \sim D_p$, then $[m]\setminus S \sim D_{\mathbf{1}-p}$, so $0 = \bar{f}_i(p) - \bar{f}_i(\mathbf{1}-p) = \mathbb{E}[f_i(S) - f_i([m]\setminus S)] = \mathbb{E}[g_i(\mathbf{1}_S - \mathbf{1}_{\bar{S}})] = G_i(2p - \mathbf{1})$. Take $x = 2p - \mathbf{1}$; its fractional coordinates are those of p. □

Equivalently, let $r \le n$ be the number of fractional coordinates of x. The law of ξ^x is a product distribution with full support on the r-dimensional subcube $\{z \in \{\pm 1\}^{[m]} : z_j = x_j \text{ for all integral } j\}$, and every g_i has mean 0 under it.

2.5 Loops and separating sets

Definition 2.9. Let $U \ne \emptyset$ be finite and let $s, r \ge 0$ be reals.

- An *s-path* in $A \subseteq \{\pm 1\}^U$ is a sequence $z_0, \dots, z_L \in A$ with $\mathrm{dist}(z_{i-1}, z_i) \le s$ for all i. It is *antipodal* if $z_L = -z_0$. The set A *contains an s-loop* if it contains an antipodal s-path.
- A set $A \subseteq \{\pm 1\}^U$ is *r-separating* if $\{\pm 1\}^U \setminus A = A^+ \sqcup A^-$ with $A^- = -A^+$ and $\mathrm{dist}(A^+, A^-) > r$. (The set $A = \{\pm 1\}^U$ is r-separating with $A^\pm = \emptyset$.) Separating sets are symmetric.

Lemma 2.10 (crossing)**.** *If g is odd and 2-Lipschitz and $z_0, \dots, z_L$ is an antipodal s-path, then $|g(z_i)| \le s$ for some i.*

Proof. Consecutive values differ by at most $2s$, and $g(z_L) = -g(z_0)$. If $g(z_0) = 0$ we are done. Otherwise the values change sign, so there are consecutive values a, b with $a > 0 \ge b$ or $a < 0 \le b$. Then $|a| + |b| = |a - b| \le 2s$, so $\min(|a|, |b|) \le s$. □

Lemma 2.11 (certificates)**.** *(a) If $A \subseteq \{\pm 1\}^U$ is r-separating, then $\varphi_A = \mathrm{dist}(\cdot, A^-) - \mathrm{dist}(\cdot, A^+)$ (and $\varphi_A \equiv 0$ if $A = \{\pm 1\}^U$) is odd and 2-Lipschitz, and $\{z : |\varphi_A(z)| \le r\} \subseteq A$.*

(b) If $A \subseteq \{\pm 1\}^U$ is nonempty and symmetric and contains no s-loop, then there is an odd 2-Lipschitz ψ on $\{\pm 1\}^U$ with $|\psi| > s$ on A.

Proof. (a) Distance functions are 1-Lipschitz, so φ_A is 2-Lipschitz. Since $A^- = -A^+$, we have $\mathrm{dist}(-z, A^-) = \mathrm{dist}(z, A^+)$, so φ_A is odd. On A^+, $\varphi_A = \mathrm{dist}(\cdot, A^-) > r$, and on A^-, $\varphi_A < -r$. (b) Consider the graph on A joining points at distance at most s. If a connected component C contained both z and $-z$, a path in C from z to $-z$ would be an s-loop. Since $z \mapsto -z$ is an isometry preserving A, it maps components to components, so the components come in pairs $\{C, -C\}$ with $C \neq -C$. Let A^+ contain one component from each pair, and let $A^- = -A^+$. Then $A = A^+ \sqcup A^-$ and $\mathrm{dist}(A^+, A^-) > s$. The function $\psi = \mathrm{dist}(\cdot, A^-) - \mathrm{dist}(\cdot, A^+)$ is odd and 2-Lipschitz as in (a), and $|\psi| > s$ on A. □

Lemma 2.12 (loops)**.** *Let $q \ge 0$ and let $\varphi_1, \dots, \varphi_q$ be odd 2-Lipschitz functions on $\{\pm 1\}^U$. Then $J = \{z : |\varphi_j(z)| \le \mathcal{K}_{q+1}$ for all $j\}$ contains a $\mathcal{K}_{q+1}$-loop.*

Proof. J is symmetric because the φ_j are odd. It is nonempty: for $q = 0$, $J = \{\pm 1\}^U$; for $q \ge 1$, the definition of $\mathcal{K}_q \le \mathcal{K}_{q+1}$ gives a point with $\max_j |\varphi_j| \le \mathcal{K}_q$. Suppose J contains no $\mathcal{K}_{q+1}$-loop. By Lemma 2.11(b) there is an odd 2-Lipschitz ψ with $|\psi| > \mathcal{K}_{q+1}$ on J. The instance $(\varphi_1, \dots, \varphi_q, \psi)$ on the item set U has $q+1$ functions, so some z satisfies $|\varphi_j(z)| \le \mathcal{K}_{q+1}$ for all j and $|\psi(z)| \le \mathcal{K}_{q+1}$. Then $z \in J$ and $|\psi(z)| \le \mathcal{K}_{q+1}$, a contradiction. □

3 Small values: $\mathcal{K}_1 = 1$ and $\mathcal{K}_2 = \mathcal{K}_3 = 2$

Theorem 1.2, first part (restated). *$\mathcal{K}_1 = 1$ and $\mathcal{K}_2 = \mathcal{K}_3 = 2$. (The value $\mathcal{K}_4 = 3$ is proved in Section 3.4.)*

3.1 Lower bounds and $\mathcal{K}_1$

Proof of $\mathcal{K}_1 = 1$ and of the lower bounds. Let g be odd and 2-Lipschitz on $\{\pm 1\}^{[m]}$ and let $x \in \{\pm 1\}^{[m]}$. Flipping the coordinates of x one at a time gives an antipodal 1-path from x to $-x$, and Lemma 2.10 yields a point with $|g| \le 1$, so $\mathcal{K}_1 \le 1$. The function $g = x_1$ is odd and 2-Lipschitz with $|g| \equiv 1$, so $\mathcal{K}_1 \ge 1$. The functions $g_1 = x_1 + x_2$ and $g_2 = x_1 - x_2$ are odd and 2-Lipschitz, and for every x one of them equals ± 2; hence $\mathcal{K}_3 \ge \mathcal{K}_2 \ge 2$. □

3.2 Upper bounds via few fractional coordinates

Let $g_1, \dots, g_n$ be odd and 2-Lipschitz on $\{\pm 1\}^{[m]}$ with $n \le 3$. By Corollary 2.8 there is x with $r \le n$ fractional coordinates and $G_i(x) = 0$ for all i. Let Q be the r-dimensional subcube of points agreeing with x on its integral coordinates. Write $w_z = \Pr[\xi^x = z]$ for $z \in Q$; these are product weights, all positive. Then

$$\sum_{z \in Q} w_z \, g_i(z) = 0 \qquad \text{for } i = 1, \dots, n, \tag{1}$$

and each g_i changes by at most 2 along every edge of Q. Call $z \in Q$ *bad for i* if $|g_i(z)| > 2$, and *good* if it is bad for no i. A good corner z is a colouring with $\max_i |g_i(z)| \le 2$, so it suffices to show that a good corner exists.

Lemma 3.1. *Let a, b be reals with $|a - b| \le 2$, and suppose $\alpha a + \beta b = 0$ for some $\alpha, \beta > 0$. Then $|a|, |b| \le 2$.*

Proof. Either $a = b = 0$, or a and b have opposite signs, say $a > 0 > b$. In the latter case $a < a - b \le 2$ and $-b < a - b \le 2$. □

Lemma 3.2 (corner lemma)**.** *Let v be a real function on the four corners of a square (a 4-cycle) such that adjacent values differ by at most 2, and let w be positive weights with $\sum_x w_x v_x = 0$. Then at most one corner x has $|v_x| > 2$.*

Proof. If $v_x > 2$, then every neighbour y of x has $v_y \ge v_x - 2 > 0$; similarly for $v_x < -2$. Two corners with $v_x > 2$ and $v_{x'} < -2$ would satisfy $|v_x - v_{x'}| > 4$, while any two corners of a square are at distance at most 2, so their values differ by at most 4. Hence any two bad corners have the same sign, say positive. If they are adjacent, each of the other two corners is adjacent to one of them; if they are opposite, each of the other two corners is adjacent to both. In either case all four values are positive, contradicting $\sum_x w_x v_x = 0$. □

Proof of $\mathcal{K}_2 \le 2$, and of $\mathcal{K}_3 \le 2$ when $r \le 2$. If $r = 0$, then $Q = \{x\}$ and $g_i(x) = G_i(x) = 0$. If $r = 1$, then $Q = \{z, z'\}$ is an edge, and by (1) and Lemma 3.1 both corners are good. If $r = 2$, then Q is a square, and by (1) and Lemma 3.2 every function has at most one bad corner; since $n \le 3 < 4$, some corner is good. For $n = 2$ we always have $r \le 2$, which proves $\mathcal{K}_2 \le 2$. □

3.3 Three fractional coordinates: the computer-assisted step

It remains to treat $n = 3$ and $r = 3$. Identify Q with $\{0,1\}^3$ through the three fractional coordinates, where a coordinate equal to $+1$ corresponds to 1. The weights are $w_y(p) = \prod_{k=1}^{3} p_k^{y_k}(1 - p_k)^{1-y_k}$ with $p \in (0,1)^3$. Write $v_y^i = g_i(y)$. Suppose, for a contradiction, that no corner is good. Assign every corner to one function for which it is bad. This gives a partition $\{0,1\}^3 = T_1 \sqcup T_2 \sqcup T_3$, with some parts possibly empty, such that every $y \in T_i$ is bad for i.

Lemma 3.3. *Each nonempty T_i is either* one-signed *($v^i > 2$ on T_i, or $v^i < -2$ on T_i) or a* mixed antipodal pair *$T_i = \{y, \bar{y}\}$ with $\bar{y} = \mathbf{1} - y$ and $v_y^i > 2 > -2 > v_{\bar{y}}^i$.*

Proof. If $v_y^i > 2$ and $v_{y'}^i < -2$, then $2\,\mathrm{dist}(y, y') \ge |v_y^i - v_{y'}^i| > 4$, so $\mathrm{dist}(y, y') = 3$, that is, $y' = \bar{y}$. Hence if T_i contains points of both signs, every positive point is antipodal to every negative point. A point has only one antipode, so T_i consists of exactly one positive point and its antipode. □

Lemma 3.4. *Let X be distributed according to the weights $w(p)$.*

(a) If $T = T_i$ is one-signed, then $\mathbb{E}[\mathrm{dist}(X, T)] > 1$.

(b) If $T_i = \{y, \bar{y}\}$ is a mixed antipodal pair with $v_y^i > 2$, then $1 < \mathbb{E}[\mathrm{dist}(X, y)] < 2$.

Proof. (a) Replacing v^i by $-v^i$ if necessary (this preserves (1)), assume $v^i > 2$ on T. For every corner z and every $y \in T$ we have $v_z^i \ge v_y^i - 2\,\mathrm{dist}(z, y) > 2 - 2\,\mathrm{dist}(z, y)$, so $v_z^i > 2 - 2\,\mathrm{dist}(z, T)$. Multiplying by $w_z > 0$, summing, and using (1) gives $0 > 2 - 2\mathbb{E}[\mathrm{dist}(X, T)]$. (b) As in (a), $v_z^i > 2 - 2\,\mathrm{dist}(z, y)$. Also $v_z^i \le v_{\bar{y}}^i + 2\,\mathrm{dist}(z, \bar{y}) < -2 + 2(3 - \mathrm{dist}(z, y)) = 4 - 2\,\mathrm{dist}(z, y)$. Summing with the weights gives $0 > 2 - 2\mathbb{E}\,\mathrm{dist}(X, y)$ and $0 < 4 - 2\mathbb{E}\,\mathrm{dist}(X, y)$. □

Since $\mathrm{dist}(X, \bar{y}) = 3 - \mathrm{dist}(X, y)$, condition (b) is symmetric in y and $\bar{y}$. A *configuration* consists of a labelling $\lambda\colon \{0,1\}^3 \to \{1, 2, 3\}$ (the parts $T_i = \lambda^{-1}(i)$) and, for every nonempty part that is an antipodal pair, a choice of type (one-signed or mixed). There are $3^8 = 6561$ labellings and 7365

configurations. Each condition in Lemma 3.4 has the form $\sum_y \rho_y w_y(p) > 0$ for an integer vector ρ: $\rho_y = \operatorname{dist}(y, T) - 1$ in case (a), and $\rho_y = \operatorname{dist}(y, y_0) - 1$ and $\rho_y = 2 - \operatorname{dist}(y, y_0)$ in case (b), where y_0 is the corner of the pair with $v^i_{y_0} > 2$. The left-hand side is a multilinear polynomial in p.

Lemma 3.5 (computer-assisted)**.** *For each of the* 7365 *configurations, the conditions of Lemma 3.4 have no common solution* $p \in [0,1]^3$.

Verification. A multilinear polynomial on a box $\prod_k [a_k, b_k]$ is the multilinear interpolation of its values at the 8 vertices. Its value at any point of the box is therefore a convex combination of the vertex values, and if it is ≤ 0 at all vertices, it is ≤ 0 on the box. For each configuration, a branch-and-bound procedure starts with the box $[0,1]^3$. A box is *cleared* if one of the configuration's polynomials is ≤ 0 at all 8 vertices, so that the corresponding strict condition fails on the whole box. Otherwise the box is bisected along its longest side. All evaluations use exact rational arithmetic. The procedure clears every configuration, with 20,613 boxes in total and subdivision depth at most 6 (Appendix A). Every point $p \in [0,1]^3$ lies in a cleared box, so some condition fails at p. □

Proof of $\mathcal{K}_3 \le 2$ *when* $r = 3$. If no corner were good, Lemmas 3.3 and 3.4 would give a configuration whose conditions all hold at the actual $p \in (0,1)^3$, contradicting Lemma 3.5. Hence a good corner exists. Together with the case $r \le 2$, this proves $\mathcal{K}_3 \le 2$ and completes the proof of Theorem 1.2. □

Remark 3.6. The product structure of the weights is essential for Lemma 3.5. If w is an arbitrary probability vector on $\{0,1\}^3$, the conditions of Lemma 3.4 can be satisfied. For instance, take $T_1 = \{y : |y| \le 1\}$, $T_2 = \{y : |y| = 2\}$, $T_3 = \{\mathbf{1}\}$, all one-signed, and put weight 0.4 on $\mathbf{0}$ and 0.6 on $\mathbf{1}$. The three expectations are then $1.2, 1.4, 1.2 > 1$, and they remain > 1 after a small perturbation making all weights positive. For four functions the corner method cannot prove $\mathcal{K}_4 \le 2$, because $\mathcal{K}_4 \ge 3$: the four additive functions $g_p(x) = \langle p, x\rangle$ on three items, $p \in \{(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)\}$, satisfy $\max_p |g_p(x)| = 3$ for every $x \in \{\pm 1\}^3$, because the eight vectors $\pm p$ are exactly the points of $\{\pm 1\}^3$. So $\mathcal{K}_4 \ge 3 > \mathcal{K}_3$ (Proposition 4.4 gives a 3-sparse witness). The matching upper bound is proved in Section 3.4.

3.4 Four functions, and sparse families of five functions

We extend the corner method to threshold c and to functions that read only some of the fractional coordinates. Let J be a finite set, let w be a product distribution on $\{\pm 1\}^J$ with full support, and let $X \sim w$.

Lemma 3.7. *Let* $c > 0$, $J' \subseteq J$, *and let* $h\colon \{\pm 1\}^{J'} \to \mathbb{R}$ *be* 2*-Lipschitz with* $\mathbb{E}\, h(X|_{J'}) = 0$. *Let* $U^+ = \{u : h(u) > c\}$ *and* $U^- = \{u : h(u) < -c\}$. *If* $U^+ \neq \emptyset$ *then* $\mathbb{E}\operatorname{dist}(X|_{J'}, U^+) > c/2$, *if* $U^- \neq \emptyset$ *then* $\mathbb{E}\operatorname{dist}(X|_{J'}, U^-) > c/2$, *and if both are nonempty then* $\operatorname{dist}(U^+, U^-) > c$. *These conditions remain true when* U^+ *and* U^- *are replaced by nonempty subsets.*

Proof. For every $u \in U^+$ and every z, $h(z) \ge h(u) - 2\operatorname{dist}(z,u) > c - 2\operatorname{dist}(z,u)$, so $h(z) > c - 2\operatorname{dist}(z, U^+)$; taking expectations gives $0 > c - 2\mathbb{E}\operatorname{dist}(X|_{J'}, U^+)$. The case of U^- is symmetric, and $h(u) - h(u') > 2c$ for $u \in U^+$, $u' \in U^-$ gives $2\operatorname{dist}(u,u') > 2c$. Shrinking the sets can only increase the distances. □

Theorem 3.8 (computer-assisted)**.** *(a)* $\mathcal{K}_4 \le 3$.

(b) Every 2*-sparse instance with at most five functions has discrepancy at most* 2.

Reduction and verification. Let $c = 3$, $n = 4$ in (a) and $c = 2$, $n \le 5$ in (b). By Corollary 2.8 there is a point x with a set J of $j \le n$ fractional coordinates at which all multilinear extensions vanish. Fix the integral coordinates and let w be the product distribution on $\{\pm 1\}^J$ with means x_e; it has full support. The restriction h_i of g_i depends only on the set $J_i \subseteq J$ of fractional items in its support, is 2-Lipschitz, and has $\mathbb{E}\, h_i = 0$. In (a), $J_i = J$; in (b), every $e \in J$ lies in J_i for at most two indices i. If no corner of $\{\pm 1\}^J$ satisfies $|h_i| \le c$ for all i, then the sets $U_i^{\pm}$ of Lemma 3.7 cover $\{\pm 1\}^J$, in the sense that every corner z has some i with $z|_{J_i} \in U_i^+ \cup U_i^-$, and they satisfy the conditions of the lemma at the actual means. Negating a coordinate of $\{\pm 1\}^J$ in all functions replaces its mean x_e by $-x_e$ and preserves all hypotheses, so we may assume $p_e = \Pr[X_e = 1] \le \frac{1}{2}$ for all e. Each condition $\mathbb{E}\,\mathrm{dist}(X|_{J_i}, U) > c/2$ is a multilinear function of p, so if it fails at all vertices of a box it fails on the whole box.

The verification runs a branch-and-bound over dyadic boxes in $[0, \frac{1}{2}]^j$ for every $j \le n$ and, in (b), every sparsity pattern (an assignment of each of the j items to a set of one or two of the five functions, up to relabelling the functions; $1, 2, 7, 22, 69, 198$ patterns for $j = 0, \dots, 5$). In each box it lists, for each function, the pairs (U^+, U^-) that satisfy the conditions at some vertex, and a depth-first search confirms that no choice of one such pair per function covers $\{\pm 1\}^J$; otherwise the box is bisected. All comparisons are exact. For (a) all boxes are cleared with 13 boxes for $j = 4$; for (b) all 299 patterns are cleared with 1213 boxes in total and subdivision depth at most 3. Two independently written programs, and two further re-verifications of all leaf boxes, agree (Appendix A). As a sanity check, the same search does not clear $(n, c) = (4, 2)$, $(2, 1.9)$, $(3, 1.9)$, $(4, 2.9)$, or the 3-sparse pattern of Proposition 4.4 at $c = 2$. In each of these cases it exhibits an interior point p at which the conditions admit a cover. Hence in (a) and (b) some corner is good, which gives the stated bounds. □

Proof of $\mathcal{K}_4 = 3$. $\mathcal{K}_4 \ge 3$ by Remark 3.6 and $\mathcal{K}_4 \le 3$ by Theorem 3.8(a). □

By Lemma 8.5, Theorem 3.8(b) says that every simple graph on at most five vertices is 2-good, so by Corollary 8.7 every 2-sparse instance whose blocks have at most five vertices has discrepancy at most 2.

3.5 Real thresholds: six functions, and $\mathcal{K}_5$, $\mathcal{K}_6$

The corner method becomes much stronger if it is run at a threshold slightly below the next integer. This rests on the following per-function version of Lemma B.1 (Appendix B), with the same proof.

Lemma 3.9. *Let $(U_v)_{v \in V}$ be supports and $D_v \ge 1$ integers. If some instance with these supports has, for every x, some v with $|g_v(x)| > D_v - 1$, then some instance with the same supports has, for every x, some v with $|g_v(x)| \ge D_v$. Consequently, if for a class of support structures every instance has a colouring with $|g_v| \le c_v$ for all v, where $c_v \in [D_v - 1, D_v)$, then every such instance has a colouring with $|g_v| \le D_v - 1$ for all v.*

Proof. Let $P_v = \{z \in \{\pm 1\}^{U_v} : g_v(z) > D_v - 1\}$. For $z, z' \in P_v$ we have $2\,\mathrm{dist}(z, -z') \ge g_v(z) + g_v(z') > 2D_v - 2$, so $\mathrm{dist}(z, -z') \ge D_v$. The function $g'_v = \mathrm{dist}(\cdot, -P_v) - \mathrm{dist}(\cdot, P_v)$ ($g'_v \equiv 0$ if $P_v = \emptyset$) is odd and 2-Lipschitz, depends only on U_v, and satisfies $|g'_v| \ge D_v$ on $P_v \cup (-P_v)$, which contains $x|_{U_v}$ for some v, for every x. The second statement follows by applying the first to an instance violating the thresholds $D_v - 1$. □

Thus Theorem 3.8(b) only needs a threshold $c < 3$, and $\mathcal{K}_n \le D - 1$ follows from $\mathcal{K}_n < D$. We run the reduction in the proof of Theorem 3.8 with real thresholds c_v: by Corollary 2.8 and Lemma 3.7, if some instance with the given supports has no colouring with $|g_v| \le c_v$ for all v,

then for some $j \le n$, some pattern of j fractional items and some $p \in (0, \frac{1}{2}]^j$ there are sets $U_v^{\pm}$ with $\operatorname{dist}(U_v^+, U_v^-) \ge \lfloor c_v \rfloor + 1$ and $\mathbb{E}\operatorname{dist}(X|_{J_v}, U_v^{\pm}) > c_v/2$ for the nonempty parts, which cover $\{\pm 1\}^j$. (At a fixed p these conditions are also sufficient for mean-zero 2-Lipschitz residuals that are bad at every corner: if both parts are nonempty, a convex combination of $c_v + \varepsilon - 2\operatorname{dist}(\cdot, U_v^+)$ and $-c_v - \varepsilon + 2\operatorname{dist}(\cdot, U_v^-)$ has mean 0, and if only U_v^+ is nonempty, subtract its mean from $c_v + \varepsilon - 2\operatorname{dist}(\cdot, U_v^+)$. So the box relaxation is the only loss of the method relative to the residuals.) The branch-and-bound is that of Theorem 3.8; with several thresholds, two functions are exchanged by the symmetry breaking only if they read the same items and have the same threshold.

Theorem 3.10 (computer-assisted)**.** *(a) Every 2-sparse instance with at most six functions has discrepancy at most 2.*

(b) $\mathcal{K}_5 = 3$ *and* $\mathcal{K}_6 = 4$*; moreover* $\mathcal{K}_8 \ge 5$*.*

Proof. (a) By Lemma 3.9 it suffices to clear the threshold $c = 2.99$. Up to relabelling the functions there are $2, 7, 23, 76, 245, 782$ sparsity patterns of $j = 1, \dots, 6$ items on six functions (these numbers agree with Burnside's lemma). All of them are cleared, with fewer than 1700 boxes in total, by two independently written programs, each run on its own enumeration of the patterns (Appendix A).

(b) Lower bounds: $\mathcal{K}_5 \ge \mathcal{K}_4 = 3$. The six additive functions $\langle a_i, x\rangle$ on six items with $a_1 = -\mathbf{1}$, $a_2 = (1,1,-1,-1,-1,-1)$, $a_3 = (1,-1,1,-1,-1,-1)$, $a_4 = (-1,1,1,1,-1,-1)$, $a_5 = (-1,1,1,-1,1,-1)$, $a_6 = (1,-1,-1,1,1,-1)$ satisfy $\max_i |\langle a_i, x\rangle| \ge 4$ for all $x \in \{\pm 1\}^6$: $|\langle a_i, x\rangle| = 6 - 2\operatorname{dist}(x, \pm a_i) \ge 4$ if and only if $\operatorname{dist}(x, \pm a_i) \le 1$, and the twelve balls of radius 1 around the $\pm a_i$ cover $\{\pm 1\}^6$. Similarly, one word from each complementary pair of the Hamming $[7, 4]$ code (which contains $\mathbf{1}$) gives eight additive functions on seven items with $\max_i |\langle a_i, x\rangle| \ge 5$, since the code is perfect of radius 1; so $\mathcal{K}_8 \ge 5$. Upper bounds: by Lemma 3.9 it suffices to clear $c = 3.99$ for five functions and $c = 4.99$ for six functions, all reading all $j \le n$ fractional items; the branch-and-bound needs $1, 1, 1, 3, 29$ and $1, 1, 1, 1, 3, 15$ boxes, respectively, for $j = 1, 2, \dots$ (two programs, identical counts). As sanity checks, the same program fails for four functions at $c = 2.99$ and for six functions at $c = 3.99$, in accordance with $\mathcal{K}_4 = 3$ and $\mathcal{K}_6 \ge 4$. □

In particular K_6, $K_{3,3}$ and the octahedron, with arbitrary bundle sizes and private items, carry only families of discrepancy at most 2, and by Corollary 8.7 so does every pattern whose blocks have at most six vertices. The values $\mathcal{K}_1, \dots, \mathcal{K}_6$ are $1, 2, 2, 3, 3, 4$.

4 Acyclic sparsity patterns

Theorem 4.1. *Let* $\mathcal{G} = (g_v, U_v)_{v \in V}$ *be a* t*-sparse instance whose bundle hypergraph is Berge-acyclic. Then* $\operatorname{disc}(\mathcal{G}) \le \mathcal{K}_t$*.*

Proof. Let $c = \mathcal{K}_t$. Functions with empty support are identically zero and are isolated in $I(\mathcal{G})$, so we drop them. Different trees of the forest $I(\mathcal{G})$ involve disjoint sets of items and functions, so we may assume that $I(\mathcal{G})$ is a tree. Root it at a function r. Every non-root function v has a parent bundle $P(v) \ni v$. Every bundle S has a parent function $p(S) \in S$, and its children are the functions in $S \setminus \{p(S)\}$, of which there are at most $t - 1$. The child bundles of a function v are the bundles containing v other than $P(v)$; this includes v's private bundle, if nonempty.

For a non-root function v, let T_v be the set of functions in the subtree of v, and let D_v be the set of items in bundles of the subtree of v, excluding the bundle $P(v)$. Every $w \in T_v$ reads only items in $D_v \cup B_{P(v)}$, and among the functions in T_v only v reads $B_{P(v)}$: another $w \in T_v$ belonging to $P(v)$ would close a cycle in $I(\mathcal{G})$. Define

$$F_v = \{y \in \{\pm 1\}^{B_{P(v)}} : \exists z \in \{\pm 1\}^{D_v} \text{ such that } |g_w(z, y)| \le c \text{ for all } w \in T_v\}.$$

For a bundle S with children $w_1, \dots, w_q$ let $G_S = \bigcap_i F_{w_i} \subseteq \{\pm 1\}^{B_S}$, and $G_S = \{\pm 1\}^{B_S}$ if $q = 0$. The subtrees of distinct children of distinct bundles use disjoint items. Hence $y \in G_S$ if and only if all subtrees hanging below S can be completed simultaneously, given the colouring y of B_S. By oddness, if z witnesses $y \in F_v$ then $-z$ witnesses $-y \in F_v$, so the sets F_v and G_S are symmetric. We prove, by induction from the leaves up:

(i) for every non-root function v, F_v is c-separating; **(ii)** for every bundle S, G_S contains a c-loop.

(i) implies (ii). Let S have children $w_1, \dots, w_q$, where $0 \le q \le t-1$. If $q = 0$, then $G_S = \{\pm 1\}^{B_S}$ contains the 1-loop obtained by flipping coordinates one at a time, and $1 = \mathcal{K}_1 \le c$. If $q \ge 1$, let $\varphi_i = \varphi_{F_{w_i}}$ be the functions from Lemma 2.11(a), so that $\{|\varphi_i| \le c\} \subseteq F_{w_i}$. By Lemma 2.12 the set $J = \{|\varphi_i| \le \mathcal{K}_{q+1}\ \forall i\}$ contains a $\mathcal{K}_{q+1}$-loop. Since $q+1 \le t$, we have $\mathcal{K}_{q+1} \le c$, so $J \subseteq \bigcap_i \{|\varphi_i| \le c\} \subseteq G_S$, and a $\mathcal{K}_{q+1}$-loop is a c-loop.

(ii) implies (i). Let v be a non-root function with child bundles $S_1, \dots, S_p$ ($p \ge 0$), and let $\Gamma_i \subseteq G_{S_i}$ be the point set of a c-loop. Put $X_i = \Gamma_i \cup (-\Gamma_i) \subseteq G_{S_i}$ (recall that G_{S_i} is symmetric) and $X = \prod_i X_i$; for $p = 0$, X is a single point. Each X_i is symmetric and *c-connected*: any two of its points are joined by a c-path in X_i, because Γ_i and $-\Gamma_i$ are c-connected and share the points $\pm z_0$. Changing one coordinate block at a time shows that X is symmetric and c-connected.

Let $y \notin F_v$ and $z \in X$. If $|g_v(z, y)| \le c$, then completing the subtrees below each S_i according to $z_i \in G_{S_i}$ would show $y \in F_v$, because the items involved are disjoint. Hence $|g_v(z, y)| > c$ for all $z \in X$. Along a step of Hamming length at most c, $g_v(\cdot, y)$ changes by at most $2c$, so it cannot jump from above c to below $-c$. By c-connectivity, $g_v(\cdot, y)$ has a constant sign $\sigma(y) \in \{\pm 1\}$ on X. Since $X = -X$, $F_v = -F_v$ and g_v is odd, we get $\sigma(-y) = -\sigma(y)$. If $y, y' \notin F_v$ and $\operatorname{dist}(y, y') \le c$, then for a fixed $z \in X$ the values $g_v(z, y)$ and $g_v(z, y')$ differ by at most $2c$ and both exceed c in absolute value, so $\sigma(y) = \sigma(y')$. Thus $A^{\pm} = \{y \notin F_v : \sigma(y) = \pm 1\}$ satisfy $A^- = -A^+$ and $\operatorname{dist}(A^+, A^-) > c$, and F_v is c-separating.

The root. Let $S_1, \dots, S_p$ be the bundles containing r, and define X as above using (ii). Choose $z \in X$. By symmetry and c-connectivity there is an antipodal c-path in X from z to $-z$. Since g_r reads only items in $B_{S_1}, \dots, B_{S_p}$, Lemma 2.10 gives z^* on this path with $|g_r(z^*)| \le c$. Since $z^*_i \in G_{S_i}$, all subtrees can be completed, and every function has $|g_w| \le c$. □

Proof of Theorem 1.3. The upper bound is Theorem 4.1 applied to $\mathcal{G}(\mathcal{F})$. For tightness, fix $\varepsilon > 0$ and take an instance $g_1, \dots, g_t$ on some $\{\pm 1\}^{[m]}$ with $\min_x \max_i |g_i(x)| > \mathcal{K}_t - \varepsilon/2$. Lemma 2.6 makes every coordinate relevant to every function at a cost of at most $\varepsilon/2$. Via Lemma 2.1(b), the result is a t-sparse family whose only bundle is $B_{[t]} = [m]$; its incidence graph is a star, and its discrepancy exceeds $\mathcal{K}_t - \varepsilon$. □

Proof of Corollary 1.4. Combine Theorem 1.3 with Lemma 2.5 and Theorem 1.2. Tightness for $t \in \{2, 3\}$ is witnessed by the family f_1, f_2 with imbalances $x_1 + x_2$ and $x_1 - x_2$. Both items are relevant to both functions, so there is a single bundle, and the discrepancy is 2. □

Fixing items. The following lemma symmetrizes an instance after some items have been fixed.

Lemma 4.2. *Let g be odd and 2-Lipschitz on $\{\pm 1\}^{F \sqcup W}$, depending only on the coordinates in U, and let $s \in \{\pm 1\}^F$. The function $\tilde{g}(x) = \frac{1}{2}(g(s, x) - g(s, -x))$ on $\{\pm 1\}^W$ is odd and 2-Lipschitz, depends only on the coordinates in $U \cap W$, and satisfies $|g(s, x) - \tilde{g}(x)| \le |F \cap U|$ for all x.*

Proof. Oddness, the Lipschitz property and the dependence are immediate. By oddness of g, $g(s, -x) = -g(-s, x)$, so $g(s, x) - \tilde{g}(x) = \frac{1}{2}(g(s, x) - g(-s, x))$. The points (s, x) and $(-s, x)$ differ only in coordinates of F, and only those in $F \cap U$ matter. Hence the difference is at most $\frac{1}{2} \cdot 2|F \cap U|$. □

Corollary 4.3. *Let $\mathcal{G} = (g_v, U_v)_{v \in V}$ be a t-sparse instance and $F \subseteq [m]$. Suppose that the instance on the items $[m] \setminus F$ with supports $U_v \setminus F$ has a Berge-acyclic bundle hypergraph. Then* $\operatorname{disc}(\mathcal{G}) \le \mathcal{K}_t + \max_v |F \cap U_v|$.

Proof. Fix any $s \in \{\pm 1\}^F$ and form $\tilde{g}_v$ as in Lemma 4.2. The instance $(\tilde{g}_v, U_v \setminus F)_v$ is t-sparse and Berge-acyclic, so by Theorem 4.1 some x has $|\tilde{g}_v(x)| \le \mathcal{K}_t$ for all v. Then $|g_v(s,x)| \le \mathcal{K}_t + |F \cap U_v|$. □

Proof of Corollary 1.5. Apply Corollary 4.3 to $\mathcal{G}(\mathcal{F})$. For $t = 2$, if F is a union of bundles whose removal leaves a forest, then the bundles outside F are unchanged, so the remaining bundle hypergraph is that forest; and $\mathcal{K}_2 = 2$. □

For example, if the item multigraph is a cycle in which some bundle has b items, then Corollary 1.5 gives $\operatorname{disc} \le 2 + b$. This is independent of the other bundle sizes, but Theorem 1.6 gives the uniform bound 4, and Corollary 8.22 gives the tight bound 2.

Proposition 4.4. *Index four items by the 3-subsets of $\{0,1,2,3\}$, so that function i reads the three items whose index contains i, and let*

$$g_0 = x_{012} + x_{013} + x_{023}, \quad g_1 = x_{012} - x_{013} + x_{123}, \quad g_2 = x_{012} - x_{023} - x_{123}, \quad g_3 = x_{013} - x_{023} + x_{123}.$$

This family is 3-sparse, its bundle hypergraph is the complete 3-uniform hypergraph on four vertices, and every colouring has $\max_i |g_i(x)| = 3$. Hence $\operatorname{disc} = 3 > \mathcal{K}_3 = 2$.

Proof. Each item lies in exactly three supports. Function g_i equals ± 3 exactly when x restricted to its three items equals $\pm$ its coefficient vector, that is, on $2 \cdot 2 = 4$ of the 16 colourings. These four sets cover all colourings. By symmetry under $x \mapsto -x$ it suffices to consider $x_{012} = 1$. Then g_0, g_1 and g_2 equal ± 3 exactly when $(x_{013}, x_{023}, x_{123})$ lies in $\{(1,1,\pm 1)\}$, $\{(-1,\pm 1,1)\}$ and $\{(\pm 1,-1,-1)\}$, respectively, and $g_3 = \pm 3$ exactly when it lies in $\{(1,-1,1),(-1,1,-1)\}$. These four sets partition $\{\pm 1\}^3$. □

5 Bounded bundle-degree: a parity decomposition

5.1 Bundle structures and the main statement

We state the theorem for a slightly more flexible notion of bundles, which also covers merged private items.

Definition 5.1. Let $\mathcal{G} = (g_v, U_v)_{v \in V}$ be an instance. A *bundle structure* for $\mathcal{G}$ is a family $(B_e, S_e)_{e \in \mathcal{E}}$, where the B_e are nonempty, pairwise disjoint item sets with $\bigcup_v U_v \subseteq \bigcup_e B_e$, the $S_e \subseteq V$ are nonempty, and

$$U_v \subseteq \textstyle\bigcup_{e \in N(v)} B_e \qquad \text{for every } v, \qquad \text{where } N(v) = \{e : v \in S_e\}.$$

Write $d_v = |N(v)|$. The structure is *t-bounded* if $|S_e| \le t$ for all e.

The bundles of Definition 2.3, with $S_e = S$ for $e = B_S$, form a bundle structure. We may also *merge* the private bundle $B_{\{v\}}$ of a function v into one of the non-private bundles $e \in N(v)$, keeping S_e. The containment condition still holds, because the supports of the other members of S_e only appear larger, and the numbers d_w do not change.

Theorem 5.2. *Let $\mathcal{G} = (g_v, U_v)_{v\in V}$ be an instance with a bundle structure $(B_e, S_e)_{e\in\mathcal{E}}$, and assume $d_v \geq 1$ for all v. (If $d_v = 0$, then $U_v = \emptyset$ and $g_v \equiv 0$.) Put*

$$k_e = \sum_{v\in S_e} \max(1, 2^{d_v-2}) \qquad (e \in \mathcal{E}).$$

Then there is $x \in \{\pm 1\}^{[m]}$ with $|g_v(x)| \leq \sum_{e\in N(v)} \min(k_e, |B_e|)$ for every v. In particular, if the structure is t-bounded and $d_v \leq \Delta$ for all v, then $\operatorname{disc}(\mathcal{G}) \leq t\Delta \max(1, 2^{\Delta-2})$.

Proof of Theorem 1.6 from Theorem 5.2. Start from the bundles of $\mathcal{G}(\mathcal{F})$. Functions sharing no item with others are handled separately (with $|g_v| \leq \mathcal{K}_1 = 1$). Every other function v lies in some hyperedge of $H(\mathcal{F})$, and we merge its private bundle, if nonempty, into one of these hyperedges. The resulting bundle structure is t-bounded, its sets S_e are the hyperedges of $H(\mathcal{F})$, and d_v is the number of hyperedges containing v. Theorem 5.2 gives $\operatorname{disc}(\mathcal{F}) \leq \max_v \sum_{e\ni v} k_e$, and each k_e is at most $t\max(1, 2^{\Delta-2})$. For $t = 2$ and $\Delta \geq 2$ this is $2\Delta 2^{\Delta-2} = \Delta 2^{\Delta-1}$. For a cycle, $d_v = 2$ for all v, so $k_e = 1 + 1 = 2$ and the bound is $2 + 2 = 4$. □

The rest of this section proves Theorem 5.2. Fix the instance and the bundle structure. Items outside $\bigcup_e B_e$ are irrelevant and are ignored.

5.2 Necklace spheres

For an integer $k \geq 1$ we use the ℓ_1-sphere $S^k = \{w \in \mathbb{R}^{k+1} : \sum_{i=0}^{k} |w_i| = 1\}$, which is homeomorphic to the standard k-sphere, with the antipodal map $w \mapsto -w$. For $w \in S^k$ let $a_i(w) = \sum_{l<i} |w_l|$ for $i = 0, \ldots, k+1$, so that $a_0 = 0$ and $a_{k+1} = 1$. Let $c_w \colon [0,1] \to \{\pm 1\}$ be the step function that equals $\operatorname{sign}(w_i)$ on $[a_i, a_{i+1})$; intervals of length zero play no role. Thus c_w is a ± 1-colouring of $[0,1]$ with at most k sign changes, all located among the points $a_1, \ldots, a_k$. This is the configuration space of the Hobby–Rice theorem [HR65, Pin76] and of consensus halving [SS03].

Lemma 5.3 (necklace map)**.** *Let $B = \{j_1, \ldots, j_M\}$ be an ordered item set and $k \geq 1$. Define $x_B \colon S^k \to [-1,1]^B$ by*

$$x_B(w)_{j_a} = M \int_{(a-1)/M}^{a/M} c_w(s)\, ds \qquad (a = 1, \ldots, M).$$

Then x_B is continuous and odd, and $x_B(w)$ has at most $\min(k, M)$ fractional coordinates for every w.

Proof. Oddness. We have $a_i(-w) = a_i(w)$ and $c_{-w} = -c_w$, so $x_B(-w) = -x_B(w)$.

Continuity. Let $w, w' \in S^k$ and $\eta = \|w - w'\|_1$. Then $|a_i(w) - a_i(w')| \leq \eta$ for all i. A point s with $c_w(s) \neq c_{w'}(s)$ either lies in the symmetric difference of the intervals $[a_i(w), a_{i+1}(w))$ and $[a_i(w'), a_{i+1}(w'))$ for some i, which has measure at most 2η, or lies in both i-th intervals while $\operatorname{sign}(w_i) \neq \operatorname{sign}(w'_i)$. In the latter case $|w_i| + |w'_i| = |w_i - w'_i|$, so the interval has length at most $|w_i - w'_i|$. Summing over i, the set $\{c_w \neq c_{w'}\}$ has measure at most $(2(k+1)+1)\eta$, and therefore $|x_B(w)_j - x_B(w')_j| \leq 2M(2k+3)\eta$ for every j.

Fractional coordinates. If no point $a_i(w)$ with $1 \leq i \leq k$ lies in the open interval $((a-1)/M, a/M)$, then c_w is constant on it except at finitely many points, so $x_B(w)_{j_a} = \pm 1$. Each of the k points $a_1, \ldots, a_k$ lies in at most one of these M disjoint open intervals. □

Order each bundle arbitrarily and let $X = \prod_{e\in\mathcal{E}} S^{k_e}$. For $w = (w_e)_e \in X$ let $x(w) \in [-1,1]^{\bigcup_e B_e}$ be the point whose restriction to B_e is $x_{B_e}(w_e)$. Let G_v be the multilinear extension of g_v and put $\widehat{G}_v(w) = G_v(x(w))$. By Lemmas 2.7 and 5.3, $\widehat{G}_v$ is continuous and depends only on $(w_e)_{e\in N(v)}$, since $U_v \subseteq \bigcup_{e\in N(v)} B_e$.

5.3 The parity decomposition

For $T \subseteq \mathcal{E}$ let $\mathrm{flip}_T \colon X \to X$ negate the coordinates w_e with $e \in T$. Then $\mathrm{flip}_T \mathrm{flip}_{T'} = \mathrm{flip}_{T \triangle T'}$.

Lemma 5.4 (parity decomposition)**.** *For $v \in V$ and $S \subseteq N(v)$ define*

$$\widehat{G}_v^S(w) = 2^{-d_v} \sum_{T \subseteq N(v)} (-1)^{|S \cap T|}\, \widehat{G}_v(\mathrm{flip}_T w).$$

Then:

(a) $\widehat{G}_v = \sum_{S \subseteq N(v)} \widehat{G}_v^S$*;*

(b) $\widehat{G}_v^S(\mathrm{flip}_U w) = (-1)^{|S \cap U|} \widehat{G}_v^S(w)$ *for all* $U \subseteq \mathcal{E}$*;*

(c) $\widehat{G}_v(\mathrm{flip}_{N(v)} w) = -\widehat{G}_v(w)$*, and consequently* $\widehat{G}_v^S \equiv 0$ *whenever* $|S|$ *is even.*

Hence $\widehat{G}_v = \sum_{S \subseteq N(v),\, |S| \text{ odd}} \widehat{G}_v^S$*, a sum of* $2^{d_v - 1}$ *terms.*

Proof. (a) $\sum_S \widehat{G}_v^S(w) = 2^{-d_v} \sum_T \widehat{G}_v(\mathrm{flip}_T w) \sum_{S \subseteq N(v)} (-1)^{|S \cap T|}$, and the inner sum equals 2^{d_v} if $T = \emptyset$ and 0 otherwise.

(b) Since $\widehat{G}_v^S$ depends only on the coordinates in $N(v)$ and $S \subseteq N(v)$, it suffices to take $U \subseteq N(v)$. Substituting $T' = T \triangle U$ gives

$$\widehat{G}_v^S(\mathrm{flip}_U w) = 2^{-d_v} \sum_{T' \subseteq N(v)} (-1)^{|S \cap (T' \triangle U)|} \widehat{G}_v(\mathrm{flip}_{T'} w),$$

and $|S \cap (T' \triangle U)| \equiv |S \cap T'| + |S \cap U| \pmod 2$.

(c) The map $\mathrm{flip}_{N(v)}$ negates $x(w)_j$ for every item j of every bundle in $N(v)$, hence every coordinate in U_v, because each x_{B_e} is odd. Since G_v depends only on these coordinates and is odd (Lemma 2.7), $\widehat{G}_v(\mathrm{flip}_{N(v)} w) = -\widehat{G}_v(w)$. Applying this identity inside the definition gives $\widehat{G}_v^S(\mathrm{flip}_{N(v)} w) = -\widehat{G}_v^S(w)$, because $\mathrm{flip}_{N(v)}$ commutes with every flip_T. By (b), however, $\widehat{G}_v^S(\mathrm{flip}_{N(v)} w) = (-1)^{|S|} \widehat{G}_v^S(w)$. For even $|S|$ the two identities force $\widehat{G}_v^S \equiv 0$. Finally, $N(v) \neq \emptyset$ has $2^{d_v - 1}$ subsets of odd size. □

Lemma 5.5 (equation count)**.** *For every $e \in \mathcal{E}$, the number of pairs (v, S) with $S \subseteq N(v)$, $|S|$ odd and $e \in S$ equals k_e.*

Proof. Such pairs have $v \in S_e$. For fixed $v \in S_e$: if $d_v = 1$, the only choice is $S = \{e\}$. If $d_v \geq 2$, then $S \setminus \{e\}$ ranges over the even-size subsets of $N(v) \setminus \{e\}$, of which there are $2^{d_v - 2}$. In both cases the count is $\max(1, 2^{d_v - 2})$. □

5.4 Tools from equivariant topology

Let $\Gamma = (\mathbb{Z}_2)^{\mathcal{E}}$, whose elements we identify with subsets $U \subseteq \mathcal{E}$ under symmetric difference. Γ acts on X by $U \cdot w = \mathrm{flip}_U w$. This action is *free*: $\mathrm{flip}_U w = w$ forces $w_e = -w_e$ for $e \in U$, which is impossible on a sphere unless $U = \emptyset$. For $S \subseteq \mathcal{E}$ let $\chi_S(U) = (-1)^{|S \cap U|}$, and let $\mathbb{R}_{\chi_S}$ be the real line on which U acts by multiplication by $\chi_S(U)$. All cohomology has coefficients in $\mathbb{F}_2$.

The Fadell–Husseini index. For a Γ-space Y, let $H^*_\Gamma(Y) = H^*(E\Gamma \times_\Gamma Y)$ be its Borel equivariant cohomology. The *Fadell–Husseini index* [FH88] is the kernel of the map $H^*(B\Gamma) \to H^*_\Gamma(Y)$ induced by $Y \to \mathrm{pt}$:

$$\mathrm{Index}_\Gamma(Y) = \ker\big(H^*(B\Gamma) \to H^*_\Gamma(Y)\big) \;\subseteq\; H^*(B\Gamma) = \mathbb{F}_2[t_e : e \in \mathcal{E}].$$

Here $B\Gamma = (\mathbb{R}\mathrm{P}^\infty)^{\mathcal{E}}$, the ring structure follows from the Künneth theorem, and t_e (of degree 1) is the pullback of the generator of $H^1(\mathbb{R}\mathrm{P}^\infty)$ under the e-th projection. Under $H^1(B\Gamma) \cong \mathrm{Hom}(\Gamma, \mathbb{Z}_2)$, the class $\sum_{e\in S} t_e$ corresponds to the character χ_S. We use the following standard facts.

Fact 5.6 (monotonicity [FH88])**.** *If there is a Γ-equivariant map $Y \to Y'$, then* $\mathrm{Index}_\Gamma(Y') \subseteq \mathrm{Index}_\Gamma(Y)$.

Reason. The map $Y \to \mathrm{pt}$ factors as $Y \to Y' \to \mathrm{pt}$, so $H^*(B\Gamma) \to H^*_\Gamma(Y)$ factors through $H^*_\Gamma(Y')$.

Fact 5.7 (products of spheres)**.** *For $X = \prod_{e\in\mathcal{E}} S^{k_e}$ with the factorwise antipodal action,* $\mathrm{Index}_\Gamma(X) = (t_e^{k_e+1} : e \in \mathcal{E})$.

Reason. The action is free and X is a compact manifold, so $H^*_\Gamma(X) \cong H^*(X/\Gamma)$, and $X/\Gamma = \prod_e \mathbb{R}\mathrm{P}^{k_e}$. The principal Γ-bundle $X \to X/\Gamma$ is the product of the double covers $S^{k_e} \to \mathbb{R}\mathrm{P}^{k_e}$, which are classified by the inclusions $\mathbb{R}\mathrm{P}^{k_e} \subseteq \mathbb{R}\mathrm{P}^\infty$. Since $H^*(\mathbb{R}\mathrm{P}^k) = \mathbb{F}_2[t]/(t^{k+1})$ [Hat02], the Künneth theorem identifies $H^*(B\Gamma) \to H^*(X/\Gamma)$ with the quotient map $\mathbb{F}_2[t_e : e \in \mathcal{E}] \to \bigotimes_e \mathbb{F}_2[t_e]/(t_e^{k_e+1})$. Its kernel is the ideal generated by the $t_e^{k_e+1}$. For one sphere this is the basic example of [FH88]; computations of this kind for products of spheres are standard in the mass-partition literature (cf. [MLVŽ06, BZ17]).

Fact 5.8 (Euler class of a sum of characters)**.** *Let $\mathbb{V} = \bigoplus_{i=1}^N \mathbb{R}_{\chi_{S_i}}$ with nonempty $S_i \subseteq \mathcal{E}$, and let $S(\mathbb{V})$ be its unit sphere. Then*

$$e(\mathbb{V}) \;:=\; \prod_{i=1}^{N} \Big(\sum_{e\in S_i} t_e\Big) \;\in\; \mathrm{Index}_\Gamma(S(\mathbb{V})).$$

Reason. The vector bundle $\xi = E\Gamma \times_\Gamma \mathbb{V} \to B\Gamma$ is the Whitney sum of the line bundles $L_i = E\Gamma \times_\Gamma \mathbb{R}_{\chi_{S_i}}$. The first Stiefel–Whitney class of the line bundle of a character χ is the class in $H^1(B\Gamma) \cong \mathrm{Hom}(\Gamma, \mathbb{Z}_2)$ corresponding to χ, so $w_1(L_i) = \sum_{e\in S_i} t_e$. By the Whitney product formula [MS74], the top Stiefel–Whitney class (the mod-2 Euler class) is $w_N(\xi) = \prod_i w_1(L_i) = e(\mathbb{V})$. The Gysin sequence of the sphere bundle $\pi\colon E\Gamma \times_\Gamma S(\mathbb{V}) \to B\Gamma$ [MS74] contains the segment $H^0(B\Gamma) \xrightarrow{\cup w_N(\xi)} H^N(B\Gamma) \xrightarrow{\pi^*} H^N_\Gamma(S(\mathbb{V}))$, which is exact, so $\pi^*(e(\mathbb{V})) = 0$; that is, $e(\mathbb{V}) \in \mathrm{Index}_\Gamma(S(\mathbb{V}))$. (See [FH88, BZ17] for this use of Euler classes.)

Fact 5.9 (monomial ideals)**.** *Let $I \subseteq \mathbb{F}_2[t_e : e \in \mathcal{E}]$ be an ideal generated by monomials. A polynomial lies in I if and only if every monomial occurring in it with nonzero coefficient is divisible by one of the generators.*

Reason. As an $\mathbb{F}_2$-vector space, I is spanned by the monomials divisible by some generator, and distinct monomials are linearly independent.

5.5 Proof of Theorem 5.2

Consider the map

$$\Phi\colon X \to \mathbb{V} = \bigoplus_{v\in V} \bigoplus_{S\subseteq N(v),\ |S| \text{ odd}} \mathbb{R}_{\chi_S}, \qquad \Phi(w) = \big(\widehat{G}_v^S(w)\big)_{v,S}.$$

By Lemma 5.4(b), $\Phi(\mathrm{flip}_U w) = U \cdot \Phi(w)$, so Φ is Γ-equivariant. It is continuous because each $\widehat{G}_v$ is.

Claim: Φ has a zero. Suppose not. Then $w \mapsto \Phi(w)/\|\Phi(w)\|$ is a Γ-equivariant map $X \to S(\mathbb{V})$, since Γ acts on $\mathbb{V}$ by coordinatewise sign changes, which preserve the Euclidean norm. By Facts 5.8 and 5.6, $e(\mathbb{V}) \in \mathrm{Index}_\Gamma(S(\mathbb{V})) \subseteq \mathrm{Index}_\Gamma(X)$, and by Fact 5.7, $\mathrm{Index}_\Gamma(X) = (t_e^{k_e+1} : e \in \mathcal{E})$. Now

$$e(\mathbb{V}) = \prod_v \prod_{S \subseteq N(v),\ |S| \text{ odd}} \Big(\sum_{e \in S} t_e\Big).$$

This is a product of nonzero linear forms in the integral domain $\mathbb{F}_2[t_e]$, so $e(\mathbb{V}) \neq 0$. Expanding the product, every monomial is obtained by choosing one variable t_e, $e \in S$, from each factor (v, S). The exponent of t_e in such a monomial is at most the number of factors containing t_e, which is k_e by Lemma 5.5. Hence every monomial occurring in $e(\mathbb{V})$ with nonzero coefficient (there is at least one) has t_e-exponent at most k_e for all e, and is not divisible by any $t_e^{k_e+1}$. By Fact 5.9, $e(\mathbb{V}) \notin (t_e^{k_e+1})$, a contradiction.

Rounding. Let $\Phi(w^*) = 0$ and $x^* = x(w^*)$. By Lemma 5.4, $G_v(x^*) = \widehat{G}_v(w^*) = \sum_{|S| \text{ odd}} \widehat{G}_v^S(w^*) = 0$ for every v. By Lemma 5.3, x^* has at most $\min(k_e, |B_e|)$ fractional coordinates in each bundle B_e. Hence at most $\sum_{e \in N(v)} \min(k_e, |B_e|)$ of them lie in U_v, since $U_v \subseteq \bigcup_{e \in N(v)} B_e$. Rounding every fractional coordinate to a nearest sign gives $x' \in \{\pm 1\}^{[m]}$ with $|g_v(x')| \leq \sum_{e \in N(v)} \min(k_e, |B_e|)$ for all v, by Lemma 2.7(d). The final bound follows from $k_e \leq t \max(1, 2^{\Delta-2})$ and $d_v \leq \Delta$. □

5.6 Cycles, and why the diagonal symmetry is not enough

Example 5.10 (cycles)**.** Let the bundles $e_0, \ldots, e_{n-1}$ form a cycle with $S_{e_i} = \{v_{i-1}, v_i\}$ (indices mod n), so that v_i reads $a = e_i$ and $b = e_{i+1}$. The two odd parity pieces of $\widehat{G}_{v_i}$ are $E_i = \widehat{G}_{v_i}^{\{a\}}$ and $O_i = \widehat{G}_{v_i}^{\{b\}}$; explicitly, $E_i(a,b) = \frac{1}{2}(\widehat{G}_{v_i}(a,b) + \widehat{G}_{v_i}(a,-b))$ and $O_i(a,b) = \frac{1}{2}(\widehat{G}_{v_i}(a,b) - \widehat{G}_{v_i}(a,-b))$. Bundle e_i receives exactly two equations, $E_i = 0$ (odd in e_i) from v_i and $O_{i-1} = 0$ (odd in e_i) from v_{i-1}. The Euler class is $\prod_i t_{e_i}^2$, $k_e = 2$, and $X = (S^2)^n$. The proof of Theorem 5.2 uses only the joint oddness and the locality of the maps $\widehat{G}_{v_i}$. It therefore shows that any continuous $F_i \colon S^2 \times S^2 \to \mathbb{R}$ satisfying $F_i(-a,-b) = -F_i(a,b)$ have a common zero of the functions $(w_0, \ldots, w_{n-1}) \mapsto F_i(w_i, w_{i+1})$ on $(S^2)^n$. We call this a *local Borsuk–Ulam theorem* for cycles. Proposition 6.1 shows that it fails for S^1.

Remark 5.11 (the diagonal action)**.** Each $\widehat{G}_v$ is equivariant for the diagonal subgroup $\{\emptyset, \mathcal{E}\} \cong \mathbb{Z}_2$ of Γ. The diagonal symmetry alone cannot force common zeros of more than $\min_e k_e$ functions. Indeed, if $n > k_{e_0} = \min_e k_e$, then $w \mapsto (w_{e_0}, 0, \ldots, 0) \in \mathbb{R}^{k_{e_0}+1} \subseteq \mathbb{R}^n$ is a continuous map, odd under the diagonal action, without zeros. (Its components are not of the local form $\widehat{G}_v$, so this does not contradict Theorem 5.2.) Equivalently, the Fadell–Husseini index of X for the diagonal $\mathbb{Z}_2$ is $(t^{\min_e k_e+1})$, while the n equations have Euler class t^n. The locality of the $\widehat{G}_v$, which the parity decomposition converts into a larger symmetry group, is what makes the argument work.

6 One cut per bundle is not enough

With $k_e = 1$ for every bundle of a cycle, the configuration space is a product of circles. The corresponding topological statement is false.

Proposition 6.1. *For every $n \geq 2$ there are continuous functions $f_i \colon S^1 \times S^1 \to \mathbb{R}$ ($i \in \mathbb{Z}_n$), each odd under the joint antipodal map $(\theta, \theta') \mapsto (\theta + \pi, \theta' + \pi)$, such that the system $f_i(\theta_i, \theta_{i+1}) = 0$ ($i \in \mathbb{Z}_n$) has no solution $(\theta_0, \ldots, \theta_{n-1}) \in (S^1)^n$.*

Proof. We use functions $f_i(\theta, \theta') = \sin(p_i\theta' - q_i\theta - c_i)$ with positive integers p_i, q_i such that $p_i - q_i$ is odd. Shifting both arguments by π adds $(p_i - q_i)\pi$ to the argument of the sine and therefore negates f_i. Since p_i, q_i are integers, f_i is well defined on $S^1 \times S^1$. Moreover, $f_i(\theta_i, \theta_{i+1}) = 0$ holds if and only if $p_i\theta_{i+1} = q_i\theta_i + c_i + a_i\pi$ for some integer a_i, where angles are represented by real numbers.

Even n. Take $(p_i, q_i) = (1, 2)$ for even i and $(2, 1)$ for odd i. Suppose there is a solution, and fix real representatives $\theta_0, \dots, \theta_{n-1}$ and $\theta_n = \theta_0 + 2\pi b$ for some integer b. From $\theta_1 = 2\theta_0 + c_0 + a_0\pi$ and $2\theta_2 = \theta_1 + c_1 + a_1\pi$ we get $\theta_2 = \theta_0 + \frac{c_0+c_1}{2} + \frac{(a_0+a_1)\pi}{2}$. Iterating over consecutive pairs, $\theta_n = \theta_0 + \frac{1}{2}\sum_i c_i + \frac{\pi}{2}\sum_i a_i$. Hence $\frac{1}{2}\sum_i c_i \in \frac{\pi}{2}\mathbb{Z}$. Choosing $\sum_i c_i = \pi/2$ makes this impossible.

Odd $n = 3 + 2r$. Take $(p_0, q_0) = (1, 4)$, $(p_1, q_1) = (p_2, q_2) = (2, 1)$, followed by r pairs $(1, 2), (2, 1)$; all differences $p_i - q_i$ are odd. As above, $\theta_1 = 4\theta_0 + c_0 + a_0\pi$, $\theta_2 = 2\theta_0 + \frac{c_0+c_1}{2} + \frac{(a_0+a_1)\pi}{2}$, and $\theta_3 = \theta_0 + \frac{c_0+c_1+2c_2}{4} + \frac{(a_0+a_1+2a_2)\pi}{4}$. Each subsequent pair $(1, 2), (2, 1)$ adds $\frac{c+c'}{2} + \frac{(a+a')\pi}{2}$, as in the even case. Hence $\theta_n - \theta_0 \in \gamma + \frac{\pi}{4}\mathbb{Z}$, where $\gamma = \frac{c_0+c_1+2c_2}{4} + \frac{1}{2}\sum_{i\geq 3} c_i$, and $\theta_n - \theta_0 \in 2\pi\mathbb{Z}$. Choosing the c_i with $\gamma = \pi/8$ makes this impossible. □

Interpretation. Proposition 6.1 concerns the topological core of the method, not discrepancy. It shows that no argument using only the joint oddness and locality of the maps $\widehat{G}_{v_i}(w_i, w_{i+1})$ can work with one cut per bundle. This is consistent with Section 5: for $k_e = 1$ the Euler class $\prod_e t_e^2$ lies in the index ideal (t_e^2), so the obstruction vanishes. With two cuts per bundle (Example 5.10) the statement is true. Whether the realizable maps $\widehat{G}_v$ (multilinear extensions composed with necklace maps) always have common zeros with one cut per bundle is a different question, which we do not address.

7 Combining the two mechanisms: bounded block-degree

Throughout this section $\mathcal{G} = (g_v, U_v)_{v\in V}$ is a t-sparse instance with bundles B_S as in Definition 2.3 and bundle hypergraph $H = H(\mathcal{G})$. As before, d_v is the number of hyperedges of H containing v, and v is *heavy* if $d_v > \Delta$.

Definition 7.1 (lobes and ports)**.** Let $v \in V$. Two functions $u, u' \in V \setminus \{v\}$ are *linked avoiding* v if there are $u = u_0, u_1, \dots, u_r = u'$ in $V \setminus \{v\}$ such that u_{i-1} and u_i lie in a common hyperedge of H for each i. A *lobe* of v is an equivalence class of this relation that contains a function sharing a hyperedge with v. The *ports* of a lobe L are the hyperedges $P_L = \{S \in E(H) : v \in S,\ S \setminus \{v\} \subseteq L\}$. The *port items* of L are $B(P_L) = \bigcup_{S\in P_L} B_S$, and the *inner items* of L are $I(L) = \bigcup\{B_S : \emptyset \neq S \subseteq L\}$.

Lemma 7.2. *Let $v \in V$, and let W be the set of functions that are neither v nor in a lobe of v.*

(a) Every hyperedge $S \ni v$ is a port of exactly one lobe of v.

(b) If $v \notin S$ and S meets a lobe L, then $S \subseteq L$. Consequently every $u \in L$ reads only items in $I(L) \cup B(P_L)$; the sets $I(L) \cup B(P_L)$ of distinct lobes are pairwise disjoint; and g_v reads only items in $B_{\{v\}} \cup \bigcup_L B(P_L)$.

(c) If S meets W, then $S \subseteq W$.

Proof. (a) Since $|S| \geq 2$, the set $S \setminus \{v\}$ is nonempty, its elements are pairwise linked avoiding v (through S), and they share the hyperedge S with v. Hence $S \setminus \{v\}$ lies in exactly one lobe. (b) The elements of S are pairwise linked avoiding v, so they all lie in the class of an element of L, namely L. A bundle B_S read by $u \in L$ has $u \in S$. If $v \in S$, then $S \setminus \{v\}$ contains u, so by (a) S is a port of L. If $v \notin S$, then $S \subseteq L$ and $B_S \subseteq I(L)$. Inner items of distinct lobes are disjoint by the first claim, port items of distinct lobes are disjoint by (a), and inner items are never port items, since

$v \notin S$ for the former and $v \in S$ for the latter. Finally, g_v reads only bundles B_S with $v \in S$; such an S is either $\{v\}$ or a hyperedge, and hence a port. (c) If $v \in S$, then S is a hyperedge or $\{v\}$; a hyperedge would be a port by (a), so $S \setminus \{v\}$ would lie in a lobe, contradicting $S \cap W \neq \emptyset$. Hence $v \notin S$, the elements of S are pairwise linked avoiding v, and since one of them is in no lobe, none of them is. □

Definition 7.3 (closing instance)**.** Let L be a lobe of v and let ψ be an odd 2-Lipschitz function on $\{\pm 1\}^{B(P_L)}$. The *closing instance* $\mathcal{G}_L^\psi$ is the instance on the item set $I(L) \cup B(P_L)$ consisting of the functions (g_u, U_u) for $u \in L$ (which read only these items, by Lemma 7.2(b)) and one new *closing function* c_L with $g_{c_L} = \psi$ and support $B(P_L)$.

Lemma 7.4. *Let L be a lobe of v and ψ as above, and let $H' = H(\mathcal{G}_L^\psi)$.*

(a) $\mathcal{G}_L^\psi$ is t-sparse. The hyperedges of H' are the hyperedges $S \subseteq L$ of H, with the same bundles, and the sets $S^c = (S \setminus \{v\}) \cup \{c_L\}$ for $S \in P_L$, with bundle B_S.

(b) $d_{c_L} = |P_L|$ in H', and d_u is the same in H' as in H for every $u \in L$.

(c) For $u \in L$, every lobe of u in H' has at most as many ports as some lobe of u in H.

Proof. (a) An item of B_S with $S \subseteq L$ lies in exactly the supports of the members of S. An item of B_S with $S \in P_L$ lies in the supports of the members of $S \setminus \{v\}$ and of c_L. Moreover $|S^c| = |S| \leq t$. (b) By Lemma 7.2(b), the hyperedges of H at $u \in L$ are the hyperedges $S \subseteq L$ containing u and the ports containing u. They correspond bijectively to the hyperedges of H' at u. The hyperedges at c_L are the S^c, $S \in P_L$. (c) Let $\iota \colon L \cup \{c_L\} \to L \cup \{v\}$ be the identity on L with $\iota(c_L) = v$. By (a), ι maps the hyperedges of H' injectively to hyperedges of H. Hence functions linked avoiding u in H' are mapped to functions linked avoiding u in H. Let K be a lobe of u in H'. Then $\iota(K)$ lies in one class $\tilde{K}$ of the relation "linked avoiding u" in H. This class is a lobe of u in H, because K contains a function sharing a hyperedge S' of H' with u, and $\iota(S')$ is a hyperedge of H containing u. If S' is a port of K, then $\iota(S') \ni u$ and $\iota(S') \setminus \{u\} = \iota(S' \setminus \{u\}) \subseteq \tilde{K}$, so $\iota(S')$ is a port of $\tilde{K}$. Since ι is injective on hyperedges, K has at most as many ports as $\tilde{K}$. □

Lemma 7.5 (closing-function lemma)**.** *Let $C \geq 0$, and let L be a lobe of v such that $\operatorname{disc}(\mathcal{G}_L^\psi) \leq C$ for every odd 2-Lipschitz ψ on $\{\pm 1\}^{B(P_L)}$. Then the set*

$$T_L = \{y \in \{\pm 1\}^{B(P_L)} : \exists z \in \{\pm 1\}^{I(L)} \text{ such that } |g_u(z, y)| \leq C \text{ for all } u \in L\}$$

is symmetric and contains a C-loop.

Proof. T_L is symmetric because all g_u are odd. For every ψ, a colouring (z, y) of $\mathcal{G}_L^\psi$ with all values at most C in absolute value shows that $y \in T_L$ and $|\psi(y)| \leq C$. Taking $\psi \equiv 0$ shows $T_L \neq \emptyset$. If T_L contained no C-loop, Lemma 2.11(b) would give an odd 2-Lipschitz ψ with $|\psi| > C$ on T_L, a contradiction. (Note that $B(P_L) \neq \emptyset$, because a lobe has at least one port.) □

Theorem 7.6. *Let $\Delta \geq 2$ and $C = t\Delta \max(1, 2^{\Delta-2})$. Let $\mathcal{G}$ be a t-sparse instance such that*

$$\text{every heavy function has at most } \Delta \text{ ports in each of its lobes.} \qquad (\star)$$

Then $\operatorname{disc}(\mathcal{G}) \leq C$.

Proof. We use induction on the number $h(\mathcal{G})$ of heavy functions, over all t-sparse instances satisfying $(\star)$.

Base case $h(\mathcal{G}) = 0$. Functions lying in no hyperedge read only private items. They form separate instances on disjoint items and can be given $|g_v| \le \mathcal{K}_1 = 1 \le C$. For the remaining functions, merge private bundles as in the proof of Theorem 1.6. This gives a t-bounded bundle structure with $d_v \le \Delta$, and Theorem 5.2 gives $|g_v| \le \sum_{e \in N(v)} k_e \le \Delta \cdot t \max(1, 2^{\Delta-2}) = C$.

Induction step. Let $h(\mathcal{G}) \ge 1$ and choose a heavy function v. Let $L_1, \dots, L_p$ be its lobes and W the set of remaining functions other than v.

The lobes. Fix i and an odd 2-Lipschitz ψ on $\{\pm 1\}^{B(P_{L_i})}$. By Lemma 7.4, $\mathcal{G}^{\psi}_{L_i}$ is t-sparse. Its closing function is light, because $d_{c_{L_i}} = |P_{L_i}| \le \Delta$ by $(\star)$ applied to v. Its heavy functions are exactly the heavy functions of $\mathcal{G}$ in L_i, which are fewer than $h(\mathcal{G})$ since $v \notin L_i$. It satisfies $(\star)$ by Lemma 7.4(c), because a heavy $u \in L_i$ is heavy in $\mathcal{G}$ and satisfies $(\star)$ there. By induction, $\mathrm{disc}(\mathcal{G}^{\psi}_{L_i}) \le C$ for every ψ. By Lemma 7.5, T_{L_i} contains a C-loop Γ_i.

The rest. By Lemma 7.2(c), the functions in W read only items of bundles B_S with $S \subseteq W$, and these items are read by no other function. The instance $\mathcal{G}_W = (g_w, U_w)_{w \in W}$ on these items has as hyperedges the hyperedges of H contained in W. Every hyperedge meeting the class of $w \in W$ lies in W, so the lobes, ports and degrees of every $w \in W$ are the same in $\mathcal{G}_W$ as in $\mathcal{G}$. Hence $\mathcal{G}_W$ satisfies $(\star)$, and $h(\mathcal{G}_W) < h(\mathcal{G})$ since $v \notin W$. By induction, the items of $\mathcal{G}_W$ can be coloured with $|g_w| \le C$ for all $w \in W$.

The heavy function v. Let $X = \prod_{i=1}^{p} (\Gamma_i \cup (-\Gamma_i)) \times \{\pm 1\}^{B_{\{v\}}}$, omitting the last factor if v has no private items. Each factor is symmetric and C-connected (for the last factor, $1 \le C$). Hence X is symmetric and C-connected and contains an antipodal C-path. By Lemma 7.2(b), g_v reads only the coordinates of X, so Lemma 2.10 gives $z^* \in X$ with $|g_v(z^*)| \le C$. Since $z^*_i \in \Gamma_i \cup (-\Gamma_i) \subseteq T_{L_i}$, the inner items $I(L_i)$ can be coloured so that $|g_u| \le C$ for all $u \in L_i$.

Assembly. Every item lies in some bundle B_S. If $v \in S$, the item is a port item or a private item of v. If $v \notin S$ and S meets a lobe, it is an inner item of that lobe. Otherwise $S \subseteq W$. By Lemma 7.2, these item sets are pairwise disjoint, and every function reads only items from its own part (together with port items, for the functions of a lobe). The colourings constructed above therefore combine into one colouring with $|g_w| \le C$ for every function w. □

For $t = 2$ we translate $(\star)$ into graph language. Recall that the blocks of a graph (its maximal connected subgraphs without a cut vertex: maximal 2-connected subgraphs, bridges and isolated vertices) partition its edge set, and that two distinct edges lie in the same block if and only if they lie on a common cycle.

Lemma 7.7. *Let H be a simple graph and v a vertex. The map sending a lobe L of v to the block containing the edges from v to L is a bijection between the lobes of v and the blocks containing an edge at v. The ports of L are exactly the edges at v of the corresponding block.*

Proof. The hyperedges at v are the edges va, and each lobe contains a neighbour of v. For distinct neighbours a, b of v, the functions a and b are linked avoiding v if and only if $H - v$ contains an a–b path. If such a path P exists, then $va \cup P \cup bv$ is a cycle, so va and vb lie in the same block. Conversely, if va and vb lie in a common block, they lie on a common cycle, and removing v from it leaves an a–b path in $H - v$. Hence two edges at v go to the same lobe if and only if they lie in the same block. □

Proof of Theorem 1.8. The first statement is Theorem 7.6 applied to $\mathcal{G}(\mathcal{F})$. For $t = 2$ and $\Delta \ge 2$ we have $C = 2\Delta 2^{\Delta-2} = \Delta 2^{\Delta-1}$. If every block of $H(\mathcal{F})$ has maximum degree at most Δ, then by

Lemma 7.7 every function has at most Δ ports in each lobe, so ($\star$) holds. For a cactus every block is an edge or a cycle, hence has maximum degree at most 2, and the bound is $2 \cdot 2^1 = 4$. □

Corollary 7.8. *Let $t = 2$ and $\Delta \geq 2$. If no function with more than Δ distinct neighbours lies on a cycle of the simple graph underlying the item multigraph, then* $\operatorname{disc}(\mathcal{F}) \leq \Delta 2^{\Delta-1}$.

Proof. A vertex on no cycle has degree 1 in every block containing it: a block with at least two edges is 2-connected, and every vertex of a 2-connected graph lies on a cycle. Hence every heavy function has exactly one port in each lobe (Lemma 7.7), so ($\star$) holds, and Theorem 7.6 applies. □

Remark 7.9. Theorem 7.6 contains Theorem 1.6 (the case without heavy functions, up to the finer bound $\max_v \sum_{e \ni v} k_e$). It also contains Theorem 1.3 with the weaker constant $2t$ instead of $\mathcal{K}_t$: take $\Delta = 2$, and note that in a Berge-acyclic pattern every lobe of a function has exactly one port, since two ports of the same lobe would close a cycle in the incidence graph. The reduction makes no progress at a function of unbounded degree inside a block, such as the hub of a wheel. There the function has a single lobe whose ports are all of its edges, so the closing function is as heavy as the function it replaces. Meanwhile the parity decomposition costs $2^{d_v - 1}$ equations at a function of degree d_v. For $t = 2$, Section 8 removes this obstacle for patterns of bounded degeneracy.

8 Bounded degeneracy for $t = 2$: hub transfer and near-loops

The reduction of Section 7 makes no progress at a function of large degree inside a block (Remark 7.9). In this section we show that for $t = 2$ large degrees are harmless as long as the graph is sparse in the sense of degeneracy. A graph is *d-degenerate* if its vertices can be ordered $v_1, \dots, v_n$ so that every v_j has at most d neighbours v_k with $k > j$ (its *later neighbours*); the *degeneracy* $\operatorname{dgn}(G)$ is the least such d. Throughout, $\mathcal{G} = (g_v, U_v)_{v \in V}$ is a 2-sparse instance, $H = H(\mathcal{G})$ is the simple graph underlying its item multigraph, and $B_{uv} = B_{\{u,v\}}$ for an edge uv of H.

Roadmap. Section 8.3 introduces the coned 2-skeleton of the cube; the next subsection proves the cube statements behind the hub-transfer lemma (Lemmas 8.12 and 8.13) and the lemma itself (Lemma 8.15, Corollary 8.16), which gives Theorem 8.1. The leveled deletions of Section 8.6 handle constraints at level 1. Section 8.7 proves the main result: the near-dichotomy and near-cover lemmas (Lemmas 8.24 and 8.25) give a transfer without loss (Corollary 8.27), hence cheap deletions (Lemma 8.28) and schedules (Lemma 8.30), and from these Theorems 8.29, 8.31 and 8.32, Corollary 8.33 and Theorem 1.9.

Theorem 8.1. *Let $\mathcal{G}$ be a 2-sparse instance such that the vertices of $H(\mathcal{G})$ can be ordered so that every vertex has at most d later neighbours. Then* $\operatorname{disc}(\mathcal{G}) \leq d + 1$.

The bound does not depend on the vertex degrees or the bundle sizes. The proof deletes the vertices from last to first. When a vertex h is deleted, every earlier neighbour r of h receives a new *local* constraint on its remaining items, which guarantees that whatever colouring is later found for these items, the bundle B_{hr} can still move along a loop; h then sees a product of loops and is satisfied by the crossing lemma. The new constraint must be of the same kind as the old one, one level lower, and this is the content of the hub-transfer lemma (Lemma 8.15). We prove it on the 2-skeleton of the cube, coned off at the barycentres of its squares, into which a two-cut necklace sphere maps. Section 8.7 then shows that a weaker demand costs no level at all, which gives Theorem 1.9.

8.1 Coarse sets and constraint systems

For $U = \emptyset$, the cube $\{\pm 1\}^{\emptyset}$ has a single element, which is its own antipode; the one-term sequence is an antipodal s-path for every $s \geq 0$, and the only odd function on $\{\pm 1\}^{\emptyset}$ is 0.

Definition 8.2. Let U be finite and $c \geq 1$. A set $Z \subseteq \{\pm 1\}^U$ is *c-coarse* if it is symmetric and $Z \supseteq \{z : |\varphi(z)| \leq c\}$ for some odd 2-Lipschitz φ on $\{\pm 1\}^U$.

A c-coarse set is c'-coarse for every $1 \leq c' \leq c$. By Lemma 2.11(a), every c-separating set is c-coarse.

Lemma 8.3. *Let U be finite, $c \geq 1$, and $Z \subseteq \{\pm 1\}^U$ symmetric.*

(a) If Z is c-coarse, then Z meets every antipodal s-path in $\{\pm 1\}^U$ with $s \leq c$.

(b) If Z meets every antipodal c-path in $\{\pm 1\}^U$, then Z is c-coarse.

Proof. (a) This is Lemma 2.10 (trivial for $U = \emptyset$). (b) If $Z = \{\pm 1\}^U$, take $\varphi \equiv 0$. Otherwise $U \neq \emptyset$, and $Y = \{\pm 1\}^U \setminus Z$ is nonempty, symmetric, and contains no c-loop, since such a loop would avoid Z. By Lemma 2.11(b) there is an odd 2-Lipschitz ψ with $|\psi| > c$ on Y, so $\{|\psi| \leq c\} \subseteq Z$. □

Definition 8.4 (constraint systems)**.** A *2-sparse constraint system* on the item set $[m]$ consists of supports $(U_v)_{v \in V}$, every item lying in at most two of them, and symmetric sets $Z_v \subseteq \{\pm 1\}^{U_v}$. A *solution* is an $x \in \{\pm 1\}^{[m]}$ with $x|_{U_v} \in Z_v$ for all v. Bundles and the graph H are defined from the supports as in Definition 2.3. A simple graph G with vertex set V is *c-good* if every 2-sparse constraint system with vertex set V, with H a subgraph of G, and with c-coarse constraints Z_v, has a solution.

Lemma 8.5. *A simple graph G with vertex set V is c-good if and only if every 2-sparse instance $(g_v, U_v)_{v \in V}$ whose graph H is a subgraph of G has discrepancy at most c. A c-good graph is c'-good for every $c' \geq c$.*

Proof. If G is c-good and (g_v, U_v) is such an instance, the sets $Z_v = \{|g_v| \leq c\}$ are c-coarse, and a solution is a colouring with all $|g_v| \leq c$. Conversely, given a system with $Z_v \supseteq \{|\varphi_v| \leq c\}$, the instance $(\varphi_v, U_v)_{v \in V}$ has the same supports, and a colouring x with all $|\varphi_v(x)| \leq c$ is a solution. The last claim holds because c'-coarse sets are c-coarse. □

Thus Theorem 8.1 says that every graph with an ordering in which each vertex has at most d later neighbours is $(d+1)$-good.

8.2 Gluing along a cut vertex

Lemma 8.6 (1-sums)**.** *Let $c \geq 1$, and let $G = G_1 \cup G_2$ be simple graphs with $V(G_1) \cap V(G_2) = \{h\}$ and $E(G_1) \cap E(G_2) = \emptyset$. If G_1 and G_2 are c-good, then G is c-good.*

Proof. Take a constraint system on G as in Definition 8.4, and let $V_j = V(G_j) \setminus \{h\}$. Let E_j be the union of the bundles B_{hu} with $u \in V_j$, let I_j be the set of items all of whose supports lie in V_j, and let $R = B_{\{h\}}$. Every relevant item lies in exactly one of E_1, E_2, I_1, I_2, R; the functions of V_j read only items of $I_j \cup E_j$, and $U_h = E_1 \cup E_2 \cup R$. Let $\Phi_j \subseteq \{\pm 1\}^{E_j}$ be the set of y such that some $z \in \{\pm 1\}^{I_j}$ satisfies $(z, y)|_{U_u} \in Z_u$ for all $u \in V_j$; it is symmetric.

We claim that Φ_j contains a c-loop. For every odd 2-Lipschitz ψ on $\{\pm 1\}^{E_j}$, the constraint system on the items $I_j \cup E_j$ with vertex set $V(G_j)$, in which $u \in V_j$ keeps (U_u, Z_u) and h has support E_j and constraint $\{|\psi| \leq c\}$, is 2-sparse, its graph is a subgraph of G_j, and its constraints

are c-coarse. Since G_j is c-good, it has a solution, so Φ_j meets $\{|\psi| \le c\}$. With $\psi \equiv 0$ this shows $\Phi_j \neq \emptyset$, which proves the claim if $E_j = \emptyset$. Otherwise, if Φ_j contained no c-loop, Lemma 2.11(b) would give an odd 2-Lipschitz ψ with $|\psi| > c$ on Φ_j, a contradiction.

Let λ_j be a c-loop in Φ_j. Moving along λ_1 (with the coordinates in $E_2 \cup R$ fixed), then along λ_2, and then flipping the items of R one at a time gives an antipodal c-path in $\Phi_1 \times \Phi_2 \times \{\pm 1\}^R$. By Lemma 8.3(a), Z_h contains a point (y_1, y_2, r) of this path. Choosing witnesses $z_j \in \{\pm 1\}^{I_j}$ for $y_j \in \Phi_j$, the colouring (z_1, z_2, y_1, y_2, r) is a solution. □

A graph is c-good if and only if each of its connected components is (components share no items, so their constraint systems are independent; a single vertex is c-good for $c \ge 1$, because a c-coarse set meets the antipodal 1-path that flips the coordinates one at a time, Lemma 8.3(a)), and a connected graph with at least two blocks is the union of a leaf block of its block–cut tree and the rest, which share only a cut vertex. By induction on the number of blocks we obtain:

Corollary 8.7. *If every block B of a simple graph G is c-good (as a graph on $V(B)$), then G is c-good. In particular, for every 2-sparse instance, $\operatorname{disc}(\mathcal{G}) \le \max_B c_B$, where c_B is any number such that the block B of $H(\mathcal{G})$ is c_B-good.*

8.3 The coned 2-skeleton of the cube

Let S be a finite set with $|S| \ge 3$, and regard $\{\pm 1\}^S$ as the vertex set of the cube $[-1,1]^S$; we call its elements *corners*. A *square* is a 2-dimensional face q of the cube. Its corner set $V(q)$ is a 4-cycle of the Hamming graph, and any two corners at Hamming distance 2 lie in exactly one common square. Let c_q be the barycentre of q. Let $K = K_S$ be the simplicial complex whose vertices are the corners and the barycentres, whose edges are the cube edges $[z, z']$ ($\operatorname{dist}(z, z') = 1$) and the segments $[c_q, z]$ ($z \in V(q)$), and whose triangles are the $[c_q, z, z']$ with $z, z' \in V(q)$ adjacent. Each square is the union of its four triangles, so $|K|$ is the union $X^{(2)}$ of the closed faces of $[-1,1]^S$ of dimension at most 2. No two barycentres are adjacent. The map $x \mapsto -x$ is a simplicial automorphism of K ($z \mapsto -z$, $c_q \mapsto c_{-q}$); we write $\bar{v} = -v$ for vertices and $\bar{A} = -A$ for vertex sets.

(K1) *Freeness.* Every point of $|K|$ has at least $|S| - 2 \ge 1$ coordinates equal to ± 1, so $x \mapsto -x$ has no fixed point on $|K|$; in particular $\bar{v} \neq v$ for every vertex.

(K2) *Connectivity.* The 1-skeleton of K is connected.

(K3) *Simple connectivity.* $X^{(2)}$ is the 2-skeleton of the CW structure of $[-1,1]^S$ by open faces, and attaching cells of dimension at least 3 does not change the fundamental group [Hat02, Proposition 1.26]; so $\pi_1(|K|) \cong \pi_1([-1,1]^S) = 1$.

(K4) *A sphere maps in.* The necklace map $x_S \colon S^2 \to [-1,1]^S$ of Lemma 5.3 with $k = 2$ is continuous and odd, and $x_S(w)$ has at most two fractional coordinates, so it lies in a closed face of dimension at most 2. Hence x_S is a continuous odd map $S^2 \to |K|$.

A labelling $\Lambda \colon V(K) \to \{P, M, N\}$ is *admissible* if no edge of K joins a P-vertex to an N-vertex. A *component* is the vertex set of a connected component of the subgraph of the 1-skeleton induced by one label class. The *component graph* Γ has the components as nodes, two of them adjacent if some edge of K joins them. It is connected by (K2), and it is bipartite between the M-components and the $(P \cup N)$-components, since two components with the same label joined by an edge would be one component, and there are no P–N edges. For a vertex set Y we write $K[Y]$ for the subgraph of the 1-skeleton induced by Y, and $K - Y$ for $K[V(K) \setminus Y]$.

Lemma 8.8 (component tree)**.** *For every admissible labelling of K, the component graph Γ is a tree.*

Proof. Suppose Γ has a cycle. As Γ is bipartite, the cycle contains an M-component D, with neighbours $X \neq Y$ on the cycle. Every vertex outside D adjacent to D lies in $P \cup N$ (an adjacent M-vertex would belong to D); let κ be 0 on such a vertex if its component is X, and 1 otherwise. Two vertices of $P \cup N$ that are equal or adjacent have the same label and lie in the same component, so they have the same κ-value whenever it is defined. For a closed edge path γ (a cyclic sequence of vertices, consecutive ones equal or adjacent) let $c(\gamma) \in \mathbb{Z}_2$ be the number of maximal runs of γ inside D whose predecessor and successor have different κ-values ($c(\gamma) = 0$ if γ lies in D or avoids D); the predecessor and successor of a run lie outside D and are adjacent to D, so κ is defined on them. We claim that c is invariant under the elementary moves $\dots uvw \dots \leftrightarrow \dots uw \dots$, where $\{u, v, w\}$ spans a simplex of K (repetitions allowed). We apply c to a based closed edge path $e_0 \dots e_n$ ($e_n = e_0$) through its cyclic sequence $(e_0, \dots, e_{n-1})$; every elementary move of based paths is an elementary move of the cyclic sequence, and the constant path has $c = 0$.

- $u, w \in D$, $v \notin D$: the move merges two runs into one with the same outer predecessor and successor, and c is unchanged because $[\kappa_1 \neq \kappa_3] \equiv [\kappa_1 \neq \kappa(v)] + [\kappa(v) \neq \kappa_3] \pmod 2$. If v is the only vertex of the path outside D, the two runs are one run with predecessor and successor v, which contributes 0, and after the move the path lies in D.
- $v \in D$, $u, w \notin D$: the move deletes the run (v), whose predecessor and successor u, w are equal or adjacent, so this run contributes 0.
- $u \in D$, $w \notin D$: if $v \in D$, the run through u only loses v; if $v \notin D$, its successor changes from v to w, which are equal or adjacent vertices outside D adjacent to u, so they have the same κ. The case $u \notin D$, $w \in D$ is symmetric.
- $u, v, w \in D$ or $u, v, w \notin D$: the runs and their neighbours do not change.

These moves generate the equivalence of closed edge paths, and since $|K|$ is simply connected by (K3), every closed edge path is equivalent to a constant one [Spa66, Chapter 3, Section 6], so $c \equiv 0$. But the closed edge path that runs once around the cycle of Γ, inside each component from its entry vertex to its exit vertex, meets D in a single run, entered from a vertex of X and left to a vertex of Y, so $c = 1$, a contradiction. □

Lemma 8.9 (nearly antipodal component)**.** *For every admissible labelling of K there are a component D and vertices $u, v \in D$ such that $u = \bar{v}$ or $u\bar{v}$ is an edge.*

Proof. Map every vertex to its component. The vertices of a simplex are pairwise equal or adjacent, so their components are pairwise equal or adjacent, and pairwise adjacent nodes of a tree number at most two (Lemma 8.8). So this is a simplicial map ϕ from K to the 1-dimensional complex of Γ, whose realization is a tree and embeds in $\mathbb{R}^2$. By the Borsuk–Ulam theorem [Mat03] applied to $\phi \circ x_S \colon S^2 \to \mathbb{R}^2$ (see (K4)) there is $w \in S^2$ with $\phi(x) = \phi(-x)$, where $x = x_S(w)$ and $-x = x_S(-w)$. Let σ be the support simplex of x, so that $x = \sum_{v \in \sigma} \lambda_v v$ with all $\lambda_v > 0$ and $-x = \sum_{v \in \sigma} \lambda_v \bar{v}$. If all vertices of σ map to one node D, then $\phi(-x) = \phi(x) = D$, so all vertices of $\bar{\sigma}$ map to D as well, and $v, \bar{v} \in D$ for $v \in \sigma$. Otherwise $\phi(x)$ is an interior point of an edge $\{D, D'\}$ of Γ, and D occurs among the images of the vertices of σ and among those of $\bar{\sigma}$: there are $v, v' \in \sigma$ with $v \in D$ and $\bar{v'} \in D$. Since v, v' are equal or adjacent, so are $\bar{v}, \bar{v'}$; put $u = \bar{v'}$. □

Lemma 8.10 (free involutions)**.** *Let G be a finite connected graph and ι an automorphism of G with $\iota^2 = \mathrm{id}$ and $\iota(v) \neq v$ for all v. Let $D \subseteq V(G)$ be nonempty with $G[D]$ connected and $D \cap \iota(D) = \emptyset$. Then $\iota(D)$ lies in a single component Z of $G - D$, and $R = V(G) \setminus Z$ satisfies $R \cap \iota(R) = \emptyset$.*

Proof. $G[\iota(D)]$ is connected and disjoint from D, so $\iota(D)$ lies in one component Z of $G-D$. Suppose that $z, \iota(z) \in R$. If $z \in D$, then $\iota(z) \in \iota(D) \subseteq Z$, and if $\iota(z) \in D$, then $z \in Z$; both are impossible. So $z \in T$ and $\iota(z) \in T'$ for components T, T' of $G-D$ other than Z. The set $\iota(T)$ is a component of $G - \iota(D)$ containing $\iota(z)$, and T' is connected, contains $\iota(z)$ and avoids $\iota(D) \subseteq Z$; hence $T' \subseteq \iota(T)$. Symmetrically $T \subseteq \iota(T')$, so $T' = \iota(T)$. Every vertex outside T' adjacent to T' lies in D (as T' is a component of $G-D$) and in $\iota(D)$ (as T' is a component of $G - \iota(D)$), hence in $D \cap \iota(D) = \emptyset$. As G is connected, $T' = V(G)$, which contradicts $\emptyset \neq D \subseteq V(G) \setminus T'$. □

By (K1) and (K2), Lemma 8.10 applies to the 1-skeleton of K with $\iota(v) = \bar{v}$. For an admissible labelling and an M-node D of the tree Γ, the components of $K - D$ are exactly the unions of the nodes of the components of $\Gamma - D$, since adjacent nodes are joined by an edge of K and nodes in different components of $\Gamma - D$ are not. Each component of $\Gamma - D$ contains exactly one neighbour of D, and we call the corresponding component of $K - D$ the component *through* that neighbour.

The last lemma relates components in K to components in the cube. For $X \subseteq \{\pm 1\}^S$ let $Q[X]$ be the subgraph of the Hamming graph induced by X; then X contains a 1-loop if and only if some component of $Q[X]$ contains a pair $z, -z$.

Lemma 8.11. *Let $X \subseteq \{\pm 1\}^S$, and let Y consist of X together with some barycentres c_q such that $|V(q) \cap X| \geq 3$ for each of them. Then two corners of X lie in the same component of $K[Y]$ if and only if they lie in the same component of $Q[X]$, and some component of $K[Y]$ contains a pair $v, \bar{v}$ if and only if X contains a 1-loop.*

Proof. The edges of $K[Y]$ between corners are the edges of $Q[X]$, and a barycentre $c_q \in Y$ is adjacent in $K[Y]$ exactly to the corners in $V(q) \cap X$, which induce a path or a cycle in $Q[X]$. So adding the barycentres attaches each of them to a single component of $Q[X]$ and merges no components, which proves the first claim and the "if" part of the second. Conversely, let a component $\hat{C}$ of $K[Y]$ contain v and $\bar{v}$. If v is a corner, then v and $-v$ lie in one component of $Q[X]$ by the first claim. If $v = c_q$, then $\bar{v} = c_{-q}$, and $\hat{C}$ contains $A = V(q) \cap X$ and $B = V(-q) \cap X$. Both A and $-B$ consist of at least three of the four elements of $V(q)$, so some $z \in A$ has $-z \in B$, and we are in the previous case. □

8.4 The hub-transfer lemma

Call a set $B \subseteq \{\pm 1\}^S$ *thick* if every antipodal 1-path in $\{\pm 1\}^S$ meets B, that is, if $\{\pm 1\}^S \setminus B$ contains no 1-loop. Thickness is invariant under $z \mapsto -z$, and the empty set is not thick. The hub-transfer lemma rests on two statements about a single cube.

Lemma 8.12 (dichotomy)**.** *Let $|S| \geq 3$ and $\{\pm 1\}^S = P \sqcup M \sqcup N$ with $\operatorname{dist}(P, N) \geq 3$ (where $\operatorname{dist}(\cdot, \emptyset) = \infty$). If M contains no 1-loop, then P or N is thick.*

Lemma 8.13 (separation)**.** *Let $|S| \geq 3$. Any two thick subsets of $\{\pm 1\}^S$ are at distance at most 1.*

The hypothesis $\operatorname{dist}(P, N) \geq 3$ cannot be weakened to $\operatorname{dist}(P, N) \geq 2$: for even $|S| \geq 4$ and $\sigma(s) = \sum_j s_j$, the sets $P = \{\sigma \geq 2\}$, $M = \{\sigma = 0\}$ and $N = \{\sigma \leq -2\}$ satisfy $\operatorname{dist}(P, N) = 2$, M contains no 1-loop, and neither P nor N is thick.

Proof of Lemma 8.12. Label the corners by P, M, N, and label a barycentre c_q by P if $V(q)$ meets P, by N if $V(q)$ meets N, and by M otherwise. This is well defined because a square has diameter $2 < \operatorname{dist}(P, N)$, and it is admissible: adjacent corners are at distance 1, a barycentre labelled P (or N) is adjacent only to corners of its square, none of which lies in N (or P), and no two barycentres are adjacent. Let Y_M, Y_{MN} and Y_{MP} be the sets of vertices whose label lies in $\{M\}$, $\{M, N\}$ and

$\{M, P\}$, respectively. The barycentres in Y_{MN} are those with $V(q) \subseteq M \cup N$, and similarly for the other two sets, so Lemma 8.11 applies to (M, Y_M), $(M \cup N, Y_{MN})$ and $(M \cup P, Y_{MP})$. Since M contains no 1-loop, every M-component D satisfies $D \cap \bar{D} = \emptyset$.

Suppose that neither P nor N is thick, so that $M \cup N$ and $M \cup P$ contain 1-loops. By Lemma 8.11 there are a component Y of $K[Y_{MN}]$ and a component Y' of $K[Y_{MP}]$ with $Y \cap \bar{Y} \neq \emptyset$ and $Y' \cap \bar{Y}' \neq \emptyset$. Each component of $K[Y_{MN}]$ is a union of M- and N-components. Let Γ_Y be the set of nodes contained in Y, and define $\Gamma_{Y'}$ (nodes labelled M or P) likewise; both are subtrees of Γ. If a node labelled M or N is adjacent to a node of Γ_Y, then it belongs to Γ_Y; likewise for $\Gamma_{Y'}$ and the labels M, P.

Step 1: there is an M-node $D \in \Gamma_{Y'}$ such that either (a) $D \in \Gamma_Y$, or (b) Y is contained in the component of $K - D$ through some N-neighbour E of D. If Γ_Y and $\Gamma_{Y'}$ share a node, that node is labelled M, and (a) holds. Otherwise take a shortest path $X_0, \ldots, X_k$ in Γ from $\Gamma_{Y'}$ to Γ_Y; then $k \geq 1$ and $X_1 \notin \Gamma_{Y'}$. If X_0 were a P-node, then X_1 would be an M-node adjacent to $\Gamma_{Y'}$, hence in $\Gamma_{Y'}$; so X_0 is an M-node, and we put $D = X_0$. Its neighbour X_1 is not a P-node (it would lie in $\Gamma_{Y'}$), so $E = X_1$ is an N-node. The path $X_1, \ldots, X_k$ and the subtree Γ_Y avoid D, so Γ_Y lies in the component of $\Gamma - D$ containing E, and (b) holds.

Step 2. Apply Lemma 8.10 to D: $\bar{D}$ lies in a component Z of $K - D$, and $R = V(K) \setminus Z$ satisfies $R \cap \bar{R} = \emptyset$. Let X^* be the neighbour of D such that Z is the component through X^*. The neighbours of D in $\Gamma_{Y'}$ are P-nodes, so Y' is contained in the union of D and the components of $K - D$ through P-neighbours of D; similarly, if $D \in \Gamma_Y$, then Y is contained in the union of D and the components through N-neighbours of D. In case (b), if the component through E were not Z, then $Y \subseteq R$ and $Y \cap \bar{Y} = \emptyset$; so $X^* = E$ is an N-node, hence $Y' \subseteq R$ and $Y' \cap \bar{Y}' = \emptyset$. In case (a), $Y' \subseteq R$ if X^* is an N-node, and $Y \subseteq R$ if X^* is a P-node. Each case contradicts the choice of Y and Y'. □

Lemma 8.14 (edge covers)**.** *Let $|S| \geq 3$. There are no sets $U, V \subseteq \{\pm 1\}^S$, neither of which contains a 1-loop, such that every edge of the Hamming graph has both endpoints in U or both endpoints in V.*

Proof. Suppose U, V are such sets. Every corner lies on an edge, so $\{\pm 1\}^S = A \sqcup B \sqcup C$ with $A = U \setminus V$, $B = V \setminus U$ and $C = U \cap V$, and no edge joins A to B. If a square contains a corner of A and a corner of B, then these are opposite, and the other two corners, each adjacent to both, lie in C. Label the corners in A, B, C by P, N, M, and label a barycentre c_q by P if $V(q)$ meets A but not B, by N if $V(q)$ meets B but not A, and by M otherwise. This labelling is admissible: no corner of A is adjacent to a corner of B, a barycentre labelled P is adjacent only to corners in $A \cup C$, and one labelled N only to corners in $B \cup C$. Let Y_{PM} and Y_{NM} be the sets of vertices labelled P or M, and N or M, respectively. The corners in Y_{PM} form U, and a barycentre c_q lies in Y_{PM} if and only if either $V(q) \subseteq U$, or $V(q)$ meets both A and B, in which case $V(q) \cap U$ consists of three corners. So Lemma 8.11 applies to (U, Y_{PM}), and likewise to (V, Y_{NM}); since neither U nor V contains a 1-loop, no component of $K[Y_{PM}]$ or of $K[Y_{NM}]$ contains a pair $v, \bar{v}$.

Let D, u and v be as in Lemma 8.9. Suppose that D is labelled P or M, and let $\hat{Y}$ be the component of $K[Y_{PM}]$ containing D. If $u = \bar{v}$, or if $u\bar{v}$ is an edge and $\bar{v} \in Y_{PM}$, then $v, \bar{v} \in \hat{Y}$. Otherwise $\bar{v}$ is labelled N and adjacent to u, so u is not labelled P; then D is an M-component, and the component of $K[Y_{NM}]$ containing D contains v and $\bar{v}$. If D is labelled N, exchange the roles of P and N. Each case contradicts the previous paragraph. □

Proof of Lemma 8.13. Let B_1, B_2 be thick with $\mathrm{dist}(B_1, B_2) \geq 2$, and let $U = \{\pm 1\}^S \setminus B_1$ and $V = \{\pm 1\}^S \setminus B_2$; neither contains a 1-loop. Let zz' be an edge. If z or z' lies in B_1, then both

are within distance 1 of B_1, hence not in B_2, so $z, z' \in V$; otherwise $z, z' \in U$. This contradicts Lemma 8.14. □

Lemma 8.15 (hub transfer)**.** *Let $S \neq \emptyset$ and T be disjoint finite sets, let $\psi\colon \{\pm1\}^{S\cup T} \to \mathbb{R}$ be odd and 2-Lipschitz, let $a \geq 0$, and let $\rho^0, \dots, \rho^L$ be an antipodal a-path in $\{\pm1\}^T$. If $T \neq \emptyset$, then for some τ the set*

$$W(\rho^\tau) = \{s \in \{\pm1\}^S : \ |\psi(s, \rho^\tau)| \leq a + 1\}$$

contains a 1-loop. If $T = \emptyset$, then $\{s \in \{\pm1\}^S : |\psi(s)| \leq 2\}$ contains a 1-loop.

Proof. Padding. If $|S| \leq 2$, replace S by $S \sqcup D$ with $|S \sqcup D| = 3$ and ψ by $\psi'(s, y, \rho) = \psi(s, \rho)$, which is odd and 2-Lipschitz. The new sets are the products of the old ones with $\{\pm1\}^D$, and deleting the D-coordinates from a 1-loop in such a product gives a 1-loop in the old set (steps of length 0 are allowed). So we may assume $|S| \geq 3$.

The case $T \neq \emptyset$. Then $\rho^L = -\rho^0 \neq \rho^0$, so $a \geq 1$. For $0 \leq \tau \leq L$ let

$$P_\tau = \{s : \psi(s, \rho^\tau) > a + 1\}, \qquad N_\tau = \{s : \psi(s, \rho^\tau) < -a - 1\},$$

so that $\{\pm1\}^S = P_\tau \sqcup W(\rho^\tau) \sqcup N_\tau$. We have: (i) $\operatorname{dist}(P_\tau, N_\tau) \geq 3$, since $2\operatorname{dist}(s, s') \geq \psi(s, \rho^\tau) - \psi(s', \rho^\tau) > 2a + 2 \geq 4$ for $s \in P_\tau$ and $s' \in N_\tau$; (ii) $\operatorname{dist}(P_\tau, N_{\tau+1}) \geq 2$ and $\operatorname{dist}(P_{\tau+1}, N_\tau) \geq 2$, since for $s \in P_\tau$ and $s' \in N_{\tau+1}$ we have $2(\operatorname{dist}(s, s') + a) \geq \psi(s, \rho^\tau) - \psi(s', \rho^{\tau+1}) > 2a + 2$, and symmetrically; (iii) $P_L = -N_0$ and $N_L = -P_0$, since $\psi(s, \rho^L) = -\psi(-s, \rho^0)$. Suppose that no $W(\rho^\tau)$ contains a 1-loop. By (i) and Lemma 8.12, for every τ one of P_τ, N_τ is thick, and by (i) and Lemma 8.13 not both; let $\sigma(\tau) \in \{P, N\}$ name the thick one. By (iii), P_L is thick if and only if N_0 is, so $\sigma(L) \neq \sigma(0)$, and there is $\tau < L$ with $\sigma(\tau) \neq \sigma(\tau + 1)$. Then P_τ and $N_{\tau+1}$, or N_τ and $P_{\tau+1}$, are thick, which contradicts (ii) and Lemma 8.13.

The case $T = \emptyset$. Let $P = \{\psi > 2\}$, $N = \{\psi < -2\} = -P$ and $M = \{|\psi| \leq 2\}$. Then $\operatorname{dist}(P, N) \geq 3$ as in (i). If M contained no 1-loop, Lemma 8.12 would make P or $N = -P$ thick, hence both, and Lemma 8.13 would give $\operatorname{dist}(P, N) \leq 1$. □

Corollary 8.16. *Let $S \neq \emptyset$ and T be disjoint finite sets, let $\ell \geq 2$, and let $Z \subseteq \{\pm1\}^{S\cup T}$ be ℓ-coarse. Then*

$$\mathcal{G}_Z = \{\rho \in \{\pm1\}^T : \ \{s \in \{\pm1\}^S : (s, \rho) \in Z\} \text{ contains a 1-loop}\}$$

is $(\ell - 1)$-coarse.

Proof. $\mathcal{G}_Z$ is symmetric because Z is. Let $Z \supseteq \{|\psi| \leq \ell\}$ with ψ odd and 2-Lipschitz. If $T = \emptyset$, then $Z \supseteq \{|\psi| \leq 2\}$ contains a 1-loop by Lemma 8.15, so $\mathcal{G}_Z = \{\pm1\}^\emptyset$, which is c-coarse for every c. If $T \neq \emptyset$, then for every antipodal $(\ell - 1)$-path (ρ^τ) in $\{\pm1\}^T$, Lemma 8.15 with $a = \ell - 1$ gives τ such that $\{s : (s, \rho^\tau) \in Z\} \supseteq W(\rho^\tau)$ contains a 1-loop, that is, $\rho^\tau \in \mathcal{G}_Z$. By Lemma 8.3(b) with $c = \ell - 1 \geq 1$, $\mathcal{G}_Z$ is $(\ell - 1)$-coarse. □

8.5 Proof of Theorem 8.1 and consequences

Proof of Theorem 8.1. By Lemma 8.5 it suffices to show that $H = H(\mathcal{G})$ is C-good for $C = d + 1$ (every subgraph of H on the same vertex set has the same ordering property). Fix a constraint system with supports (U_v), graph a subgraph of H (which we again call H), and C-coarse constraints, and an ordering $v_1, \dots, v_n$ with at most d later neighbours per vertex. Let $H_k = H[\{v_1, \dots, v_k\}]$, let the items of H_k be those all of whose supports lie in $\{v_1, \dots, v_k\}$, and let $E_k(v)$ be the set of items of v in H_k. Let $m_k(v)$ be the number of neighbours of v among $v_{k+1}, \dots, v_n$.

We define constraint systems $\mathcal{S}_k$ on H_k, for $k = n, n-1, \dots, 0$, in which the constraint $Z_v^{(k)} \subseteq \{\pm 1\}^{E_k(v)}$ of v is $(C - m_k(v))$-coarse. Let $\mathcal{S}_n$ be the given system. Given $\mathcal{S}_k$, let $h = v_k$. For each neighbour r of h in H_k let $S_r = B_{hr} \neq \emptyset$ and $T_r = E_k(r) \setminus S_r = E_{k-1}(r)$, and let $Z_r^{(k-1)} = \mathcal{G}_{Z_r^{(k)}}$ as in Corollary 8.16; the other constraints are restricted unchanged. Since h is a later neighbour of r not counted in $m_k(r)$, we have $m_k(r) \leq d - 1$, so $Z_r^{(k)}$ is ℓ-coarse with $\ell = C - m_k(r) \geq 2$, and $Z_r^{(k-1)}$ is $(C - m_{k-1}(r))$-coarse by Corollary 8.16, since $m_{k-1}(r) = m_k(r) + 1$.

We show by induction on k that $\mathcal{S}_k$ has a solution; $\mathcal{S}_0$ is empty. Let x solve $\mathcal{S}_{k-1}$ and let $h = v_k$. For each neighbour r of h in H_k we have $x|_{T_r} \in Z_r^{(k-1)}$, so there is a 1-loop $\lambda_r \subseteq \{\pm 1\}^{S_r}$ with $(s, x|_{T_r}) \in Z_r^{(k)}$ for all $s \in \lambda_r$. The set $E_k(h)$ is the disjoint union of the bundles S_r and the private items of h. Moving along the loops λ_r one after the other and then flipping the private items one at a time gives an antipodal 1-path in $\{\pm 1\}^{E_k(h)}$. Since $Z_h^{(k)}$ is $(C - m_k(h))$-coarse and $C - m_k(h) \geq C - d = 1$, it contains a point of this path (Lemma 8.3(a)); extend x by this point. Then h satisfies $Z_h^{(k)}$, every neighbour r of h in H_k satisfies $Z_r^{(k)}$ because $x|_{S_r}$ lies on λ_r, and the other constraints are unaffected. A solution of $\mathcal{S}_n$ is a solution of the given system. □

The bound is best possible for $d \leq 1$: a single function and two functions sharing a bundle are 2-sparse families with $d = 0$ and $d = 1$, and their worst discrepancies are $\mathcal{K}_1 = 1$ and $\mathcal{K}_2 = 2$ (Theorem 1.2). The same bookkeeping lets one delete only some of the vertices and finish with another theorem.

Corollary 8.17. *Let $\mathcal{G}$ be 2-sparse and $V = V_1 \sqcup V_2$. Suppose V_1 can be ordered so that every vertex of V_1 has at most d_1 later neighbours in V_1, that $H[V_2]$ is β-good for some $\beta \geq 1$, and that every vertex of V_2 has at most q neighbours in V_1. Then* $\operatorname{disc}(\mathcal{G}) \leq \max(d_1 + 1, \beta + q)$.

Proof. Let $C = \max(d_1 + 1, \beta + q)$ and run the deletion process of the proof of Theorem 8.1 for the vertices of V_1 only, latest first. A vertex of V_2 is modified at most q times and has level at least $\beta + 1 \geq 2$ before each modification; a vertex of V_1 is modified at most d_1 times, has level at least 2 before each modification, and has level at least 1 when it is deleted. After all of V_1 is deleted, the constraints on V_2 are β-coarse and the graph is a subgraph of $H[V_2]$, so there is a solution, which extends to the vertices of V_1 one at a time as before. □

Theorem 8.1 alone gives, for example, the bounds 6 for planar patterns, $k + 1$ for treewidth k and $\Delta + 1$ when every block has maximum degree at most Δ; Section 8.7 improves all of them. The hub-transfer lemma uses only that every item of the hub is read by at most one other function, so high-degree functions can also be removed from patterns with larger t. Call a t-sparse support structure on a set V_2 of functions *β-good*, where $\beta \geq 1$, if every instance with these supports has discrepancy at most β (for example by Theorem 4.1 or 5.2).

Corollary 8.18 (pairwise hubs)**.** *Let $\mathcal{G}$ be t-sparse and $V = V_1 \sqcup V_2$, where every item in the support of a function of V_1 lies in at most two supports. Suppose V_1 can be ordered so that every function of V_1 shares items with at most d_1 later functions of V_1, that every function of V_2 shares items with at most q functions of V_1, and that the supports $U_v \setminus \bigcup_{u \in V_1} U_u$ $(v \in V_2)$ form a β-good structure. Then* $\operatorname{disc}(\mathcal{G}) \leq \max(d_1 + 1, \beta + q)$.

Proof. Define constraint systems and c-coarse constraints for arbitrary t-sparse supports as in Definition 8.4; the correspondence of Lemma 8.5 holds verbatim. Let $C = \max(d_1 + 1, \beta + q)$ and delete the functions of V_1, latest first, maintaining the invariant that every remaining function v has a $(C - j_v)$-coarse constraint on its remaining items, where j_v is the number of deleted functions of V_1 that shared items with v. When $h \in V_1$ is deleted, its remaining items are the disjoint union of

its private items and the sets $S_r = U_h \cap U_r \neq \emptyset$ for the remaining functions r sharing items with h; these sets are read only by h and r. Corollary 8.16 (whose proof never uses how the other items of r are shared) replaces the constraint of each such r by a constraint on $U_r \setminus S_r$ one level lower; the levels before each replacement are at least 2 as in Corollary 8.17, since $\beta \geq 1$. After all of V_1 is deleted, the remaining constraints are β-coarse with supports $U_v \setminus \bigcup_{u \in V_1} U_u$, so there is a solution, and the functions of V_1 are added back in order exactly as in the proof of Theorem 8.1. □

For example, any number of functions of unbounded degree, no two sharing an item and each sharing its items pairwise, can be added to a family covered by Theorem 1.6 at an additive cost of q, the largest number of added functions that share items with a single function of the family.

8.6 Leveled deletions

The deletion process of Theorem 8.1 does not have to run to the end. It can stop at a forest, or recurse on blocks, provided one keeps track of the individual levels. In a forest, vertices of level 1 are harmless as long as no two of them are adjacent. The only new analytic ingredient is an asymmetric version of $\mathcal{K}_2 = 2$.

Lemma 8.19. *For any odd 2-Lipschitz φ, ψ on $\{\pm 1\}^{[m]}$ there is x with $|\varphi(x)| \leq 1$ and $|\psi(x)| \leq 2$. Consequently every 1-coarse set contains a 2-loop, and if $\ell_1, \ell_2 \geq 1$ with $\max(\ell_1, \ell_2) \geq 2$, every ℓ_1-coarse set contains an ℓ_2-loop.*

Proof. Take x, Q and the positive weights w as in Section 3, with $r \leq 2$ fractional coordinates (Corollary 2.8). For $r = 0$, $\varphi(x) = \psi(x) = 0$. For $r = 1$, $|\psi| \leq 2$ at both corners (Lemma 3.1), and either φ vanishes at a corner or it takes values $a > 0 > b$ with $a - b \leq 2$, so $\min(a, |b|) \leq 1$. For $r = 2$, at most one corner has $|\psi| > 2$ (Lemma 3.2). Suppose no corner has $|\varphi| \leq 1$ and $|\psi| \leq 2$. Then at most one corner b has $|\varphi(b)| \leq 1$. Adjacent corners with $|\varphi| > 1$ have the same sign of φ, so if there were no such b, φ would have constant sign, contradicting $\sum w_z \varphi(z) = 0$. Let a, c be the neighbours of b and d the opposite corner. We may assume $\varphi > 1$ on $\{a, c, d\}$ and $\psi(b) > 2$. Then $w_b \geq w_b |\varphi(b)| > w_a + w_c + w_d$, and $\psi(a), \psi(c) > 0$, $\psi(d) > -2$, so $\sum w_z \psi(z) > 2w_b - 2w_d > 0$, a contradiction.

If $Z \supseteq J = \{|\varphi| \leq 1\}$ and J contained no 2-loop, Lemma 2.11(b) would give ψ with $|\psi| > 2$ on J, contradicting the first part ($J \neq \emptyset$ by $\mathcal{K}_1 = 1$; for $U = \emptyset$ there is nothing to prove). A 2-coarse set contains a 1-loop by the case $T = \emptyset$ of Lemma 8.15. The last claim follows. □

The lemma is sharp: $(1, 1)$ fails for $x_1 + x_2$, $x_1 - x_2$.

A *leveled graph* (G, ℓ) is a simple graph G on V with $\ell \colon V \to \{1, 2, \dots\}$. A *leveled constraint system* on it is a 2-sparse constraint system with graph $H \subseteq G$ in which Z_v is ℓ_v-coarse, and (G, ℓ) is *good* if every such system has a solution. Thus G is c-good if and only if (G, c) is good, and goodness is preserved by passing to spanning subgraphs and by raising levels.

Lemma 8.20 (rules)**.** *Let (G, ℓ) be a leveled graph.*

(a) (leaf) *If w has at most one neighbour v in G, $\max(\ell_v, \ell_w) \geq 2$ when v exists, and $(G - w, \ell)$ is good, then (G, ℓ) is good.*

(b) (deletion) *If every neighbour of h has level at least 2 and $(G - h, \ell')$ is good, where ℓ' lowers the levels of the neighbours of h by one, then (G, ℓ) is good.*

(c) (1-sum) *If $G = G_1 \cup G_2$ with $V(G_1) \cap V(G_2) = \{h\}$, $E(G_1) \cap E(G_2) = \emptyset$, and both (G_j, ℓ) are good, then (G, ℓ) is good; likewise for disjoint unions.*

Proof. (b) is the inductive step of the proof of Theorem 8.1: neighbours r of h in H receive $\mathcal{G}_{Z_r}$, which is $(\ell_r - 1)$-coarse by Corollary 8.16; neighbours of h in G but not in H keep their ℓ_r-coarse, hence $(\ell_r - 1)$-coarse, constraints; and h then sees the antipodal 1-path obtained by moving along the 1-loops of its bundles and flipping its private items, which Z_h ($\ell_h \geq 1$) meets. (c) is the proof of Lemma 8.6, with the threshold c in $\{|\psi| \leq c\}$ replaced by ℓ_h and "G_j is c-good" replaced by "(G_j, ℓ) is good".

(a) If w has no neighbour in H, its items are private, and $Z_w \neq \emptyset$. Otherwise let $S = B_{vw}$, $P = B_{\{w\}}$ and $T = U_v \setminus S$. The projection $F = \{s : \exists p,\ (s,p) \in Z_w\} \subseteq \{\pm 1\}^S$ is ℓ_w-coarse: extend an antipodal path in $\{\pm 1\}^S$ by flipping the coordinates of P one at a time, and use Lemma 8.3. By Lemma 8.19, F contains an ℓ_v-loop $\gamma_0, \dots, \gamma_K$. Put $Z'_v = \{\rho \in \{\pm 1\}^T : \exists s \in F,\ (s,\rho) \in Z_v\}$. If $T = \emptyset$, Z_v meets the ℓ_v-loop, so $Z'_v = \{\pm 1\}^\emptyset$. Otherwise, for an antipodal ℓ_v-path (ρ^τ) in $\{\pm 1\}^T$, the sequence $(\gamma_0, \rho^0), \dots, (\gamma_K, \rho^0), (-\gamma_0, \rho^1), \dots, (-\gamma_0, \rho^L)$ is an antipodal ℓ_v-path, so Z_v contains one of its points; since $\pm\gamma_0, \gamma_k \in F$, some ρ^τ lies in Z'_v. So Z'_v is ℓ_v-coarse (Lemma 8.3). Solve the system on $G - w$ in which v has support T and constraint Z'_v, and extend the solution through F and Z_w. □

Call (G, ℓ) *reducible* if it can be reduced to the empty graph by the operations of Lemma 8.20 (removing a leaf, deleting a vertex, or splitting a 1-sum or a disjoint union into smaller pieces that are reduced separately). By Lemma 8.20, reducible leveled graphs are good. By Theorem 3.8(b) and Lemma 8.5, a leveled graph on at most five vertices with all levels ≥ 2 is good, and it may be used as an additional starting point (computer-assisted). By induction along the reduction, no reducible leveled graph has two adjacent vertices of level 1; this is necessary for goodness ($Z_u = \{|x_1 + x_2| \leq 1\}$, $Z_v = \{|x_1 - x_2| \leq 1\}$ on a common bundle).

Theorem 8.21. *(a) If G is a forest and no two vertices of level 1 are adjacent, then (G, ℓ) is good.*

(b) Let G be triangle-free such that no vertex has two neighbours on a cycle avoiding it. If every block of G contains at most one vertex of level 1, then (G, ℓ) is good. In particular such graphs are 2-good; they include all triangle-free outerplanar graphs and all triangle-free graphs in which no vertex has two neighbours of degree ≥ 3 (e.g. every graph obtained from a simple graph by subdividing each edge at least twice).

(c) If $D \subseteq V(G)$ is a set of vertices no two of which are adjacent or have a common neighbour, $G - D$ is a forest, and $N(D)$ is independent, then G is 2-good.

(d) Let $C \geq 2$ and $V = V_1 \sqcup V_2$, where V_1 has an ordering with at most $C - 1$ later neighbours in V_1 for each vertex, and each $v \in V_2$ has $q_v \leq C - 1$ neighbours in V_1. If $(G[V_2], \ell)$ with $\ell_v = C - q_v$ is good, then G is C-good.

Proof. (a) Remove leaves one at a time (Lemma 8.20(a)).

(b) The class of such graphs is closed under subgraphs, and the blocks of the two pieces of a 1-sum are blocks of G. Induction on $|V|$; we may assume $\ell \leq 2$ (raising levels preserves goodness). Graphs with at most one vertex are trivial; disconnected graphs and graphs with a cut vertex are split by Lemma 8.20(c), and an edge is handled by (a). Let G be 2-connected with at least three vertices, and let y be its level-1 vertex, or any vertex if there is none; all neighbours of y have level 2. Delete y (Lemma 8.20(b)); the vertices of $N(y)$ get level 1. If a block B of $G - y$ contained two vertices $s, t \in N(y)$, then either $B = st$ and $y\,s\,t$ is a triangle, or, by Whitney's theorem, s, t lie on a cycle of the 2-connected graph B, which avoids y. So $G - y$ satisfies the hypothesis, and the induction applies. For the examples: in a triangle-free outerplanar graph, a vertex y with two neighbours s, t on a cycle $C \not\ni y$ would give a subdivision of $K_{2,3}$ (the arcs of C between s, t have

length ≥ 2); and in general such s, t have degree ≥ 3. A graph obtained from a simple graph by subdividing every edge at least twice has girth at least 9, its vertices of degree ≥ 3 are original vertices, and no vertex is adjacent to two of them.

(c) Delete the vertices of D; each other vertex loses at most one level, and the result satisfies (a).

(d) Delete the vertices of V_1, latest first. A remaining V_1-neighbour of the deleted vertex has lost at most $C-2$ levels and a V_2-neighbour at most $q_v - 1 \leq C-2$, so all deletions are allowed, and afterwards $v \in V_2$ has level $C - q_v$. □

Corollary 8.22. *For 2-sparse families with arbitrary bundle sizes and private items:*

(i) if $H(\mathcal{F})$ is a cycle, a cactus, a triangle-free outerplanar graph, or a graph obtained from a simple graph by subdividing every edge at least twice, then $\operatorname{disc}(\mathcal{F}) \leq 2$*, which is tight;*

(ii) if $H(\mathcal{F})$ is a subgraph of $K_{p,n}$, then $\operatorname{disc}(\mathcal{F}) \leq \max(2, p)$*; in particular $K_{2,n}$ gives 2.*

Proof. (i) Theorem 8.21(b), Lemma 8.20(c) for the blocks of a cactus, and Theorem 8.29 below for triangles. Tightness: take $g_0 = x_1 + x_2$, $g_1 = x_1 - x_2$ on one bundle and $g_v = 0$ otherwise, and replace each g_v ($U_v \neq \emptyset$) by $(1-\delta)g_v + \delta|U_v|^{-1}\sum_{j\in U_v} x_j$, which is odd, 2-Lipschitz, makes the items of U_v relevant for all but finitely many δ (as in Lemma 2.6, averaging over U_v only to keep the pattern), and has discrepancy at least $2 - 3\delta$. (ii) For $p \leq 1$ the pattern is a forest. For $p \geq 2$ use Theorem 8.21(d) with $C = p$ and V_1 all but one vertex a_p of the side of size p: $G[V_2]$ is a star centred at a_p (level $p \geq 2$) plus isolated vertices, so (a) applies. □

Theorem 8.21(d) strengthens Corollary 8.17, which requires the final graph to have a uniform level. A fan with at least six vertices is not reducible with these rules, even with the five-vertex base (it has no leaf and no cut vertex, it is too large for the base, and every deletion creates two adjacent vertices of level 1); it is 2-degenerate, however, and so is covered by Theorem 8.29 below, which also contains parts (i) and (ii) of Corollary 8.22. As evidence only (small bundle sizes; UNSAT answers not independently certified), SAT computations with the leveled version of Lemma B.1 (threshold $D_v = \ell_v + 1$) found no counterexample to the statement that a leveled graph is good whenever no two vertices of level 1 are adjacent (e.g. C_4 with levels $(2, 1, 2, 1)$, which is not reducible with these rules but is good by Lemma 8.28 below: delete a vertex of level 2 with J both of its neighbours, and then the other vertex of level 2 likewise); this would strengthen Conjecture 1.12.

8.7 Near-loops: discrepancy $\max(2, d)$, and 3 for planar patterns

By Remark 8.38(1) below, demanding a 1-loop from a neighbour of a deleted vertex costs that neighbour one level. We now show that a slightly weaker demand costs *nothing*, at every level, and that a deleted vertex h can afford this weaker demand from up to ℓ_h of its neighbours. This improves the bound $d+1$ of Theorem 8.1 to d for $d \geq 2$ (Theorem 8.29); for $d \leq 1$ the bound $d+1$ remains, and is tight. Mixing cheap deletions with ordinary ones gives $k+1$ for patterns with an orientation of maximum in-degree k (Theorem 8.31), and 3 for planar patterns, using canonical orderings (Theorem 8.32).

Near-loops. Let S be finite. The 2-*step graph* $Q_S^{(2)}$ has vertex set $\{\pm 1\}^S$, and $z \neq z'$ are adjacent if $\operatorname{dist}(z, z') \leq 2$. A *near-loop* in $A \subseteq \{\pm 1\}^S$ is a 2-path $z_0, \ldots, z_k$ in A with $\operatorname{dist}(z_k, -z_0) \leq 1$. Every 1-loop and every 2-loop is a near-loop, and A contains a near-loop if and only if some component Y of the induced subgraph $Q_S^{(2)}[A]$ contains z, w with $\operatorname{dist}(w, -z) \leq 1$; we then call Y *near-antipodal*. Negating a near-loop in A gives a near-loop in $-A$. A set $\Omega \subseteq \{\pm 1\}^S$ is *near-thick* if $\{\pm 1\}^S \setminus \Omega$

contains no near-loop; Ω is near-thick if and only if $-\Omega$ is. For $S = \emptyset$ every nonempty set contains a near-loop.

Lemma 8.23 (near pairs)**.** *Let G, ι be as in Lemma 8.10, and let $\approx$ be a reflexive symmetric relation on $V(G)$ such that $u \approx v$ implies $\iota(u) \approx \iota(v)$, and $u = v$ or $uv \in E(G)$. Let $D \subseteq V(G)$ be nonempty with $G[D]$ connected and with no $d, d' \in D$ such that $d \approx \iota(d')$. Then $\iota(D)$ lies in a single component Z of $G - D$, and $R = V(G) \setminus Z$ contains no r, r' with $r \approx \iota(r')$.*

Proof. By reflexivity $D \cap \iota(D) = \emptyset$, so Z exists as in Lemma 8.10. Note that $r \approx \iota(r')$ implies $\iota(r) \approx r'$. Let $r, r' \in R$ with $r \approx \iota(r')$. Both in D is excluded. If $r \in D \not\ni r'$, then $\iota(r) \in Z$ and $\iota(r) \approx r'$, so $r' \in Z$ (as $r' \notin D$ is equal or adjacent to a vertex of Z), a contradiction; the case $r \notin D \ni r'$ is symmetric. Otherwise $r \in T$, $r' \in T'$ for components $T, T' \neq Z$ of $G - D$. The connected set T avoids $\iota(D) \subseteq Z$, and r is equal or adjacent to $\iota(r') \notin \iota(D)$, so T lies in the component $\iota(T')$ of $G - \iota(D)$. Symmetrically $T' \subseteq \iota(T)$, so $T' = \iota(T)$, and the end of the proof of Lemma 8.10 applies: every vertex adjacent to T' from outside lies in $D \cap \iota(D) = \emptyset$, so $T' = V(G)$, which contradicts $D \neq \emptyset$. □

Lemma 8.24 (near-dichotomy)**.** *Let $S \neq \emptyset$ and $\{\pm 1\}^S = P \sqcup W \sqcup N$ with $\operatorname{dist}(P, N) \geq 2$. If W contains no near-loop, then P or N is near-thick.*

Proof. Padding. For a set D' disjoint from S, a set $A \subseteq \{\pm 1\}^S$ contains a near-loop if and only if $A \times \{\pm 1\}^{D'}$ does: deleting the D'-coordinates of a near-loop gives one, and a near-loop $z \to w$ in A followed by flipping the D'-coordinates one at a time from y to $-y$ gives one in $A \times \{\pm 1\}^{D'}$. Replacing P, W, N by their products with $\{\pm 1\}^{D'}$ preserves the hypotheses, so we may assume $|S| \geq 3$ and work in the complex K of Section 8.3.

Labelling. Label the corners of P, W, N by P, M, N. Label a barycentre c_q by M if $V(q)$ contains two opposite corners of W; otherwise by P if $V(q)$ meets P, and by N if $V(q)$ meets N (if $V(q) \subseteq W$, the first case applies). If $V(q)$ meets both P and N, a corner of P and one of N in $V(q)$ are opposite (as $\operatorname{dist}(P, N) \geq 2$), and the other two corners are adjacent to both, hence lie in W; so such c_q are labelled M. Thus a barycentre labelled P (or N) has no corner in N (or P), and since adjacent corners are not in P and N, the labelling is admissible. By Lemma 8.8, Γ is a tree.

Components. Let $X \in \{N, P\}$. We claim that the corner sets of the components of the vertices labelled M or X are exactly the components of $Q_S^{(2)}[W \cup X]$, and that every such component contains a corner; likewise for the label M alone and $Q_S^{(2)}[W]$. Each barycentre labelled M has a corner in W and each barycentre labelled X has one in X, and a barycentre is adjacent only to the corners of its square, which are pairwise at distance at most 2. Conversely, corners of $W \cup X$ at distance 1 are adjacent in K; let $a, b \in W \cup X$ with $\operatorname{dist}(a, b) = 2$, opposite in the square q. If $V(q)$ meets the label class other than M and X, then a, b are adjacent to a corner of that class, so they lie in W and c_q is labelled M; otherwise c_q is labelled M or X (if $V(q)$ meets neither P nor N, then $V(q) \subseteq W$ and c_q is labelled M). In both cases $a\, c_q\, b$ is a path inside the label set. For M alone, opposite corners $a, b \in W$ make c_q an M-vertex.

The argument. Suppose that $W \cup N$ and $W \cup P$ contain near-loops. By the claim there are a component Y of the vertices labelled M or N and a component Y' of the vertices labelled M or P whose corner sets are near-antipodal. Each of them is a union of single-label components; let Γ_Y and $\Gamma_{Y'}$ be the sets of nodes of Γ contained in Y and Y'. They are subtrees of Γ, and a node labelled M or N (respectively M or P) that is adjacent to a node of Γ_Y (respectively $\Gamma_{Y'}$) belongs to it.

Step 1: there is an M-node $D \in \Gamma_{Y'}$ such that either (a) $D \in \Gamma_Y$, or (b) Y is contained in the component of $K - D$ through some N-neighbour E of D. If Γ_Y and $\Gamma_{Y'}$ share a node, that node is

labelled M, and (a) holds. Otherwise take a shortest path $X_0, \dots, X_k$ in Γ from $\Gamma_{Y'}$ to Γ_Y; then $k \geq 1$ and $X_1 \notin \Gamma_{Y'}$. If X_0 were a P-node, then X_1 would be an M-node adjacent to $\Gamma_{Y'}$, hence in $\Gamma_{Y'}$; so X_0 is an M-node, and we put $D = X_0$. Its neighbour X_1 is not a P-node (it would lie in $\Gamma_{Y'}$), so $E = X_1$ is an N-node. The path $X_1, \dots, X_k$ and the subtree Γ_Y avoid D, so Γ_Y lies in the component of $\Gamma - D$ containing E, and (b) holds.

Step 2. Apply Lemma 8.23 to the 1-skeleton of K, $\iota(v) = \bar{v}$, and the relation in which two corners u, v satisfy $u \approx v$ if $\operatorname{dist}(u, v) \leq 1$, and any other two vertices (in particular a corner and a barycentre) only if they are equal. Its hypothesis holds for D: by the claim, the corners of D form a component of $Q_S^{(2)}[W]$, which is not near-antipodal; and if c_q and c_{-q} both lie in D, then D contains an opposite pair w, w' of W-corners of q (which are adjacent to c_q) and an opposite pair of W-corners of $-q$, which is $\{-w, -w'\}$ or $\{-u, -u'\}$ for the other diagonal $\{u, u'\}$ of q; then $w, -w \in D$ or $w, -u \in D$ with $\operatorname{dist}(w, u) = 1$, which is again a near pair of corners in D. Hence $-D$ lies in a component Z of $K - D$, and $R = V(K) \setminus Z$ contains no corners r, r' with $\operatorname{dist}(r, -r') \leq 1$, so neither Y nor Y' is contained in R.

Let X^* be the neighbour of D in Γ such that Z is the component of $K - D$ through X^*. The neighbours of D in $\Gamma_{Y'}$ are P-nodes, so Y' is contained in the union of D and the components of $K - D$ through P-neighbours of D; similarly, if $D \in \Gamma_Y$, then Y is contained in the union of D and the components through N-neighbours of D. In case (b), if the component through E were not Z, then $Y \subseteq R$; so $X^* = E$ is an N-node, and hence $Y' \subseteq R$. In case (a), $Y' \subseteq R$ if X^* is an N-node, and $Y \subseteq R$ if X^* is a P-node. Each case is a contradiction. □

Lemma 8.25 (near-cover lemma)**.** *Let $S \neq \emptyset$ and $U \cup V = \{\pm 1\}^S$. Then U or V contains a near-loop. Consequently any two near-thick subsets of $\{\pm 1\}^S$ intersect.*

Proof. Let Θ be the bipartite graph whose nodes are the components of $Q_S^{(2)}[U]$ (*U-components*) and the components of $Q_S^{(2)}[V]$ (*V-components*), taken as distinct nodes, a U-component C and a V-component C' being adjacent if $\operatorname{dist}(C, C') \leq 2$. For a corner v let $C(v)$ and $C'(v)$ be the U- and V-components containing v, when defined. Let $p(v) \in |\Theta|$ be $C(v)$ if $v \in U \setminus V$, $C'(v)$ if $v \in V \setminus U$, and the midpoint of the edge $C(v)C'(v)$ if $v \in U \cap V$. The corners of a face F of $[-1, 1]^S$ of dimension at most 2 have pairwise distances at most 2, so its U-corners lie in one U-component C_F and its V-corners in one V-component C'_F. Let $e_F \subseteq |\Theta|$ be the closed edge $C_F C'_F$ if F has corners in U and in V, and the node C_F or C'_F otherwise. Then $p(v) \in e_F$ for every corner v of F, and $e_{F'} \subseteq e_F$ for faces $F' \subseteq F$.

For $x \in [-1, 1]^S$ let $\lambda_v(x) = \prod_{j \in S} \frac{1 + v_j x_j}{2}$ be the weight of the corner v under the product distribution with mean x. These weights are continuous, sum to 1, satisfy $\lambda_{-v}(-x) = \lambda_v(x)$, and are positive exactly on the corners of the smallest face $F(x)$ containing x. On the union $X^{(2)}$ of the faces of dimension at most 2, let $\Phi(x) = \sum_v \lambda_v(x) p(v)$, the convex combination being taken in $e_{F(x)}$. On each face F of dimension at most 2 this is the continuous map $x \mapsto \sum_{v \in V(F)} \lambda_v(x) p(v)$ into e_F, so $\Phi \colon X^{(2)} \to |\Theta|$ is continuous.

Let $x_S \colon S^2 \to [-1, 1]^S$ be the odd necklace map of Lemma 5.3 with two cuts; as in (K4), $x_S(S^2) \subseteq X^{(2)}$, so $f = \Phi \circ x_S \colon S^2 \to |\Theta|$ is continuous. Since S^2 is simply connected, f lifts to the universal cover of the component of $|\Theta|$ containing $f(S^2)$, a tree [Hat02]; the image of the lift lies in a finite subtree, which embeds in $\mathbb{R}^2$. By the Borsuk–Ulam theorem [Mat03] there is w with equal lifts at w and $-w$, so $f(w) = f(-w)$. Let $x = x_S(w)$, $F = F(x)$ and $\nu = V(F)$; then $x_S(-w) = -x$, $F(-x) = -F$, and $\Phi(-x) = \sum_{v \in \nu} \lambda_v(x) p(-v)$, with all $\lambda_v(x) > 0$ for $v \in \nu$.

If $\Phi(x)$ is a node Ξ, then all points $p(v)$ and $p(-v)$, $v \in \nu$, equal Ξ; so Ξ contains v and $-v$, and U or V contains a near-loop. Otherwise $\Phi(x) = \Phi(-x)$ is an interior point of an edge CC', so $e_F = e_{-F} = CC'$: all corners of $\nu \cup (-\nu)$ in U lie in C, and those in V lie in C'. If some $v \in \nu$

has $v, -v \in C$ or $v, -v \in C'$, we are done. Otherwise no $v \in \nu$ lies in $C \cap C'$, and ν splits into $\nu_C = \nu \cap C$ and $\nu_{C'} = \nu \cap C'$, with $-v \in C' \setminus C$ for $v \in \nu_C$ and $-v \in C \setminus C'$ for $v \in \nu_{C'}$. A corner of $\nu \cup (-\nu)$ in $C \setminus C'$ is not in V, so p maps it to C; likewise corners in $C' \setminus C$ map to C'. Hence the weight of C in $\Phi(x)$ is $\sum_{v \in \nu_C} \lambda_v(x)$ and in $\Phi(-x)$ it is $\sum_{v \in \nu_{C'}} \lambda_v(x)$; both equal $\frac{1}{2}$, so $\nu_C, \nu_{C'} \neq \emptyset$. The corners of F span a connected subgraph of the cube, so some $v \in \nu_C$ and $v' \in \nu_{C'}$ are adjacent. Then $v, -v' \in C$ and $\operatorname{dist}(-v', -v) = 1$, so C is near-antipodal and U contains a near-loop.

If Ω_1, Ω_2 are disjoint near-thick sets, $U = \{\pm 1\}^S \setminus \Omega_1$ and $V = \{\pm 1\}^S \setminus \Omega_2$ contradict the first part. □

Lemma 8.26 (near-loop transfer)**.** *Let $S \neq \emptyset$ and T be disjoint finite sets, let $\psi \colon \{\pm 1\}^{S \cup T} \to \mathbb{R}$ be odd and 2-Lipschitz, let $\ell \geq 1$, and let $\rho^0, \dots, \rho^L$ be an antipodal ℓ-path in $\{\pm 1\}^T$. If $T \neq \emptyset$, then for some τ the slice $W_\tau = \{s \in \{\pm 1\}^S : |\psi(s, \rho^\tau)| \leq \ell\}$ contains a near-loop. If $T = \emptyset$, then $\{s : |\psi(s)| \leq \ell\}$ contains a 2-loop.*

Compared with Lemma 8.15, the threshold is ℓ instead of $\ell + 1$.

Proof. For $T = \emptyset$, $\{|\psi| \leq \ell\}$ is 1-coarse and contains a 2-loop by Lemma 8.19. Let $T \neq \emptyset$, and let $P_\tau = \{s : \psi(s, \rho^\tau) > \ell\}$ and $N_\tau = \{s : \psi(s, \rho^\tau) < -\ell\}$. Then (i) $\operatorname{dist}(P_\tau, N_\tau) \geq 2$, since $2 \operatorname{dist}(s, s') > 2\ell \geq 2$ for $s \in P_\tau$, $s' \in N_\tau$; (ii) $P_\tau \cap N_{\tau+1} = N_\tau \cap P_{\tau+1} = \emptyset$, since $|\psi(s, \rho^\tau) - \psi(s, \rho^{\tau+1})| \leq 2\ell$; (iii) $P_L = -N_0$ and $N_L = -P_0$. Suppose that no W_τ contains a near-loop. By (i) and Lemma 8.24, one of P_τ, N_τ is near-thick, and by Lemma 8.25 not both; let $\sigma(\tau) \in \{P, N\}$ name it. By (iii), $\sigma(L) \neq \sigma(0)$, so for some τ the sets $P_\tau, N_{\tau+1}$ or $N_\tau, P_{\tau+1}$ are near-thick, and they intersect by Lemma 8.25, contradicting (ii). □

Corollary 8.27. *Let $S \neq \emptyset$ and T be disjoint finite sets, let $\ell \geq 1$, and let $Z \subseteq \{\pm 1\}^{S \cup T}$ be ℓ-coarse. Then*

$$Z^\natural = \{\rho \in \{\pm 1\}^T : \ \{s \in \{\pm 1\}^S : (s, \rho) \in Z\} \text{ contains a near-loop}\}$$

is ℓ-coarse.

Proof. $Z^\natural$ is symmetric, since the slice at $-\rho$ is the negative of the slice at ρ. If $T = \emptyset$, then $Z^\natural = \{\pm 1\}^\emptyset$ by Lemma 8.26. Otherwise Lemma 8.26 shows that $Z^\natural$ meets every antipodal ℓ-path, and Lemma 8.3(b) applies. □

Lemma 8.28 (cheap deletion)**.** *Let (G, ℓ) be a leveled graph, h a vertex, and $J \subseteq N_G(h)$ with $|J| \leq \ell_h$, where $\ell_h \geq 2$ if $J \neq \emptyset$, such that every vertex of $N_G(h) \setminus J$ has level at least 2. Let ℓ' lower the levels of the vertices of $N_G(h) \setminus J$ by one. If $(G - h, \ell')$ is good, then (G, ℓ) is good.*

Proof. Take a leveled constraint system on (G, ℓ) with graph H. For $r \in N_H(h)$ let $S_r = B_{hr}$ and $T_r = U_r \setminus S_r$, and give r the support T_r and the constraint $Z_r^\natural$ if $r \in J$, which is ℓ_r-coarse by Corollary 8.27, and $\mathcal{G}_{Z_r}$ if $r \notin J$, which is $(\ell_r - 1)$-coarse by Corollary 8.16 (if $T_r = \emptyset$, the new constraint is $\{\pm 1\}^\emptyset$). The other vertices of $G - h$ keep their supports and constraints; a neighbour $r \notin J$ of h in G but not in H has an ℓ_r-coarse, hence $(\ell_r - 1)$-coarse, constraint, since $\ell_r \geq 2$. This is a leveled system on $(G - h, \ell')$; let x be a solution. For $r \in N_H(h)$ the slice $F_r = \{s \in \{\pm 1\}^{S_r} : (s, x|_{T_r}) \in Z_r\}$ contains a near-loop $s_0^r, \dots, s_{k_r}^r$ if $r \in J$, and a 1-loop (a near-loop with $s_{k_r}^r = -s_0^r$) if $r \notin J$. Let $Q = B_{\{h\}}$ and $y_0 \in \{\pm 1\}^{U_h}$ with $y_0|_{S_r} = s_0^r$. Moving the blocks S_r along these paths one after the other and then flipping the coordinates of Q one at a time leads, with steps of length at most 2 (at most 1 if $J = \emptyset$), to a point y_1 with $\operatorname{dist}(y_1, -y_0) \leq |J \cap N_H(h)| \leq \ell_h$. Followed by $-y_0$, this is an antipodal ℓ_h-path whose points lie in $A = \prod_r F_r \times \{\pm 1\}^Q$, except the last, which lies in $-A$. By Lemma 8.3(a) and the symmetry of Z_h, some $y \in Z_h \cap A$ exists, and extending x by y satisfies all constraints. □

For $J = \emptyset$ this is Lemma 8.20(b); with $J = N_G(h)$ and $|N_G(h)| = 1$ it contains Lemma 8.20(a) when $\ell_h \geq 2$. Like Corollary 8.16, Corollary 8.27 never uses how the other items of r are shared.

Theorem 8.29. *Let (G, ℓ) be a leveled graph with all levels at least 2, and suppose $V(G)$ can be ordered so that every vertex v has at most ℓ_v later neighbours. Then (G, ℓ) is good. Consequently, every 2-sparse instance $\mathcal{G}$ has* $\operatorname{disc}(\mathcal{G}) \leq \max(2, \operatorname{dgn}(H(\mathcal{G})))$.

Proof. Induction on $|V(G)|$: the first vertex v_1 has at most ℓ_{v_1} neighbours; delete it by Lemma 8.28 with $J = N_G(v_1)$, which changes no level, and apply induction to $G - v_1$ with the induced ordering. The second statement follows with $\ell \equiv \max(2, d)$ and Lemma 8.5. □

The same bookkeeping can mix cheap deletions and payments. For an ordering $v_1, \ldots, v_n$ of $V(G)$ in which every edge is marked *absorbed* or *paid*, let a_v be the number of absorbed edges joining v to later vertices and π_v the number of paid edges joining v to earlier vertices.

Lemma 8.30 (schedules)**.** *Suppose that $\ell_v \geq \pi_v + \max(1, a_v)$ for every vertex v, and $\ell_v \geq \pi_v + 2$ for every vertex v with $a_v \geq 1$. Then (G, ℓ) is good.*

Proof. Induction on $|V(G)|$. Delete v_1 by Lemma 8.28, with J the set of its neighbours joined to it by absorbed edges. Since $p_{v_1} = 0$, we have $|J| = a_{v_1} \leq \ell_{v_1}$, and $\ell_{v_1} \geq 2$ if $J \neq \emptyset$. A neighbour u joined by a paid edge has $\pi_u \geq 1$, hence $\ell_u \geq \pi_u + 1 \geq 2$. These neighbours lose one level. For $G - v_1$ with the induced ordering and marks, a_u is unchanged and π_u and ℓ_u both drop by one if $v_1 u$ is paid; so the hypotheses are inherited, and induction applies. □

If all edges are paid and the ordering is the reverse of an ordering with at most d later neighbours, this is Theorem 8.1 ($\ell \equiv d + 1$); if all edges are absorbed, it contains Theorem 8.29.

Theorem 8.31. *If $H(\mathcal{G})$ has an orientation in which every vertex has in-degree at most k, then* $\operatorname{disc}(\mathcal{G}) \leq k + 1$. *Consequently* $\operatorname{disc}(\mathcal{G}) \leq p(H) + 1$, *where $p(H) = \max_{H'} \lceil |E(H')|/|V(H')| \rceil$ over the nonempty subgraphs H' of $H = H(\mathcal{G})$; in particular, if every function has at most Δ distinct neighbours, then* $\operatorname{disc}(\mathcal{G}) \leq \lceil \Delta/2 \rceil + 1$.

Proof. Take any ordering, and mark an edge absorbed if it is oriented towards its earlier endpoint and paid otherwise. Then $\pi_v + a_v$ is the in-degree of v, at most k; so $\pi_v + \max(1, a_v) \leq k + 1$, and $\pi_v + 2 \leq k + 1$ if $a_v \geq 1$. Lemma 8.30 with $\ell \equiv k + 1$ and Lemma 8.5 give the first claim. By Hakimi's theorem [Hak65], H has an orientation with all in-degrees at most $p(H)$. A graph of maximum degree Δ has an orientation with all in-degrees at most $\lceil \Delta/2 \rceil$: join a new vertex to all vertices of odd degree and orient an Euler circuit of each component. □

A graph of degeneracy d has a subgraph of minimum degree d, so $p(H) \geq \lceil d/2 \rceil$: Theorem 8.31 can improve Theorem 8.29 by up to a factor of two, for example for regular patterns, but it does not give $o(\operatorname{dgn})$, and for some patterns (e.g. d-trees) Theorem 8.29 is better.

Theorem 8.32. *Let $c \geq 2$. Every graph that is the union of a 2-degenerate graph and $c-2$ forests is c-good; so* $\operatorname{disc}(\mathcal{G}) \leq c$ *whenever $H(\mathcal{G})$ is a subgraph of such a graph. In particular, planar patterns have* $\operatorname{disc}(\mathcal{G}) \leq 3$.

Proof. Write $G = D \cup F_1 \cup \cdots \cup F_{c-2}$ with pairwise edge-disjoint parts (assign every edge to one part), where D is 2-degenerate, fix an ordering $v_1, \ldots, v_n$ of $V(G)$ in which every vertex has at most two later D-neighbours, and root every component of every F_i. Mark every edge of D absorbed. An edge of F_i joins a vertex x to its parent q; mark it absorbed if x precedes q and paid otherwise. Then every edge is counted by exactly one of the numbers a_v, π_v of Lemma 8.30: an edge uw of

D with u earlier only by a_u (at w it is an absorbed edge to an earlier vertex), and an edge of F_i only by a_x or π_x at its child x. Hence $a_v = \delta_v + f_v^+$ and $\pi_v = f_v^-$, where $\delta_v \le 2$ is the number of later D-neighbours of v, and f_v^+ and f_v^- count the forests in which v has a parent that comes later, respectively earlier; $f_v^+ + f_v^- \le c-2$. So $\pi_v + a_v \le c$, $\pi_v + 1 \le c-1$ and $\pi_v + 2 \le c$, and Lemma 8.30 with $\ell \equiv c$ shows that G is c-good. The second claim follows by Lemma 8.5.

For planar patterns we may, by Lemma 8.5, replace $H = H(\mathcal{G})$ by a maximal planar graph on the same $n \ge 3$ vertices (for $n \le 2$ the pattern is a forest). Fix a plane embedding with outer face $u_1u_2u_n$ and let $u_1, \dots, u_n$ be a canonical ordering [dFPP90]: for $3 \le k \le n$, the plane graph G_{k-1} induced by $u_1, \dots, u_{k-1}$ has an outer cycle C_{k-1} through the edge u_1u_2 (for $k = 3$, read C_2 as the edge u_1u_2), and the neighbours of u_k in G_{k-1} form a path $w_0w_1 \cdots w_s$ with $s \ge 1$ in $C_{k-1} - u_1u_2$. Then C_k arises from C_{k-1} by replacing $w_0 \cdots w_s$ with $w_0u_kw_s$, and all later vertices lie in the outer face of G_k, so the inner vertices $w_1, \dots, w_{s-1}$ lie on none of the cycles $C_{k'}$, $k' \ge k$. Let D consist of the edge u_1u_2 and the edges u_kw_0 and u_kw_s ($3 \le k \le n$), and let F consist of the remaining edges u_kw_j ($0 < j < s$); every edge u_iu_k with $i < k$ is assigned at step k, so D and F partition the edges. In the ordering $u_n, \dots, u_1$ every vertex has at most two later D-neighbours, so D is 2-degenerate. A vertex is an inner vertex of the path of at most one u_k, because it leaves the outer cycle at that step; so every vertex has at most one F-neighbour of larger index. Orienting the edges of F towards the larger index gives out-degrees at most one and no directed cycle, so F is a forest, and the first part applies with $c = 3$. □

Within schedules of cheap deletions, the 2-degenerate part cannot be replaced by a union of two forests: for $c = 2$, an exhaustive search over deletion sequences with Lemma 8.28 finds no schedule at level 2 for the square of the 12-cycle with the edges $\{0,1\}$ and $\{6,7\}$ removed, although this graph is the union of two forests (it is 3-good by Theorem 8.31). We do not know whether every graph of arboricity 3 has a schedule at level 3; random unions of three spanning trees always had one in our experiments. As a sanity check of the proof, the marking above was verified by computer to satisfy the hypotheses of Lemma 8.30 at level 3 for 400 random triangulations with 5 to 120 vertices and for the octahedron and the icosahedron. The search code and logs are in the ancillary files.

Corollary 8.33. *For 2-sparse families with arbitrary bundle sizes and private items:*

(i) if $H(\mathcal{F})$ is 2-degenerate or a wheel, then $\operatorname{disc}(\mathcal{F}) \le 2$*, which is tight; this includes outerplanar and series–parallel patterns, planar patterns of girth at least 6, cacti, fans and K_4;*

(ii) planar patterns have $\operatorname{disc}(\mathcal{F}) \le 3$*, and patterns of treewidth k have* $\operatorname{disc}(\mathcal{F}) \le \max(2, k)$*;*

(iii) if every block of $H(\mathcal{F})$ has maximum degree at most $\Delta \ge 2$, then $\operatorname{disc}(\mathcal{F}) \le \lceil \Delta/2 \rceil + 1$*, and if the functions with more than Δ distinct neighbours form an independent set, then* $\operatorname{disc}(\mathcal{F}) \le \max(2, \Delta)$*;*

(iv) every 2-sparse family of $n \ge 3$ functions has $\operatorname{disc}(\mathcal{F}) \le \lceil n/2 \rceil$*, and* $\operatorname{disc}(\mathcal{F}) \le \lfloor n/2 \rfloor$ *if $n \ge 4$.*

Only the second bound in (iv) uses the computer-assisted Theorem 3.8; in particular the bound 2 for cacti, outerplanar patterns with triangles and K_4 no longer needs it.

Proof. (i) The 2-degenerate case is Theorem 8.29. For the wheel with hub h and rim $r_0, \dots, r_{m-1}$ ($m \ge 3$), use Lemma 8.30 with $\ell \equiv 2$, the ordering $r_1, \dots, r_{m-1}, r_0, h$, the edges r_0r_1 and r_0h paid, and all other edges absorbed: then (π, a) is $(0,2)$ for $r_1, \dots, r_{m-1}$, $(1,0)$ for r_0 and for h. Tightness is shown as in Corollary 8.22(i). Of the examples in (i), $K_4 = W_3$ is a wheel; outerplanar graphs (in particular fans), series–parallel graphs and cacti are 2-degenerate, and so is a planar graph of girth at least 6, since on $n' \ge 3$ vertices it has fewer than $\frac{3}{2}n'$ edges. (ii) The planar bound is

Theorem 8.32, and graphs of treewidth k are k-degenerate. (iii) A block of maximum degree Δ has an orientation with all in-degrees at most $\lceil \Delta/2 \rceil$ (proof of Theorem 8.31), so it is $(\lceil \Delta/2 \rceil + 1)$-good by the proof of Theorem 8.31, and Corollary 8.7 combines the blocks; for the second bound order the high-degree functions last and apply Theorem 8.29. (iv) Every pattern on n vertices is a subgraph of K_{2L} with $L = \lceil n/2 \rceil$, and of K_{2L+1} with $L = \lfloor n/2 \rfloor$. For $L \geq 3$, K_{2L} with all levels L reduces to K_{2L-2} with all levels at least $L-1$ by two applications of Lemma 8.28: delete a vertex with $|J| = L$ (the other $L-1$ vertices drop to $L-1 \geq 2$), then one of these with J consisting of the other $L-2$ of them and one vertex of the first J (the remaining $L-1$ vertices drop to $L-1$). Likewise K_{2L+1} with all levels L reduces to K_{2L-1} with all levels $L-1$: delete a vertex with $|J| = L$, then one of the L vertices that dropped to $L-1$, with J the other $L-1$ of them. The bases are K_4 at level 2, a wheel (part (i)), and K_5 at level 2 (Theorem 3.8(b) and Lemma 8.5). □

For small n, Section 3.5 gives better bases: every 2-sparse family of at most six functions has discrepancy at most 2 (Theorem 3.10), and the leveled version below gives 3 for eight functions.

Theorem 8.34 (computer-assisted)**.** *Let (G, ℓ) be a leveled graph in which no two vertices of level 1 are adjacent. If G has at most five vertices, or six vertices of which at most two have level 1, then (G, ℓ) is good. Consequently every 2-sparse family of eight functions has discrepancy at most 3. Every 2-sparse family has discrepancy at most 2 if its pattern becomes a graph on at most six vertices after repeatedly deleting vertices with at most two neighbours, or if its pattern is a graph on seven vertices other than the eight graphs $K_7 - E(F)$ with F isomorphic to a subgraph of C_4, $3K_2$ or K_3.*

Proof. As for Lemma 8.5, (G, ℓ) is good if and only if every 2-sparse instance with graph in G has a colouring with $|g_v| \leq \ell_v$ for all v; by Lemma 3.9 with $D_v = \ell_v + 1$ it suffices to clear the thresholds $\ell_v + 0.99$ in the corner method of Section 3.5. Since higher levels only help, it suffices to take $G = K_n - E(K_I)$ with I the (independent) set of vertices of level 1 and all other levels 2 ($|I| = 0$ is Theorem 3.10(a), and $|I| = n$ is trivial); the patterns use only edges of G and are enumerated up to $\mathrm{Aut}(G, \ell)$. All cases are cleared (3966 patterns for $n \leq 5$; 4755 and 7297 patterns for $n = 6$ and $|I| = 1, 2$), by two independently written programs (Appendix A).

For eight functions: in K_8 with all levels 3, delete a vertex by Lemma 8.28 with $|J| = 3$, leaving K_7 with levels $(2, 2, 2, 2, 3, 3, 3)$; then delete a vertex of level 2 with J two other vertices of level 2, leaving K_6 with levels $(1, 2, 2, 2, 2, 2)$, which is good. For the second claim, delete the peeled vertices in order by Lemma 8.28 with $J = N(h)$ at level 2, and use Theorem 3.10(a) for the rest. For seven vertices, all at level 2: a vertex h of degree at most 2 is deleted with $J = N(h)$, and one of degree 3 with $|J| = 2$, leaving six vertices of which at most one has level 1. A graph on seven vertices with at least eight non-edges has a vertex of degree at most 3. The 146 graphs with at most seven non-edges were checked by an exhaustive search over sequences of applications of Lemma 8.28 ending in the bases of the first part; it succeeds for all of them except the eight graphs K_7 minus a subgraph of C_4, $3K_2$ or K_3. Two independently written searches agree. □

For six vertices and three vertices of level 1 the corner method fails, although the leveled graphs in question may well be good: with $I = \{0, 1, 2\}$ and items $2i, 2i+1$ read by i and by 3, the mean-zero residuals $h_i = z_{2i} + z_{2i+1}$ ($i \in I$) and $h_3 = 3 - 2|\{i : z_{2i} = z_{2i+1}\}|$ under the uniform measure violate the thresholds $(1, 1, 1, 2)$ at every corner, and hence also the thresholds $\ell_v + 0.99$ used above, since $|h_i| \in \{0, 2\}$ and $|h_3| \in \{1, 3\}$; yet the star they live on is good by Theorem 8.21(a).

Remark 8.35. (1) Both relaxations in Corollary 8.27 are needed. With steps of length 1 instead of 2 the statement fails: for $|S| = 3$, $T = \{t_1, t_2\}$ and $\psi = t_1 + t_2 + s_1 s_2 s_3$, the slice of $\{|\psi| \leq 2\}$ at $(1, 1)$ is $\{s_1 s_2 s_3 = -1\}$, four corners at pairwise distance 2, at distance 3 from their antipodes, and the

slice at $(-1,-1)$ is its negative; so the variant with 1-steps contains neither point of the antipodal 2-path $(1,1),(-1,-1)$, while $Z^\natural = \{\pm 1\}^T$. With deficiency 0 (that is, 2-loops) it fails by the example of Remark 8.38(1), whose slice $\{s_0 = -1\}$ contains no antipodal pair. (2) A deleted vertex cannot make more than ℓ_h neighbours cheap with demands of this product form: let $q = \ell_h + 1$ and $Z_{r_i} = \{|\psi_i| \le \ell\}$ with $\psi_i = \sum_{j\in S_i} s_j + \sum_{j\in T_i} \rho_j$, $|S_i|$ odd and $|T_i| = \ell$ $(i = 1,\dots,q)$. At $\rho_i = \mathbf{1}$ the slice is $\{-2\ell \le \sum_{S_i} s \le 0\}$, which contains a near-loop, so the transferred constraints allow $\rho_i = \mathbf{1}$ for all i; but $\varphi_h = \sum_i \sum_{j\in S_i} s_j$ is at most $-q$ on the product of these slices, so $Z_h = \{|\varphi_h| \le \ell_h\}$ misses it. (3) *A counting limit.* Consider any sequence of applications of Lemma 8.28 (which includes the ordinary deletions, $J = \emptyset$) that deletes all vertices of a leveled graph (G,ℓ). Let $a_v = |J|$ when v is deleted and let π_v be the number of levels v has lost before. Every edge is counted once, by a at the endpoint deleted first or by π at the other one, and $a_v \le \ell_v - \pi_v$ since $|J|$ is at most the current level. Hence $|E(H')| \le \sum_{v\in V(H')} \ell_v$ for every subgraph H' of G, and a uniform level ℓ must satisfy $\ell \ge \max_{H'} |E(H')|/|V(H')|$, i.e. $\ell \ge p(G)$; Theorem 8.31 is within one of this limit. For K_n this gives $\ell \ge \lfloor n/2 \rfloor$, and $\ell \ge \lceil n/2 \rceil$ if no base case is used, because the last vertex has $a_v = 0$ and $\pi_v \le \ell - 1$; so Corollary 8.33(iv) is optimal for such schedules. Schedules that end in a base case proved by other means can do better by an additive constant: Theorem 8.34 gives level 3 for K_8, where $p(K_8) = 4$, from a computer-verified six-vertex base with a vertex of level 1.

Proof of Theorem 1.9. For $d = 0$ the functions share no items and Theorem 8.1 gives 1. For $d \ge 1$ the bound $\max(2,d)$ is Theorem 8.29, and the bound $p(H)+1$ is Theorem 8.31. Tightness for $d \le 2$: a pattern with an edge carries a family of discrepancy arbitrarily close to 2, as in the proof of Corollary 8.22(i). Part (a) is Corollary 8.33(ii), together with the fact that K_r-minor-free graphs are $O(r\sqrt{\log r})$-degenerate [Kos84, Tho84]; part (b) is Corollary 8.33(i), and subgraphs of $K_{q,n}$ are q-degenerate (order the side of size n first); parts (c) and (d) are Corollary 8.33(iii) and (iv). □

8.8 Dense patterns

For a dense pattern such as K_n, Theorem 1.9(d) gives only $\lfloor n/2 \rfloor$, worse than the bound $\sqrt{2n(1+\ln 2n)}$ of [HMMS25]. When any two functions share few items, the logarithm can be removed by replacing the randomized rounding of [HMMS25] with a deterministic assignment of the fractional items to their readers.

Theorem 8.36. *Let $\mathcal{G}$ be an instance with n functions in which any two functions share at most s items.*

(a) If $\mathcal{G}$ is 2-sparse, then $\operatorname{disc}(\mathcal{G}) \le 1 + 2\lceil\sqrt{sn/2}\,\rceil$. *In particular every 2-sparse instance whose item multigraph is simple (for example on K_n) has* $\operatorname{disc}(\mathcal{G}) \le 1 + 2\lceil\sqrt{n/2}\,\rceil$.

(b) If $\mathcal{G}$ is t-sparse, then $\operatorname{disc}(\mathcal{G}) \le 1 + 2s\sqrt{t(t-1)n}$.

Proof. By Corollary 2.8 there is $x \in [-1,1]^m$ with a set Φ of at most n fractional coordinates such that $G_v(x) = 0$ for all v. Fix the integral coordinates to their values in x, let $r_v\colon \{\pm 1\}^{U_v\cap\Phi} \to \mathbb{R}$ be the resulting restriction of g_v, and let ξ be the product distribution on $\{\pm 1\}^\Phi$ with means x_e, $e \in \Phi$; it has full support, r_v is 2-Lipschitz, and $\mathbb{E}\, r_v(\xi) = G_v(x) = 0$.

Fractional items in no support are irrelevant and are rounded arbitrarily. Assign every other $e \in \Phi$ to one of the functions whose support contains it, its *owner*, and let $O_v \subseteq U_v \cap \Phi$ be the items owned by v and $k_v = |U_v \cap \Phi \setminus O_v|$. Let $\bar r_v(y) = \mathbb{E}\, r_v(y,\xi')$ for $y \in \{\pm 1\}^{O_v}$, where ξ' is the restriction of ξ to the remaining fractional items of v. Then $\bar r_v$ is 2-Lipschitz, $\mathbb{E}\,\bar r_v(\xi|_{O_v}) = 0$, and $|r_v(y,y') - \bar r_v(y)| \le \mathbb{E}|r_v(y,y') - r_v(y,\xi')| \le 2k_v$ for all y, y'. Since the sets O_v are disjoint, they can be coloured independently. As $\bar r_v$ has mean 0 under a distribution with full support on

$\{\pm1\}^{O_v}$, it takes a value ≥ 0 and a value ≤ 0, and by 2-Lipschitzness some point of an edge path between them has $|\bar{r}_v| \leq 1$ (if $O_v = \emptyset$, then $\bar{r}_v = 0$). Colour O_v by such a point for every v. Then $|g_v| = |r_v| \leq 1 + 2k_v$ for every v, and it remains to choose the owners so that $\max_v k_v$ is small.

(a) Let F be the multigraph on V with one edge uv for each $e \in \Phi$ read by u and v (fractional private items are owned by their reader). Choosing an owner is orienting the edge towards the non-owner, and k_v is the in-degree of v. A subgraph of F on j vertices has at most $\min(n, s\binom{j}{2})$ edges, which is at most $j\lceil\sqrt{sn/2}\,\rceil$ (for $s = 0$ there are no edges; let $s \geq 1$): if $j \leq \sqrt{2n/s}$, then $s\binom{j}{2} < sj^2/2 \leq j\sqrt{sn/2}$, and otherwise $n < j\sqrt{sn/2}$. By Hakimi's theorem [Hak65], F has an orientation with all in-degrees at most $\lceil\sqrt{sn/2}\,\rceil$.

(b) Let F be the simple graph on V in which $u \sim v$ if they share an item of Φ; it has at most $|\Phi|\binom{t}{2} \leq nt(t-1)/2$ edges, and hence degeneracy $\delta \leq \sqrt{t(t-1)n}$, since a graph of degeneracy δ has a subgraph of minimum degree δ and thus at least $\delta(\delta+1)/2$ edges. Order V so that every function has at most δ later neighbours in F, and let the owner of each item be its last reader. Then every item counted in k_v is shared with one of the at most δ later neighbours of v, so $k_v \leq s\delta$. □

Theorem 1.10 is Theorem 8.36 applied to $\mathcal{G}(\mathcal{F})$, using $1 + 2\lceil\sqrt{n/2}\,\rceil \leq \sqrt{2n} + 3$.

For arbitrary bundle sizes, a weighted union bound in the randomized rounding of [HMMS25] removes both the logarithm and the dependence on s.

Theorem 8.37. *Every t-sparse instance with n functions, $t \geq 2$, has* $\operatorname{disc}(\mathcal{G}) \leq \sqrt{2\ln(2t)\,n}$. *In particular every 2-sparse family of n functions has* $\operatorname{disc}(\mathcal{F}) < 1.666\sqrt{n}$, *for arbitrary bundle sizes.*

Proof. Let x, Φ ($|\Phi| \leq n$), ξ and r_v be as in the proof of Theorem 8.36, and let $d_v = |U_v \cap \Phi|$, so that $\sum_v d_v \leq t|\Phi| \leq tn$; if $d_v = 0$ then $r_v \equiv G_v(x) = 0$. For $d_v \geq 1$, $r_v(\xi)$ has mean 0 and bounded differences 2 in d_v independent coordinates, so $\Pr[|r_v(\xi)| \geq \lambda] \leq 2e^{-\lambda^2/(2d_v)}$ by McDiarmid's inequality [McD89]. Let $C^2 > 2\ln(2t)$ and $\lambda = C\sqrt{n}$. Since $C^2 > 2\ln(2t) \geq 2\ln 4 > 2$, the function $d \mapsto d^{-1}e^{-C^2n/(2d)}$ is increasing on $(0, n]$, so $e^{-C^2n/(2d_v)} \leq (d_v/n)e^{-C^2/2}$, and $\sum_v \Pr[|r_v(\xi)| \geq C\sqrt{n}] \leq \frac{2}{n}e^{-C^2/2}\sum_v d_v \leq 2te^{-C^2/2} < 1$. Hence some outcome of ξ gives a colouring with all $|g_v| < C\sqrt{n}$, and letting $C^2 \downarrow 2\ln(2t)$ proves the claim. □

By Lemma 3.9, applied to the class of t-sparse supports with n functions, the bound improves to $\lfloor\sqrt{2\ln(2t)\,n}\rfloor$. For K_n with arbitrary bundles this improves $\min(\lfloor n/2\rfloor, \sqrt{2n(1+\ln 2n)})$ for $n = 12$ and all $n \geq 14$; for single-item bundles Theorem 8.36(a) remains better for large n. Neither argument gives a bound independent of n, and the residual method cannot give one: two functions sharing k fractional items can have mean-zero 2-Lipschitz residuals $h_1 = a_k - |S|$ and $h_2 = S$, with $S = \sum_i z_i$ and $a_k = \mathbb{E}|S|$ under the uniform measure, and then $\max(|h_1|, |h_2|) \geq a_k/2 \approx 0.4\sqrt{k}$ at every corner. For $k = 25$ this is at least 3 at every corner (by parity), so the corner method of Sections 3.4 and 3.5 cannot prove Conjecture 1.12 for patterns with at least 25 functions and a bundle of at least 25 items, although $\mathcal{K}_2 = 2$.

Remark 8.38 (limits of the method)**.** (1) *The loss in Corollary 8.16 is optimal.* For $S = \{s_0\}$, $|T| = \ell$ and $\psi(s, \rho) = s_0 + \sum_{j\in T}\rho_j$, the set $\mathcal{G}_Z$ for $Z = \{|\psi| \leq \ell\}$ is $\{|\sum_j \rho_j| \leq \ell - 1\}$, which misses the antipodal ℓ-path $(\mathbf{1}, -\mathbf{1})$ and so is not ℓ-coarse. For $T = \emptyset$ the threshold 2 in Lemma 8.15 is tight: $\psi(s) = \sum_{j\in S} s_j$ with $|S|$ even has $\{|\psi| \leq 1\} = \{\psi = 0\}$, which contains no 1-loop. So every deletion step of the kind used in Theorem 8.1 costs a level; Section 8.7 avoids this loss for up to ℓ_h neighbours with a weaker demand.

(2) *The method is specific to pairwise-shared bundles.* For $t \geq 3$ a bundle at a hub may be read by two further functions r_1, r_2, and the hub then needs a loop inside the intersection of their slices. By Lemma 2.12 it would suffice that each slice be 2-coarse, but the corresponding transfer statement fails with unbounded loss. Let $S = S_1 \sqcup S_2$, where each S_j is split into M pairs, and let

$e_j(s)$ be the number of pairs of S_j on which s is constant minus the number of the other pairs; e_j is even and 2-Lipschitz. Let $T = T_1 \sqcup T_2$ with $|T_j| = m \geq M$, m even, $\chi_j(\rho) = \frac{1}{m}\sum_{i\in T_j}\rho_i$, and $\psi = \frac{1}{2}(e_1\chi_1 + e_2\chi_2)$, which is odd and 2-Lipschitz. There is an antipodal 1-path in $\{\pm 1\}^T$ along which (χ_1, χ_2) runs through $(1,0), (1,1), (-1,1), (-1,0)$, so that $\max(|\chi_1|, |\chi_2|) = 1$ throughout. For each ρ on it, the four sets on which every pair of S_1 is constant or every pair is non-constant, and likewise for S_2, contain 2-loops, and $\psi(\cdot,\rho) = \frac{M}{2}(\pm\chi_1 \pm \chi_2)$ on them; on one of them $\psi(\cdot,\rho) \geq \frac{M}{2}$. So $\{s : |\psi(s,\rho)| < \frac{M}{2}\}$ misses a 2-loop and is not 2-coarse, for every ρ on the path. Thus $Z = \{|\psi| < \frac{M}{2}\}$ is ℓ-coarse for every $\ell < \frac{M}{2}$, while the set of ρ whose slice of Z is 2-coarse misses an antipodal 1-path, so the loss of such a transfer statement cannot be bounded independently of M.

9 Open problems

1. **Dense patterns ($t = 2$).** The smallest open case is the complete graph K_n with arbitrary bundle sizes. Theorem 1.9 gives $\lfloor n/2 \rfloor$, which for large n is weaker than the general bound $O(\sqrt{n \log n})$ of [HMMS25]; Theorem 8.37 gives $1.666\sqrt{n}$, and Theorem 1.10 gives $\sqrt{2n} + O(1)$ for single-item bundles. Is the discrepancy of 2-sparse families on K_n bounded? Arguments that use only the fractional point of [HMMS25], the sparsity pattern and mean-zero residuals cannot give a bound on K_n independent of n when bundles are large (end of Section 8.8). A bound independent of the pattern would answer Question 1.1 for $t = 2$, and Conjecture 1.12 asks for the bound 2. Deleting several hubs in parallel would require a multi-bundle version of the hub-transfer lemma.

2. **Degeneracy versus discrepancy.** Theorem 1.9 gives $\max(2, d)$ and $p + 1$, and is tight for degeneracy $d \leq 2$. Is $\operatorname{disc}(\mathcal{F}) = o(d)$ for 2-sparse families of degeneracy d, for example $O(\sqrt{d})$ as for single-item bundles on complete graphs (Theorem 1.10)? Deletions with 1-loop and near-loop demands cannot go below the pseudoarboricity, up to the additive gain from base cases of bounded size, and it is at least $d/2$ (Remark 8.35(3)), so a different way of combining deletions is needed. Is $\operatorname{disc}(\mathcal{F}) \leq 2$ for all planar patterns, or for K_7, as Conjecture 1.12 predicts (it holds for K_6 by Theorem 3.10)? Is every leveled graph in which no two vertices of level 1 are adjacent good (Section 8.6)? This holds for at most five vertices, and for six vertices with at most two vertices of level 1 (Theorem 8.34).

3. **Bounded bundle-degree for $t \geq 3$.** Can the exponential dependence on Δ in Theorem 1.6 be removed for $t \geq 3$, where hub transfer is not available (Remark 8.38)? Even $O(t\Delta)$ would be interesting.

4. **General t.** Is $\operatorname{disc}(\mathcal{F}) = O(\mathcal{K}_t)$, or at least $O(\sqrt{t \log t})$, for every t-sparse family? Theorem 1.3 gives $\mathcal{K}_t$ for acyclic patterns, and Proposition 4.4 shows that $\mathcal{K}_t$ itself can be exceeded for $t = 3$.

5. **The constants $\mathcal{K}_n$.** What are $\mathcal{K}_7$ and $\mathcal{K}_8$ (we know $\mathcal{K}_7 \geq 4$ and $\mathcal{K}_8 \geq 5$)? What is the growth of $\mathcal{K}_n$: is $\mathcal{K}_n = O(\sqrt{n})$, which is Question 12 of [HMMS25]? A computer-free proof of $\mathcal{K}_3 = 2$ would also be welcome.

6. **Algorithms.** Theorems 1.3–1.9 are non-constructive. The tree argument is constructive given oracles for $\mathcal{K}_t$-colourings of small subproblems. The topological part relies on Fadell–Husseini index theory, and consensus halving is PPA-complete [FRG18, FRHSZ23]. Can colourings with $O_t(1)$ discrepancy be found efficiently for the patterns covered here, for example for cycles of bundles?

7. **One cut per bundle.** Do the realizable maps $\widehat{G}_v$ on cycles always have a common zero with one cut per bundle, in contrast to Proposition 6.1? The bound 2 for cycles of bundles now follows

from Corollary 8.22, so a positive answer would mainly give a topological proof of it and might extend to other patterns covered by Theorem 1.6.

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A The computer-assisted step and ancillary files

Ancillary files. The directory `ancillary/` accompanying this paper contains:

- `k3_rigorous_exact.py` (with its helper `k3_exact.py`), the verification program for Lemma 3.5, and `k3_certificate.py`, a self-contained, dependency-free re-implementation of the same procedure;
- `cover_search.py` and the drivers `run_two.py`, `run_graphs.py`, `run_random.py` and `run_named.py`, which implement the search of Appendix B;

- `small_n_check/`, the verification of Theorem 3.8: `cover_bnb.py`, `patterns.py` and the driver `main.py` (exact integer arithmetic; about 8 seconds), the independent cross-checks `brute.py` and `simple_dfs.py` (pure rational arithmetic), `check_witness.py` for the sanity checks, and a second, independently written implementation `kn_bnb_exact.py` (part (a)) and `sparse_bnb_dfs.py` (part (b)), with their logs;
- `dense3_claude/`, the verification of Theorems 3.10 and 8.34: an exact C implementation of the corner method with real and per-function thresholds (`corner_bnb_v7_claude.c` in `code_claude/`, with 128-bit integer arithmetic), its pattern enumerators, the script `run_final_claude.sh` that re-runs every computation and writes the logs in `logs_claude/final/`, and, in `leadcheck_claude/`, the second, independently written check (pattern enumeration from scratch with Burnside counts, the program `cover_bnb.py` of `small_n_check/` run unchanged, and patched textually to per-function thresholds for the leveled cases, and an independent search over deletion schedules for seven vertices);
- `corner_check.py`, an exploratory floating-point search for four functions (not used in any proof);
- the output files of the runs reported below, and copies of the raw SAT logs (`sat_logs/`).

What the program checks. For every labelling $\lambda\colon \{0,1\}^3 \to \{1,2,3\}$ and every admissible choice of types (Lemma 3.3), the program forms the integer vectors ρ of Lemma 3.4. For a one-signed part T, $\rho_y = \operatorname{dist}(y,T) - 1$. For a mixed pair with fixed corner y_0, there are two vectors, $\rho_y = \operatorname{dist}(y,y_0) - 1$ and $\rho_y = 2 - \operatorname{dist}(y,y_0)$. The program then runs a depth-first bisection of $[0,1]^3$. A box is discarded as soon as some vector ρ satisfies $\sum_y \rho_y w_y(p) \le 0$ at all eight vertices p of the box; the weights $w_y(p) = \prod_k p_k^{y_k}(1-p_k)^{1-y_k}$ are computed with Python's `fractions.Fraction`, that is, in exact rational arithmetic. A configuration counts as a failure if some box remains undischarged at depth greater than 16. The validity of discarding a whole box rests on the convex-combination property of multilinear polynomials stated in the proof of Lemma 3.5.

Output. Running `python3 -I k3_rigorous_exact.py` (Python 3.13.5, Apple M3 Pro, about 10 seconds) printed

```
choices 7365 boxes 20613 fails 0
```

and the independent re-implementation `python3 -I k3_certificate.py` printed

```
choices 7365 boxes 20613 max_depth 6 fails 0
VERIFIED: no product weight admits a bad cover
```

Thus all 7365 configurations are refuted on all of $[0,1]^3$, which proves Lemma 3.5.

Theorem 3.8. Running `python3 -I main.py` in `small_n_check/` reports, for part (a), all boxes cleared for $j \le 4$ (13 boxes for $j = 4$, seven leaves), and for part (b), for $j = 0, \dots, 5$: $1, 2, 7, 22, 69, 198$ sparsity patterns, all cleared, with $1, 2, 7, 22, 101, 1080$ boxes (1213 in total, depth at most 3). The completeness of the pattern enumeration is checked by verifying that the orbit sizes sum to $\binom{14+j}{j}$, the number of multisets of j items over the 15 nonempty subsets of size at most two of the five functions. The independent program `sparse_bnb_dfs.py` reproduces the same pattern and box counts, and `kn_bnb_exact.py` reproduces the 13 boxes of part (a). The program `simple_dfs.py`, which uses no symmetry breaking and pure rational arithmetic, re-verified all leaf boxes; the leaf boxes of each problem have total volume exactly 2^{-j}.

Theorems 3.10 and 8.34. All thresholds are the rationals 299/100, 399/100, 499/100 (and 199/100 for level 1), and all comparisons are exact. For Theorem 3.10(a), the C program and `cover_bnb.py` (run on the C program's pattern lists, and again on pattern lists generated independently in `leadcheck_claude/`) clear all 1135 patterns; the numbers of boxes depend on the chosen representatives of the patterns (391 for $j \le 5$ in all runs, and 1256 and 1232 for $j = 6$ for the two lists). For Theorem 3.10(b), both programs need $1, 1, 1, 3, 29$ boxes for $\mathcal{K}_5$ at $c = 3.99$ and $1, 1, 1, 1, 3, 15$ for $\mathcal{K}_6$ at $c = 4.99$ (with coordinates sorted, which is valid because all functions read all items); at $c = 2.99$ for four functions and $c = 3.99$ for six functions both runs end with an uncleared box. For Theorem 8.34, the pattern counts for $2 \le n \le 5$ and $1 \le |I| \le n-1$ (9; $49 + 31$; $230 + 220 + 86$; $1052 + 1302 + 776 + 211$), and for $n = 6$ and $|I| = 1, 2$ (4755, 7297), agree with Burnside's lemma, and both programs clear all patterns. For the hardest pattern, a level-1 and a level-2 function sharing all six fractional items, the patched `cover_bnb.py` was run with the coordinates sorted (valid, since all six items have the same two readers) and needs, like the C program, 316 boxes (about 70 minutes in Python). The two seven-vertex schedule searches (`seven_vertex_claude.py` and `lead_seven_claude.py`) both find exactly the eight exceptional graphs. As a sanity check, the leveled program does not clear six vertices with $|I| = 3$.

The SAT runs. The results in Table 2 come from runs of the scripts above with kissat 4.0.4. A subset of them was re-run from the ancillary copy, with identical outcomes: cycles $(n, b) \in \{(3,3), (4,3), (5,3), (3,4), (4,4)\}$, K_5, K_6, K_4 with bundles of size 2, the wheel W_5, $K_{2,4}$ with bundles of size 2, two functions sharing $m \le 12$ items, and three functions sharing $m \le 7$ items. The same holds for the sanity checks described in Appendix B.

B Computational evidence for $t = 2$

For a fixed support structure, the question whether some instance has large discrepancy is a finite combinatorial problem.

Lemma B.1. *Let $(U_v)_{v \in V}$ be supports on $[m]$ and let $D \ge 1$ be an integer. Some instance with these supports has discrepancy greater than $D-1$ if and only if there are sets $P_v \subseteq \{\pm 1\}^{U_v}$ such that*

(i) (D-agreement) $\operatorname{dist}(z, -z') \ge D$ *for all $z, z' \in P_v$, and*

(ii) (cover) for every $x \in \{\pm 1\}^{[m]}$ there is v with $x|_{U_v} \in P_v \cup (-P_v)$.

In that case some instance with these supports has discrepancy at least D.

Proof. If disc $> D-1$, let $P_v = \{z : g_v(z) > D-1\}$. For $z, z' \in P_v$ we have $2 \operatorname{dist}(z, -z') \ge g_v(z) - g_v(-z') = g_v(z) + g_v(z') > 2D - 2$, so $\operatorname{dist}(z, -z') \ge D$ because distances are integers. Every x has some v with $|g_v(x)| > D-1$, that is, $x|_{U_v} \in P_v \cup (-P_v)$. Conversely, put $g_v = \operatorname{dist}(\cdot, -P_v) - \operatorname{dist}(\cdot, P_v)$ (and $g_v \equiv 0$ if $P_v = \emptyset$). This function is odd and 2-Lipschitz, and $g_v \ge D$ on P_v by (i), so by (ii) every x has some $|g_v(x)| \ge D$. □

With $D = 3$, "no cover" means that *every* instance with the given supports has discrepancy at most 2. We encode (i) and (ii) in CNF with one Boolean variable for each pair (v, z), $z \in \{\pm 1\}^{U_v}$. Condition (i) becomes binary clauses: z and z' cannot both lie in P_v if $\operatorname{dist}(z, z') > |U_v| - D$. Condition (ii) becomes one clause for each colouring x up to global sign, which gives 2^{m-1} clauses. For all instances in Table 2, $m \le 20$, and all 2^{m-1} cover clauses are included in a single CNF. The answer UNSAT therefore certifies (modulo the correctness of the solver and of the encoding) that

Support structure (item multigraph)	Items m	Result for $D = 3$
two functions sharing all items ($t = 2$)	$m \leq 15$	no cover
three functions sharing all items ($t = 3$)	$m \leq 8$	no cover
cycle C_n with bundles of size b, for $(n, b) \in \{(3,3),(4,3),(5,3),(6,3),(3,4),(4,4),(5,4)\}$	9–20	no cover
complete graphs K_5, K_6 (simple), K_4 with bundles of size 2	10, 15, 12	no cover
wheels W_5, W_6 (rim bundles of size 1, spokes of size 2)	15, 18	no cover
$K_{2,4}$, $K_{2,5}$ with bundles of size 2 (two hubs in one block)	16, 20	no cover
random multigraphs, 3–6 functions, some private items	12–18	146 no cover, 2 time-outs

Table 2: Exhaustive searches for discrepancy-3 instances ($D = 3$ in Lemma B.1). "No cover" means that every instance with this support structure has discrepancy at most 2. Rows with at most five functions (both t-sparse rows, C_n for $n \leq 5$, K_5, K_4 with bundles of size 2, and the random instances with 3–5 functions, including both time-outs, which have three functions) are implied by Theorems 1.2 and 3.8, and all rows with six functions (C_6, K_6, W_5, $K_{2,4}$ and the random six-function instances) by Theorem 3.10. Time-outs were at 1200 seconds.

no cover exists. Models, when found, are verified independently by recomputing $\min_x \max_v |g_v(x)|$. We used the solver kissat (version 4.0.4).[1] As sanity checks, the search finds covers with $D = 2$ for two functions sharing two items and for a 4-cycle with bundles of size 2, both verified to have discrepancy 2. It finds none with $D = 2$ for a single function, in accordance with $\mathcal{K}_1 = 1$.

The results are summarized in Table 2. All rows are now covered by theorems, for all bundle sizes (computer-assisted for $\mathcal{K}_3 \leq 2$ and Theorems 3.8 and 3.10): the first two rows ($\mathcal{K}_2 = \mathcal{K}_3 = 2$), the cycles (Theorem 1.9(b)), K_5, K_4 with bundles of size 2 and the random instances with at most five functions (Theorem 3.8), $K_{2,4}$ and $K_{2,5}$ (Theorem 1.9(b)), the wheels W_5 and W_6 (Corollary 8.33(i)), and K_6 and the 43 random instances with six functions (Theorem 3.10). The SAT computations, which were done first and motivated Conjecture 1.12, are therefore only an independent consistency check of these theorems for the bundle sizes tested; no UNSAT certificates were checked independently.

[1] https://github.com/arminbiere/kissat.